

Uniform Cartier sections for Fano type contractions

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September 25, 2026

Abstract

We prove the Cartier-divisor conjecture of Birkar and Shokurov for rational boundaries in characteristic zero and for real boundaries over $\mathbb{C}$. For an ϵ -lc Fano type contraction with positive-dimensional base and nef negative log canonical class, a Cartier divisor through any prescribed base point can be chosen with pullback log canonical threshold bounded below in terms of the dimension and ϵ alone.

Introduction

Through a prescribed point of a singular variety, can one choose a Cartier hypersurface whose log canonical threshold is bounded away from zero? For a Fano type contraction $f : X \rightarrow Z$, the corresponding question asks for a hypersurface on Z whose pullback has controlled singularities along the entire fibre. The hypersurface may depend on the contraction and the point. The desired lower bound depends only on the dimension of X and a positive lower bound for its log discrepancies.

We work over an algebraically closed field of characteristic zero. A *contraction* is a projective surjective morphism of normal integral varieties with connected fibres, expressed by $f_*\mathcal{O}_X = \mathcal{O}_Z$. It is of *Fano type* if an effective rational boundary C exists such that (X, C) is klt and $-(K_X + C)$ is ample over Z . This auxiliary boundary need not equal the boundary in the following theorem.

A rational pair (X, B) has effective rational boundary B and rationally Cartier log canonical divisor $K_X + B$. If $p : W \rightarrow X$ is a smooth birational model and E is a prime divisor on W , our discrepancy convention is

$$a(E; X, B) = 1 + \text{coeff}_E(K_W - p^*(K_X + B)).$$

Thus ϵ -lc means $a(E; X, B) \geq \epsilon$ for every such E ; lc means that they are nonnegative; and klt means that they are strictly positive.

The obstruction is visible in a single ratio. If D has local equation g on the base, log canonicity after adding tf^*D requires

$$t \leq \frac{a(E; X, B)}{\text{ord}_E(f^*g)}$$

for every prime divisor E over the relevant inverse image with positive denominator. A discrepancy bound alone does not control those orders. Even on a smooth curve, the divisor $m[z]$ has threshold $1/m$. The problem is therefore to choose an equation through z for which all these inequalities hold with one uniform positive t .

Theorem 1 (Uniform Cartier sections). *For every integer $d > 0$ and every real number $\epsilon > 0$, there is a real number $\tau = \tau(d, \epsilon) > 0$ with the following property. Let (X, B) be an ϵ -lc rational pair of dimension d , and let $f : X \rightarrow Z$ be a Fano type contraction of normal integral quasi-projective varieties with $\dim Z > 0$. Suppose that $-(K_X + B)$ is nef over Z . For every closed point $z \in Z$, there are an open neighbourhood U of z and a nonzero effective Cartier divisor D on U , with $z \in \text{Supp } D$, such that*

$$(f^{-1}(U), B|_{f^{-1}(U)} + \tau f^* D) \quad \text{is log canonical.}$$

The conclusion of Theorem 1 concerns the whole inverse image of a neighbourhood of z . The constant is independent of Cartier indices, boundary denominators, and any choice of polarization. It is non-effective; the divisor and the neighbourhood may depend on the contraction and the point.

Cartier sections and complements. For the identity contraction, the question above concerns arbitrary ϵ -lc singularities. Equivalently, it asks for a uniform positive lower bound for the log canonical threshold of the maximal ideal at a closed point; the passage to a general element of that ideal is recalled in Lemma H.6. The general contraction requires the corresponding bound for the pulled-back point ideal. This is a different local question from bounding the discriminant coefficients in the canonical bundle formula: those coefficients are defined by thresholds over generic points of divisors on birational models of the base. A Cartier section through a prescribed point must control the whole fibre at once.

The complement formulation asks for a uniformly bounded complement index. Over a neighbourhood of a base point, a *monotonic klt n -complement* of (X, B) is an effective rational divisor $\Lambda \geq B$ such that (X, Λ) is klt and

$$n(K_X + \Lambda) \sim 0.$$

Here linear equivalence includes the assertion that the displayed divisor is Cartier. In Shokurov's boundedness conjecture, as formulated in [10, Conjecture 1.1], the index n depends only on the total dimension d , the discrepancy bound $\epsilon > 0$, and a fixed finite set $\mathfrak{R} \subset [0, 1] \cap \mathbb{Q}$ containing the coefficients of B . The geometric hypotheses are those of Theorem 1, with the zero-dimensional base also allowed.

Complements entered birational geometry through Shokurov's work on threefold log flips, including bounded local lc complements for surfaces [31, Section 5]. His surface theorem and bounded-complement conjectures developed the relation between mild singularities and bounded complement indices [32, Conjecture 1.3 and Main Theorem 1.4]. The stronger expectation that bounded complements preserve a positive lower bound on log discrepancies appears in his 2004 formulation [33]. Birkar proved boundedness of lc complements in arbitrary dimension for Fano type pairs with hyperstandard coefficients [3, Theorems 1.7–1.8]. Those lc complements provide the bounded initial index used below. A klt complement must have strictly positive discrepancies, and obtaining this additional property uniformly is the issue addressed here.

Several earlier results describe the reach of the Cartier-section problem. Sankaran and Santos bounded the smallest weight of an ϵ -lc weighted blowup of affine space [30, Theorem 1.3], proving the weighted-blowup case of Birkar's question. For torus-invariant boundaries with coefficients in a fixed DCC set (one with no infinite strictly decreasing sequence), Ambro constructed invariant Cartier sections through torus-fixed base points in the toric setting [2, Theorem 3.12, arXiv version 2]. Under his hyperstandard coefficient hypothesis, his Theorem 1.2 also gives bounded complements preserving a positive discrepancy bound. Birkar's control of base singularities gives the Cartier assertion over curves, together

with bounded klt complements in the curve-base case [5, Theorem 1.1 and Corollary 1.4]. These curve and toric results illustrate two distinct ways to control the orders appearing in the threshold inequalities.

Over $\mathbb{C}$, Chen proves the decisive general reduction. For each bound ℓ on the base dimension, bounded klt complements, the Cartier-section assertion, and its identity-contraction case are equivalent; on the complement side, birational contractions with zero boundary suffice [10, Theorem 1.6]. He proves the surface local assertion with coefficient $\epsilon^2/24$, and hence both conjectures for bases of dimension at most two [10, Theorem 1.7 and Corollary 1.8]. He also removes the coefficient restriction in the toric Cartier-section assertion [10, Theorem 1.11]. We use the precise local formulation of the Birkar–Shokurov conjecture in [10, Conjecture 1.3]; the relatively ample formulation also appears in [7, Conjecture 1.27].

Theorem 1 proves the rational-boundary assertion in characteristic zero, and Corollary 3 below gives the real-boundary formulation over $\mathbb{C}$. The conjecture is therefore resolved positively in these settings. The new input to Chen’s reduction is the following birational theorem.

Theorem 2 (Birational klt complements). *For each $d > 0$ and $\epsilon > 0$, there is a positive integer $n = n(d, \epsilon)$ such that the following holds. Let X be an ϵ -lc rationally Gorenstein variety of dimension d , and let $f : X \rightarrow Z$ be a birational Fano type contraction with $-K_X$ nef over Z . For every closed $z \in Z$, there is an effective rational divisor Δ over a neighbourhood of z such that (X, Δ) is klt there and $n(K_X + \Delta)$ is Cartier and linearly trivial there, after shrinking the neighbourhood.*

The companion article [27, Theorem 1.1 and Corollary 1.2] treats bounded klt complements for Fano contractions and their finite-rational-coefficient extension by an independent argument. The present proof establishes Theorem 2 directly, through its own character and pinning obstruction.

Applying Chen’s Theorem 1.6 in arXiv version 4, implication (2) $\Rightarrow$ (3), transfers Theorem 2 to Theorem 1. For a fixed total dimension d , use the base-dimension bound $\ell = d$ and the birational theorem in every dimension at most d , for every positive discrepancy gap. Implication (2) $\Rightarrow$ (1) of the same theorem also supplies the finite-rational-coefficient complement assertion stated above over $\mathbb{C}$. Its Cartier-section conclusion already allows real boundaries, giving the following extension.

Corollary 3 (Real boundaries). *Over $\mathbb{C}$, the conclusion of Theorem 1 holds for effective real boundaries B with $K_X + B$ real Cartier, with the same uniform dependence on d and ϵ .*

The reduction requires rationalization only in the zero-boundary birational case: an integral Weil divisor that is real Cartier is rationally Cartier, and a real klt log Fano witness can be chosen rational. Section H supplies these details, the reduction from arbitrary algebraically closed characteristic-zero fields to $\mathbb{C}$ for Theorem 1, and the passage between point-ideal thresholds and individual Cartier divisors. No approximation of the arbitrary nef real boundary in Corollary 3 is required.

The argument. Suppose that Theorem 2 fails. Section H uses bounded lc complements to construct a sequence of complex klt germs $x_i \in S_i$ of bounded dimension, with finite cyclic groups Γ_i fixing x_i . The quotient maps

$$S_i \longrightarrow S_i/\Gamma_i$$

are quasi-étale, meaning étale outside subsets of codimension at least two, and the quotient germs have a fixed positive discrepancy gap for primitive prime orders. On S_i , a nonzero

regular function h_i and a nowhere-vanishing canonical generator have the same character χ_i . The pair $(S_i, \text{div}(h_i))$ is lc, and a centred quasi-monomial valuation T_i satisfies

$$A_{S_i}(T_i) = T_i(h_i) = 1,$$

where A_{S_i} is the log-discrepancy function with zero boundary. Failure of bounded klt complements gives the stronger condition

$$\forall m \in \mathbb{Z}_{>0} \exists i(m) \forall i \geq i(m) : \quad T_i(s) \geq m \quad \text{for every nonzero regular } s \text{ of character } \chi_i^m.$$

Thus the order bound holds for every section in each fixed character degree, rather than for a selected sequence of sections. The reduction uses a canonical cover in the isomorphism case and a signed anti-canonical grading for other birational contractions.

Character relations become concrete after division by powers of h_i . For a section s of character χ_i^m , the rational function $Y = s/h_i^m$ is Γ_i -invariant and

$$T_i(Y) = T_i(s) - m.$$

Requiring nonnegative order for both Y and Y^{-1} forces the exact equality $T_i(Y) = 0$. The construction uses pairs of ratios with opposite orders to impose these two inequalities. The induction builds and retains relations of this kind. It optimizes valuations on the constrained h_i -lc slice, passes to finitely generated central algebras, and imposes further invariant relations while preserving the algebraic independence of the variables already obtained. Either it produces another independent variable or the remaining character direction yields a quotient prime of arbitrarily small discrepancy. Dimension bounds the first process, while the discrepancy gap excludes the second. Theorem G.1 formalizes this obstruction, and Section H uses it to prove the birational theorem before applying Chen's equivalence.

The proof requires uniform geometric estimates and exact control of the valuation constraints. Sections A–C establish the discrepancy comparisons, phase nonvanishing, and saturation statements for fields obtained by cutting off small discrepancy directions. Their geometric inputs include Birkar's boundedness of weak log Fano varieties [4, Theorem 1.1], boundedness of rationally connected generalized Calabi–Yau varieties up to isomorphism in codimension one [5, Theorem 1.7], and effective birationality for nef and big integral Weil divisors [6, Theorem 1.1]. The latter theorem also requires the polarization minus the canonical divisor to be pseudo-effective. These results apply without a prior bound on the nef divisors' Cartier indices. Filipazzi's generalized canonical bundle formula supplies the generalized-pair structures on successive bases [12, Theorem 1.4]; the uniform discrepancy comparisons needed here are established in Section A.

For the local optimization, the normalized-volume theory of Li, Blum, Xu, Li–Xu, and Xu–Zhuang supplies existence, quasi-monomiality, uniqueness, and stable degeneration for ordinary klt-pair minimizers [9, 24, 25, 36–38]. The order inequalities above impose additional constraints. Section E constructs an effective rational klt pair whose ordinary minimizer lies exactly on the constrained slice; only then is the stable-degeneration theorem applied. Section F transfers the resulting equalities and signs through later degenerations and changes of grading. Together with the character counts of Section D, these exact transfers ensure that each step of the induction in Section G retains the earlier relations.

Conventions on generalized pairs and sequences. For a generalized pair, a rationally Cartier nef b-divisor $\mathbf{M}$ is determined and nef on a model $\pi : Y \rightarrow X$. On that model the boundary is defined by

$$K_Y + B_Y + \mathbf{M}_Y = \pi^*(K_X + B + \mathbf{M}_X).$$

Discrepancies on higher models are the ordinary discrepancies of the resulting, possibly noneffective, boundary. The order of a rationally Cartier $\mathbb{Q}$ -divisor is computed after pulling it back from a determining model. Unless expressly stated otherwise, rational divisor data are used for MMPs.

Orders and discrepancies extend homogeneously to positive multiples of divisorial valuations. A lower discrepancy bound on genuine singularities is always imposed on primitive orders ord_E , not on all of their positive rescalings. This distinction is essential when valuations on covers, quotients, and subfields are compared.

The auxiliary arguments concern sequences in bounded dimension. The symbols $O(1)$ and $o(1)$ refer to those sequences, with additional fixed parameters indicated at the point of use. Sequence conditions are required on every subsequence. We pass to further subsequences when proving a contradiction and choose auxiliary growth rates slowly enough to satisfy the preceding finite sets of requirements. Each uniform conclusion keeps this order of choices explicit. In Sections E–H, germs are complex algebraic germs. An h -lc valuation means a valuation of zero discrepancy for the pair with boundary $\text{div}(h)$, using a power to descend the rational divisor through a finite quotient when necessary.

Contents

Introduction	1
A Discriminant comparison and uniform chambers	8
A.1 Conventions and the canonical bundle formula	8
A.2 The comparison statements	9
A.3 Extraction and the Mori step	10
A.4 Tailored complements and connected minimizing places	11
A.5 Uniform charts and discrepancy estimates	13
A.6 Integral approximation on the cones	20
A.7 Proofs of the comparison statements	21
B Nonvanishing from signed phases	26
B.1 Seifert bundles and two consequences of comparison	27
B.2 Proof of phase nonvanishing	30
C Cutoff fields, phase bounds, and section counts	36
C.1 Polarized discrepancy data and cutoff fields	36
C.2 A phase estimate at a varying scale	40
C.3 Exact saturation of neutral fields	45
C.4 Adjoint lattice points for integral Weil polarizations	47
C.5 Completion of the saturation count	52
D Characters and regularization on affine cones	53
D.1 Angular fields and the discrepancy normalization	53
D.2 Which divisorial lc tests occur	56
D.3 Sections and a uniform sandwich argument	57
D.4 Bounded multiples, character counts, and tilted heights	59
E Optimization on common log canonical slices	62
E.1 Retractions, jets, and the topology of the slice	63
E.2 Convex domination and strict volume comparison	66
E.3 Exact realization by a rational klt pair	69
F Degeneration, exact sign transfer, and rank refinement	73
F.1 Presentations and rational approximations	73
F.2 The canonical calculation before lc equality	74
F.3 An exact bridge for inequalities	76
F.4 Compression of two gradings	78
F.5 Balanced cones satisfying the angular order condition	79
G Pinning the gradings	80
G.1 Costs and the induction data	81
G.2 The next character block and the terminal case	83
G.3 Neutral tests and simultaneous non-cancellation	84
G.4 The count at a high scale	86
G.5 The high-scale pin	87
G.6 Finding all necessary relations	87
G.7 Rational directions and anchor size	90
G.8 Field and lattice consequences of the pin	92
G.9 Termination	94

H	From complement failure to the pinning input	94
H.1	A prime saturating the complete systems	94
H.2	The canonical cover and the signed torsor	96
H.3	The quotient gap and matching the canonical character	100
H.4	Chen's reduction and change of field	102
H.5	Conclusion of the two theorems	103

A Discriminant comparison and uniform chambers

We establish three comparison results for generalized Calabi–Yau data. Their proofs reduce to bounded generic Fano fibres, but impose no bound on the Cartier indices of the nef b-divisors. The geometric input is a family of charts with a uniform bound on angular variation of discrepancies; the contact of a trait with the parameter space is allowed to be arbitrarily large.

A.1 Conventions and the canonical bundle formula

We use the usual generalized pair notation. A nef part $\mathbf{M}$ is a $\mathbb{Q}$ -Cartier b-divisor determined and nef on some model (nef over the point unless indicated). In particular all pullbacks involved in forming generalized discrepancies are ordinary pullbacks on sufficiently high models. Write $a(v)$ for log discrepancy, extended to positive multiples of ord_E by homogeneity; unless specified, numerical conditions on divisorial valuations are imposed for *primitive* ones (ord_E). Coefficient functions of b-divisors are extended by homogeneity in the same way. Under a generically finite pullback use log pullback, so discrepancies just compose with restriction (ramification formula, characteristic zero). Restriction of a divisorial valuation to a finitely generated subfunction field is a multiple of a divisorial valuation or trivial. Indeed, let L/K be the extension, of transcendence degree q , and let v be divisorial on L , with nontrivial restriction w . Both value groups have rational rank one and

$$\text{trdeg}_{\kappa(w)} \kappa(v) \leq q, \quad \text{trdeg}_k \kappa(w) \leq \text{trdeg}_k K - 1.$$

Equality $\text{trdeg}_k \kappa(v) = \text{trdeg}_k L - 1$ forces equality in both inequalities. Thus w is divisorial. If v has integral values, its restricted value group is a subgroup of $\mathbb{Z}$, so the multiple of the primitive order is a positive integer.

Here and below an *effective generalized CY over S* (possibly not lc) on $X \rightarrow S$ means data as above with $B \geq 0$ on X and $K_X + B + \mathbf{M}_X \sim_{\mathbb{Q}} f^*L$ for some $\mathbb{Q}$ -Cartier L on S . CY will also indicate its discrepancy function. Assertions about varying pairs below need only be read on the *rational* parameter values (piecewise linear discrepancy functions can of course be extended on their intervals). For an affine discrepancy family it suffices to choose common resolutions for finitely many defining members: differences of generalized discrepancies are Cartier b-evaluations. No denominator bounds on the nef parts are assumed. Varieties in the statements below are projective normal in characteristic zero; one can work over algebraically closed fields. The bounds/sequence statements work also with fields allowed to vary: boundedness, complements and families used below are uniform in characteristic zero (or one can work with descents to characteristic zero subfields embeddable in $\mathbb{C}$). We limit the dimension of the total space by a fixed integer.

We use the klt MMP with big boundary and the extraction and Mori dream space consequences of [8]; preservation of Fano type under rational contractions [29, p. 155 and Lemma 2.8]; rational connectedness of projective klt log Fano pairs [14, Corollary 1.13][40, Theorem 1]; BAB [4, Theorem 1.1]; and ordinary bounded lc complements [3, Theorems 1.7 and 1.8]. For complements, the seed coefficients used below are only 0 and 1, the varieties are of Fano type over the relevant base, and the negative log canonical divisor is nef. The conversion to ordinary data will be proved below; no generalized complement theorem with an unstated nef-index hypothesis is used. We also use the generalized klt-trivial canonical bundle formula [12, Theorem 1.4 and Sections 4–5]. Recall the latter in the projective rational case: for a contraction $g : X \rightarrow W$, data with nef $\mathbb{Q}$ -Cartier b-part, $K_X + B + \mathbf{M}_X \sim_{\mathbb{Q}} g^*L_W$, B effective over the generic point, generalized klt generically there, the discriminant is given on every model by

$$b(P) = \min_{T|_{k(W)} = e \text{ord}_P, e > 0} a(T)/e \tag{A.1}$$

(log-discrepancy convention) and L_W minus the sum of canonical and discriminant-boundary terms (that is, $L_W - (K_W + B_W)$, on higher models using pullback of L_W) gives a b-nef $\mathbb{Q}$ -Cartier moduli part. This is the b-nefness/b-Cartier theorem for a generalized klt-trivial fibration; the generic rank condition follows by generic klt and horizontal effectivity on X (on a resolution the rounded-up negative boundary over the generic field is effective exceptional), using $g_*\mathcal{O}_X = \mathcal{O}_W$. All data here are projective with nef part over the point, NQC by rationality. We only need the generically klt case as stated here (no global klt condition). To fit the subboundary convention in the cited theorem when some vertical trace coefficients exceed one, subtract g^*G for a sufficiently large effective Cartier divisor G on the base containing the images of those components, and replace L_W by $L_W - G$. Generic klt is unchanged. The discriminant boundary changes from B_W to $B_W - G$, so the moduli b-divisor is unchanged. Thus this harmless shift permits the stated application to arbitrary vertical coefficients. Minima are finite and attained (work over the generic DVR of P). We apply this to g over S with L_W a pullback of L . If B is effective on X , the induced trace B_W is effective because any prime on W is covered by a prime in its inverse image. Discriminants compose by the minimum formula, including through crepant changes and birational contractions preserving the CY discrepancy data. In fact pushing traces in small modifications and birational contractions over S (not extracting) preserves effectivity and discrepancies; crepancy follows by writing total log canonical divisor as a pullback from S plus a rational principal divisor.

A.2 The comparison statements

Here are three sequence versions. Subscripts indexing a sequence will often be suppressed; “eventually” allows discarding finitely many indices. Every assertion about a test sequence of valuations is required on every subsequence of the original sequence as well. Bounds may depend on the fixed dimension and on the fixed parameters explicitly appearing in a statement. In particular, a condition on every test sequence is uniform against a diagonal choice of counterexamples. A subsequent diagonal passage may make any auxiliary divergent parameter grow more slowly.

Lemma A.1 (Comparison along long parameter intervals). *Let $X \rightarrow S$ be a sequence of Fano type contractions as above. Suppose a_t are effective generalized CYs over S , continuous finite rational piecewise affine families of discrepancies (with a finite subdivision of parameter per example, so the pieces can be treated on a common resolution), on $0 \leq t \leq L_i$ or $-L_i \leq t \leq L_i$ with $L_i \rightarrow \infty$. Pieces and endpoints can be taken rational. Values at each valuation are concave in t , and a_0 is klt. Assume the corresponding one of the following (one-sided for $0 \leq t \leq L_i$; two-sided for $-L_i \leq t \leq L_i$):*

- (i) *(one-sided) the discrepancy functions are nonincreasing, and whenever $a_0(T) \rightarrow 0$, $a_t(T)/a_0(T) \rightarrow 1$ for every fixed t ;*
- (ii) *(two-sided) for some fixed $\delta > 0$, whenever $a_0(T) < \delta$, $a_t(T)/a_0(T) \rightarrow 1$ for every fixed signed t .*

As throughout these lemmas, tests on sequences of valuations include along subsequences. Then on every fixed bounded parameter range the data are eventually klt over the generic point of S . Write b_t for their discriminant discrepancies on S . In case (i), they are nonincreasing and satisfy $b_t(P)/b_0(P) \rightarrow 1$ for every fixed t whenever $b_0(P) \rightarrow 0$. In case (ii), some fixed $\delta' > 0$ gives the same convergence for every fixed signed t whenever $b_0(P) < \delta'$. These conclusions include every subsequence and primitive prime orders on every birational model of S .

Lemma A.2 (Phases under pure pullback). *Let a be a sequence of klt effective generalized CYs over S in the same Fano type setting, and let H be a sequence of $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisors on the respective bases S . Suppose for some fixed positive integer m , for every $a(T) \rightarrow 0$ the phase $v_T(mf^*H) \bmod 1$ has lifts to $\mathbb{R}$ of size $O(a(T))$ bounded below by $-o(a(T))$. Then with discriminant b , for some fixed positive multiple m' , $v_P(m'H)$ satisfies the same phase condition wherever $b(P) \rightarrow 0$.*

Lemma A.3 (A uniform linear chamber). *In the same setting suppose a is an lc effective generalized CY over S , with values in $(1/p)\mathbb{Z}$ for fixed p , and $a+td$ for $0 \leq t \leq t_0$ are effective generalized CYs, klt for positive t , $d \geq 0$ on all divisorials; take L_t affine, and $t_0 > 0$ fixed rational. Then eventually there is a uniform smaller $t_1 > 0$ on which discriminants equal $a' + td'$ with a' of bounded denominator (i.e., the same p' works on all models). There is also the following version with extra data:*

- (i) $a + \lambda h + sc$ are effective generalized CYs over S for $0 < \lambda \leq \lambda_0$, $0 < s \leq \lambda\theta_0$, $\lambda_0, \theta_0 > 0$ fixed, h of bounded denominator, $c \geq 0$;
- (ii) on $a = 0$, $d \leq c \leq Cd$, $-\eta_i c \leq h \leq Cc$ with fixed C and $\eta_i \rightarrow 0^+$.

Then there are uniform $\lambda_1, \theta_1 > 0$, with $\lambda_1 \leq \lambda_0$ and $\theta_1 \leq \theta_0$, such that eventually, on the sector $0 < \lambda \leq \lambda_1$, $0 < s \leq \lambda\theta_1$, the data are klt generically over S , their discriminants are exactly $a' + \lambda h' + sc'$, and h' has bounded denominator. Moreover, eventually there is no divisorial T horizontal over S with $a(T) = 0$, $h(T) < 0$. In particular for S a point exact nonnegativity on $a = 0$ follows even though only the approximate bound $-\eta_i c$ was assumed.

In this statement the minima are lexicographic: first minimize a/e , then d/e on its minimum locus; for the extra data first minimize a/e , then h/e , then c/e . Here $e \text{ ord}_P = T|_{k(S)}$. Thus a' itself is the discriminant in discrepancy convention even if the endpoint is not klt generically. The family $a' + td'$, including its endpoint at zero, has the generalized effective log data (with nef b -part and total logarithmic divisor L_t).

The endpoint assertion follows from the positive-parameter assertion. Indeed, for positive parameters use the generalized canonical bundle formula. The moduli b -parts chosen using L_t are affine on *all* models by the affine discrepancies; thus the limit b -divisor at zero is a difference of $\mathbb{Q}$ -Cartier b -divisors by extrapolation, and nef by passage to the limit on a common determination. The same applies on any intermediate base using pullback of L_t .

We give details of the geometry and proofs of these lemmas. Boundedness of denominators here and later does not require that the b -parts of the input have bounded Cartier index.

A.3 Extraction and the Mori step

We can pass to a small $\mathbb{Q}$ -factorialization on which the families still have effective traces. Given an effective generalized klt CY A over S and Fano type, we can extract all exceptional divisors with $A < \delta_*$ ($0 < \delta_* < 1$) to a $\mathbb{Q}$ -factorial Fano type model over S , extracting no others (or just select any subset of this list). There are finitely many: on a log resolution where the nef part descends any further ones must be toroidal for the snc support (the integral reduced-log discrepancy otherwise contributes at least 1: all lc places of a reduced snc pair are monomial at the strata), and all their weights are bounded using positivity on the finitely many components. For extraction one can pass to an ordinary big klt CY over S having the selected divisors still strictly positive in log-pulled boundary. For clarity, mix the given data with small weight of auxiliary crepant sub-data with ample b -part determined on that resolution, horizontal and vertical trace on X effective and containing a general relatively ample contribution; these auxiliary data with total $\sim_{\mathbb{Q}} 0/S$ exist by taking the ample b -part sufficiently small and using a klt log Fano pair on X/S . More explicitly subtract the small pushforward of the ample divisor and a small ample divisor on X from the log Fano ample

class (twist by a pullback if needed). Sub-klt persists on a common resolution after mixing, as do effectivity on X and positive coefficients at the selected divisors. Replace the resulting ample moduli contribution on the high model by general representatives (over S , twisting by pullback), with small coefficients. Apply klt extraction; the log transform is effective and still big over S by the general ample contribution on X avoiding the finitely many centres. Big klt CY implies Fano type.

Other CY families retain effective traces on the extracted model provided their discrepancies at the extracted divisors are at most 1. Each application below verifies this condition. When the whole list is retained, the resulting *ordinary variety* is δ_* -lc. For an exceptional divisor on a higher determination, effectivity of the trace and the negativity lemma for the difference between the pullback of the nef-part trace and its nef determination give $A_X(T) \geq A(T)$. Every divisor for which $A(T) < \delta_*$ is already on the extracted model; its ordinary discrepancy there is 1. The remaining exceptional divisors have $A_X(T) \geq A(T) \geq \delta_*$. A K -MMP over S , if $\dim X > \dim S$, now ends at a δ_* -lc Fano Mori step $g : X_1 \rightarrow W$ over S with $\dim W < \dim X$, preserving the CY discrepancies/effectivity we arranged and Fano type. Here $-K_X$ is big over S , and the ordinary discrepancies stay bounded below by monotonicity in the K -MMP; the geometric generic fibre F of g (we also write F for the fibre over $k(W)$) is a bounded Fano by BAB (generic discrepancies via horizontal divisors and separable constant extensions). W is Fano type over S (image of Fano type). If instead the contraction to S is birational no further step is needed: the function fields agree and the discriminant is the original discrepancy function. This is the terminal case in each induction.

A.4 Tailored complements and connected minimizing places

Lemma A.4 (Tools at an lc place). *Let $g : X_1 \rightarrow W$ be a Mori step as above, and let A_* be an effective generalized CY discrepancy on X_1 over S .*

- (i) *Suppose that A_* is lc over the generic point of W , and that an effective generalized klt CY over S is available for perturbation. Any prescribed horizontal divisorial lc place remains an lc place of an ordinary boundary Ω , effective on the generic fibre F , with $n(K_F + \Omega_F) \sim 0$, for a bounded positive integer n .*
- (ii) *For P on a birational model of W , suppose A_* is klt generically and T computes the discriminant $b_*(P)$. There is an ordinary n -CY over the DVR, lc in its b -discrepancies and effective horizontally on F , with n bounded, keeping T as an lc place. Here n -CY means that the ordinary total log canonical divisor, multiplied by n , is linearly trivial over the trait. Moreover for the adjusted discrepancy $A_* - b_*(P)e$, any two minimizing divisorials can be joined by a chain/interpolation of lc places in the following sense: on a log resolution over the generic DVR their strict divisors (or extractions of them) belong to a connected snc floor consisting of vertical coefficient-one divisors with monomial cones at the intersections. One can include any additional finite discrepancy data in the resolution. In particular normalized evaluations a/e of such other data interpolate continuously along the chain, using rational weights to approximate intermediate values by divisorial evaluations.*

Proof. Here everything over a generic DVR can be carried out by shrinking about a generic point of a prime in a smooth model over k . Only assertions over the generic fibre and that DVR are needed. *A model retaining the selected fibre components.* Write R for the trait ring. Fix a nonempty finite collection of divisors computing $b_*(P)$ and choose a log resolution Z over $\text{Spec } R$ on which they occur. For the adjusted generalized data $A_* - b_*(P)e$, they have boundary coefficient one. We first construct an ordinary klt pair on Z whose minimal model contracts every other special-fibre component and every horizontal divisor exceptional over F .

Subtract a sufficiently small positive multiple of the fibre from the adjusted boundary. The resulting generalized subpair is sub-klt, and its coefficients on the selected components remain positive. Mix it with a sufficiently small amount of auxiliary sub-CY data having an ample b-part on Z and effective nonexceptional horizontal trace. Such data exist because F is Fano: a small pushed ample contribution can be subtracted from $-K_F$, and vertical terms can be adjusted freely over the trait. The pushforward is rationally Cartier on the $\mathbb{Q}$ -factorial total space X_1 . Replace the mixed ample b-part by general relatively free representatives. Make these choices on a common resolution and retain a general big contribution avoiding the components whose coefficients will be changed.

Increase the coefficients of all horizontal divisors exceptional over F and all special-fibre components outside the selected collection, strictly and to values in $[0, 1)$. We obtain an effective ordinary klt boundary Δ , big over the trait, and

$$K_Z + \Delta \sim_{\mathbb{Q}, R} E,$$

where $E \geq 0$ is supported precisely on the divisors just increased. In particular at least one special-fibre component is absent from $\text{Supp } E$.

Here the relative fixed part can be computed directly. For divisible m , a rational section of $\mathcal{O}_Z(mE)$ restricts on the generic fibre to a rational function with poles only along divisors exceptional over the normal projective variety F . It descends to a regular function on F , and is therefore a constant in the trait field. The section condition on a retained special-fibre component, whose coefficient in E is zero, forces that constant to have nonnegative DVR order. Conversely every element of R gives such a section. Thus the relative section module is R , and the relative fixed part of $|mE|$ is exactly mE .

The good minimal model theorem for klt pairs with big boundary now applies, after spreading over an open neighbourhood of the generic point of P . Every curve in a negative extremal ray lies in $\text{Supp } E$, so no divisor outside this support is contracted. The section ring is preserved by the MMP. On its semiample output the surviving trace of E must therefore vanish, since otherwise it would remain a nonzero fixed part. Consequently every component of E is contracted. Call the output Y . Its boundary remains big and klt and its log canonical divisor is rationally trivial over the trait, so Y is of Fano type there. Its generic fibre agrees with F in codimension one. The original adjusted generalized data are crepant through these birational operations. Their trace on Y is effective: the only special-fibre components are the retained ones, with coefficient one, and the surviving horizontal coefficients are the effective nonexceptional coefficients over F .

Connectedness of the minimizing locus. Let $\pi : \tilde{Y} \rightarrow Y$ be a log resolution on which the nef part of the original adjusted data descends. Write $B^{\log}$ for its boundary and $H = (B^{\log})^1$. The negative boundary is π -exceptional because the trace on Y is effective. Moreover $-(K_{\tilde{Y}} + B^{\log})$ is nef over Y , and is big over Y because π is birational. The birational connectedness lemma therefore makes H connected over each fibre of π .

One can see the relevant structure-sheaf assertion directly. Put $G = \lceil -(B^{\log})^{<1} \rceil$, an effective exceptional divisor. Relative Kawamata–Viehweg vanishing for $\mathcal{O}_{\tilde{Y}}(G - H)$ gives a surjection

$$\mathcal{O}_Y \longrightarrow \pi_* \mathcal{O}_H(G).$$

The image factors through $\pi_* \mathcal{O}_H$, which injects into $\pi_* \mathcal{O}_H(G)$. Hence $\pi_* \mathcal{O}_H$ is the structure sheaf of the reduced image. This image is the entire reduced special fibre of Y , since all its components have coefficient one. The special fibre is connected by Stein factorization. Thus H is connected, and the prescribed divisors lie in one connected snc coefficient-one locus. Generic klt ensures that every component of this locus is vertical. Monomial valuations along its intersections join its divisorial orders continuously; the uniformizer has positive value

throughout. Evaluations of any additional discrepancy data are linear on these cones after taking a common resolution, so their ratios to e interpolate continuously. Rational weights recover the required divisorial interpolants.

An ordinary complement preserving the chosen place. For a tailored ordinary Ω , use the model above (or horizontally, extract the horizontal place using a CY slightly perturbed towards klt as funding). Write T_0 for the resulting prime, which has coefficient one. Run a $-(K + T_0)$ -MMP; this divisor is pseudo-effective over the trait (respectively generically) by the effective generalized CY containing T_0 . On every model the pair with strict T_0 is lc by comparison with the generalized data and b-nef negativity. At the nef model take a bounded ordinary lc complement (coefficients only 0 and 1, Fano type; in the relative case use Birkar's complement over a closed point of the relevant prime/open, after appropriate shrinkings as above, containing the generic trait/field in the resulting neighbourhood). Indeed over the trait/field the final nef divisor is semiample by Fano type; generation of a multiple spreads out, as does a Fano type witness (relative ampleness and klt) and lc of the indicated pair via resolution, shrinking by projectivity to ensure the conditions for complements over an open. Its crepant pullback boundary on the model with T_0 still dominates T_0 : it adds the pull of the effective complement addition and the effective MMP discrepancy of $-(K + T_0)$. In particular contraction of T_0 in a negative step would even contradict lc. Push generically to F , preserving effective boundary there. All complement indices use only the absolute dimension bound. This proves both assertions. $\square$

A.5 Uniform charts and discrepancy estimates

Proposition A.5 (Uniform charts). *At a chosen vertical place in Lemma A.4(ii), after a bounded degree extension of $k(W)$ and a chosen extension of the DVR and of the place, there is a toroidal cone representing the place, with nonnegative coordinates*

$$(x, e_*), \quad Mx = re_*. \quad (\text{A.2})$$

Here M ranges over a finite list of integral matrices of bounded sizes, r is integral (possibly huge), e_ is value of the new uniformizer, and original uniformizer value is pe_* with positive bounded ρ . Nonzero integral solutions on the cone give divisorial valuations (not necessarily primitive after restriction); evaluations of discrepancy data always refer to homogeneous values by restriction. For horizontal places as in the first bullet one just has an orthant on a bounded generic resolution, without e_* . For any effective generalized CY data A on X_1 over S , values on these cones are finite continuous homogeneous piecewise linear, and are Lipschitz in x with uniform constant at fixed $e_* > 0$ (uniform homogeneous Lipschitz in the horizontal case). Constants depend on the boundedness in this Mori step and the complement indices, not on r , vertical coefficients, or the nef part index or the chosen data A . In particular finitely many differences at fixed parameter separations have uniform Lipschitz bounds as well.*

The proof separates the construction of the charts from the metric estimate. The first lemma keeps track of their integral lattices and logarithmic coordinates; the second converts generic-fibre degree into an angular bound. We then replace the nef part at two chosen valuations to apply that bound to arbitrary generalized CY data.

Lemma A.6 (Bounded toroidal charts). *For the prescribed place and tailored complement in Proposition A.5, there are charts with its stated contact equations and integral-valuation property. The prescribed place is monomial on one of these charts. They have rational functions $z_1, \dots, z_f$, $f = \dim F$, with uniformly bounded pole degrees on a bounded embedding of F , such that:*

- (i) *their logarithmic differentials form a relative log basis in the vertical case and an absolute log basis over the generic field in the horizontal case;*
- (ii) *their orders are linear functions of the chart coordinates with uniformly bounded slopes;*
- (iii) *every rational monomial valuation in the vertical chart with $e_* > 0$ restricts to a Gauss valuation on $K_1(z_1, \dots, z_f)$, where $K_1/k(W)$ is the bounded extension used in constructing the chart. Horizontally the same assertion holds with the trivial valuation on the generic field.*

Here Gauss means that evaluating a polynomial in the z_j gives the minimum of its coefficient-adjusted monomial weights, without cancellation among terms of equal weight.

Proof. Bounded marked fibres and descent of an embedding. The pairs $(F, \text{Supp } \Omega_F)$ are in bounded embedded families over the geometric field. Indeed boundedness gives a very ample class $\mathcal{H}$ of bounded degree and embedding dimension on each bounded F . One can use the complete system here (dimension still bounded by degree for the nondegenerate image). To bound the anticanonical degree, let $f = \dim F$ and take a smooth complete-intersection curve C of $f - 1$ general members of the very ample system $\mathcal{H}$, avoiding the singular locus. Adjunction gives

$$-K_F \cdot \mathcal{H}^{f-1} = (f - 1)\mathcal{H}^f + 2 - 2g(C) \leq (f - 1)\mathcal{H}^f + 2.$$

For $f = 1$, the same formula is the usual degree formula on the smooth curve F . The nonzero coefficients of Ω_F are at least $1/n$, so

$$\deg_{\mathcal{H}} \text{Supp } \Omega_F \leq n \deg_{\mathcal{H}} \Omega_F = -nK_F \cdot \mathcal{H}^{f-1}.$$

The support therefore has bounded degree. In the DVR case we can secure the embedding over a bounded degree extension of $k(W)$. The geometric Picard group is free abelian of bounded rank: it injects by normality into the Picard group of a smooth projective resolution, which is rationally connected (rational connectedness for klt Fano) and has bounded Betti numbers for a suitable resolution, by resolving the bounded families after stratification. We use here the usual characteristic zero Picard consequences of smooth rational connectedness (in particular trivial connected Picard and no torsion). The finite image of Galois on the discrete Picard lattice has bounded order. One way to see the uniformity is to reduce a finite subgroup of $\text{GL}_r(\mathbb{Z})$ modulo 3: the principal congruence subgroup at 3 is torsion free, so this reduction is injective, and $|\text{GL}_r(\mathbb{F}_3)|$ bounds its order. After bounded extension the chosen $\mathcal{H}$ is invariant; its complete linear system descends projectively as a projective-space form of bounded dimension (the ambiguity on section maps is scalar), which splits after another bounded extension (Severi–Brauer index bound). This suffices. Bounded-degree embedded varieties and their marked reduced divisors as above are parametrized by finitely many Chow families [23, Definition 3.21, (3.21.1)–(3.21.3)]. Here we use effective cycles and their reduced supports, with the universal cycles restricted to smooth characteristic-zero strata.

Toroidalization of the fixed families. Apply marked weak toroidalization [1, Theorem 1.1] to these families. Its source and base modifications can be taken projective and birational; allowing a fixed alteration would give the same construction after a bounded extension. Here are the uniformity and chart details we use. By compactification and noetherian stratification, for a dense open of each parameter stratum we can use a single proper toroidalized family over a modification (an alteration would also suffice) of its compactified base. Over the dense open (which can always be shrunk, treating its complement by noetherian induction) the new fibres give birational models of the original varieties. Here one works stratum by stratum starting with geometrically integral generic fibres, retaining these and birationality in the spread-out open (in case of base alteration it suffices to modify the base-changed family birationally).

The hypotheses on the fibres parametrized, such as geometric integrality and marked support of lower dimension, can be imposed by ordinary Hilbert stratification. The compactification of a stratum serves to extend traits whose *field* point is in that dense open; their closed points need not parametrize good pairs at all. We include horizontal markings as follows: first resolve the generic fibre with its marked support over this stratum, spread out over an open, and mark the strict boundary and the exceptional locus; arrange their preimage in the toroidalization to be in the boundary. Thus on the resulting generic-fibre models relevant to us any *positive* crepant coefficient of Ω_F is supported on the toroidal boundary: outside the old marking one had smooth with crepant subboundary ≤ 0 . We arrange the dense open to be interior on the toroidal base of the family. We may use smooth strict toroidal models (also by toroidal resolution); all modifications here are of fixed finite type families. Any base alteration only requires bounded degree field extension to lift the field point on the chosen dense open. Thus after bounded degree extension of $k(W)$, say to K_1 , and extension of the place, we have a lift of the whole new trait to this base by properness, with its generic point interior. Write R for the new trait ring, t_* for its uniformizer, ρ the ramification index over the original DVR (bounded). We always use log pullback over the finite extension. One may view the trait as the localization at a prime divisor in a smooth model over k , working close to the generic point of that divisor (and spreading the map to such a neighbourhood, shrinking to pull the base boundary only to the closed-point boundary there). In particular the divisorial log structure of the trait comes from a log-smooth k -pair locally.

Full character rank and the log chart. We verify log smoothness at the strata, including the case in which the map on boundary exponent groups has a kernel. In a completed toroidal description use the full character lattices of the toric models. These include invertible torus characters as well as the characters cutting out the boundary. Dominance gives an injective map of full character lattices after tensoring with $\mathbb{Q}$. In their logarithmic differential bases its matrix has constant integral entries. It therefore has full target rank at the closed stratum as well as at the generic point. Any lattice torsion is tame in characteristic zero. The unit characters are essential here: two target boundary characters can have the same boundary exponents on the source while their quotient pulls back to a nonconstant unit, whose differential supplies the missing direction.

To detail the algebraic chart check, at closed points in these smooth strict models write the target boundary equations y_j , $1 \leq j \leq s$, as pulling to $u_j x^{\alpha_j}$ with $x_1, \dots, x_m$ source boundary parameters and u_j units. The map of log differential fibres has full target rank (as above in the completed toroidal models, where boundary equations match up to units). For $\alpha : \mathbb{Z}^s \rightarrow \mathbb{Z}^m$ the exponent map, let $K = \ker(\alpha)$. The quotient $\mathbb{Z}^s/K$ is the free group $\text{im}(\alpha)$, so K is saturated and admits an integral projection $p : \mathbb{Z}^s \rightarrow K$. The unit homomorphism u from $\mathbb{Z}^s$ factors as its kernel-direction contribution via this projection, times a homomorphism on $\text{im}(\alpha)$. The latter extends to $\mathbb{Z}^m$ after adjoining unit roots étale locally, so can be absorbed by rescaling the x_ℓ by units. We now have the chart $\mathbb{N}^s \rightarrow \mathbb{N}^m \oplus K$ using α and the projection, with the free group summand mapping on the source by unit functions $u|_K$. This is injective on groupifications with tame cokernel torsion. The induced map to the product of the target with the source toric chart scheme over $\mathbb{A}^s$ is smooth near the point: besides the rescaled x_ℓ , one needs independent ordinary differentials for those kernel units and the pulled target transverse coordinates along the source stratum. Exactly these are independent there by full log differential rank (their boundary-weight entries are zero). This proves the smoothness required in Kato's chart criterion [20, Theorem 3.5 and Remark 3.6]; see also the statement and proof in [19, Theorem 4.1, Remark 4.2 and Section 6]. In particular the required rank at a closed stratum follows from the constant full-character matrix; it does not require specializing an arbitrary ordinary differential minor.

The trait and its integral contact cone. Take the fine saturated log fibre product with

the trait equipped with its closed-point divisorial log structure. Log smoothness is preserved by fs base change and composition [16, Section 1.1]. Spread the trait to the log-smooth k -pair described above and form the same fs fibre product there. Its source is log smooth over k with the trivial log structure. It therefore admits smooth local charts over affine toric varieties [19, Theorem 4.7 and Section 7]. In these charts the monomial stratum quotient is regular, and its dimension plus the characteristic-monoid rank is the ambient dimension. Thus the spread-out source is log regular; localization gives this property for the fs fibre product over the trait. Its main component is proper over the trait and gives the desired generic model. Indeed the map from the fs fibre product to the ordinary one is proper here (in charts, integralization and finite toric saturation); the trivial-log open is dense in a log regular model, and maps to the generic point of the trait. At a stratum with boundary generators from the total family and from the trait, the cone of nonnegative orders is given precisely by $Mx = re_*$, $x, e_* \geq 0$. Here rows come from the toroidal base boundary equations, pulled to total boundary monomials times units with multiplicities M , and to the trait with contacts r ; thus M , unlike r , comes from fixed chart types. These are the matching conditions from the pushout chart (the total family and its base before this log base change can have smooth strict toroidal structures, hence boundary-coordinate charts). After spreading near the trait the base has also ordinary residue/transverse coordinates, with the prime divisor as its single boundary near the generic point in use; only its boundary weight thus enters these equations. Units have weight zero; at the point used only boundary generators vanishing at its images in the factors are counted. No further independent zero conditions on weights arise from units of the saturated monoid becoming units at the point: a positive multiple of such an element comes from the original generators and if it is a unit cannot involve any generator vanishing at the point. An integral match defines an integer-valued homomorphism on the groupification of the pushout monoid. Saturation takes place in that same groupification, so this homomorphism remains integer-valued on the saturated monoid. We may work in strict étale local charts at a geometric stratum and then restrict valuations. Thus nonzero rational weights give monomial divisorial evaluations (and the zero vector the trivial evaluation); integral ones restrict to integral multiples of primitive divisorials when nontrivial. Indeed in the split toroidal chart one is taking the dual cone of the characteristic monoid at the stratum. A nonzero rational ray even on a face can be introduced in a toroidal subdivision (over the corresponding generization stratum for a face); the primitive vector for its toric lattice then gives a boundary-divisor order. Matched integral weights as just described are integral in this lattice. For $e_* > 0$ the restriction to K_1 is $e_* \text{ord}_R$. This description is equivalently obtained by fs pulling back the log-smooth monomial charts and subdividing toroidally. Use the family of monomial valuations for one chosen lift of the stratum in a split local chart (cones/subdivisions there, then restrict to the original function field); integral points give integral orders since they already do so in such a chart. No degree bound on further strict étale local charts is needed here: restriction of an integer-valued valuation still has value group a subgroup of $\mathbb{Z}$, hence is an integral multiple of a primitive divisorial order. It is never divided by the chart degree. Only the initial field extension contributes the separately bounded ramification index ρ . The rational functions used in the metric estimate below are taken on the fixed total family itself.

All special components are log boundary on this model. The subboundary of the crepantly pulled ordinary Ω is bounded above by the reduced log boundary (coefficient bound by lc, horizontal positive support marked as above). Thus its chosen lc place is a place also of the reduced-log pair, since the difference is effective $\mathbb{Q}$ -Cartier. Such lc places are monomial on the log regular model: pass to a toroidal snc subdivision and use the lc place description for an snc pair. We take a cone as above containing the extended chosen place. For a horizontal place as in the first bullet one just takes a bounded stratified log resolution of the generic

pairs $(F, \text{Supp } \Omega_F)$; same reasoning yields an ordinary snc monomial cone (over the generic field, or its extension).

Logarithmic coordinates and Gauss restriction. For the later metric estimate, choose rational log coordinates $z_1, \dots, z_f$, $f = \dim F$, on charts of the *fixed* toroidalized families, such that $d \log z_j$ form a basis of relative log differentials near the stratum in question. Indeed the latter are locally free by log smoothness, and one can choose among boundary parameters and invertible functions giving generators from the smooth total chart. Finitely many choices of genuine rational functions on opens of smooth strict charts suffice (Cartier local boundary equations and local units generate logarithmic differentials). Their weights are bounded linear forms in the cone coordinates (boundary parameters or units), and the basis property persists under fs base change and toroidal resolution. Moreover the z_j have bounded pole degrees on the embedded F 's for these charts: they are chosen on fixed families birational to the embedded ones, so on a sufficiently shrunk dense open of each stratum they give fixed bounded-degree rational maps on the embedded fibres. This shrinking can be built into the stratification (express each function by fractions of bounded-degree embedded coordinates on the generic fibre of the family and spread). Use the analogous absolute log bases in the horizontal generic case.

A rational toroidal valuation here with $e_* > 0$ restricts to a Gauss valuation on $K_1(z_1, \dots, z_f)$. To check, rescale to a primitive divisor on a toroidal subdivision. Write $\nu_j = v(z_j) \in \mathbb{Z}$, $e_* = v(t_*) > 0$. At this divisor the residues of $z_j^{e_*} t_*^{-\nu_j}$ have independent log differentials. Indeed relative log cotangent there, modulo the divisor, is identified with residue differentials over the base residue field: $t_* = \pi^{e_*} \cdot (\text{unit})$, for uniformizer π , eliminates $d \log \pi$, while base unit differentials restrict as usual. This also follows from the exact sequences for log differentials at the generic smooth divisors (and separability). The indicated unit log differentials are e_* times the chosen basis in relative log cotangent. The residues are thus algebraically independent. In particular a lattice basis of the weight ties (monomials z^γ with $\nu \cdot \gamma$ a multiple of e_* , adjusted by that power of t_*) has algebraically independent residues too, so no cancellation occurs at equal coefficient-adjusted weights. Horizontally one keeps $d \log \pi$, without a base uniformizer; for a nontrivial monomial valuation of integral rank one weights, a basis of the kernel characters of the weight gives independent residues by the same log basis calculation. This again gives Gauss, now for the trivial base valuation. $\square$

Lemma A.7 (Degrees control angular variation). *On a chart of Lemma A.6, the order of a nonzero rational function is continuous, homogeneous and finite piecewise linear, including on the faces. If its pole divisor on the embedded generic fibre has degree at most m , where $m \geq 1$, its Lipschitz constant in x at fixed $e_* > 0$ is at most Cm . Horizontally this bound holds on the entire cone. The constant C depends only on the fixed families and chart choices. A function with pole degree less than one is constant on the generic fibre and has zero angular variation.*

Proof. Piecewise linearity and faces. On these cones rational functions have continuous homogeneous finite piecewise linear values, including faces. For example subdivide a split chart toroidally to snc (subfaces using the corresponding generization strata in the subdivision) and use the monomial order formula at the generic stratum (after splitting a chart if necessary), by expanding at the snc parameters: minima on power series supports are finite polyhedral by monomial noetherianity. The same minima using weights zero off a subcollection of parameters give the values on its face, i.e. at the more general stratum. Indeed the monomial ideals in the retained parameters test order there; membership is still tested in the power series ring localized along the prime of those parameters (faithful flatness and the monomial formula; a positive-threshold monomial ideal for the retained parameters only is primary to

their prime, as seen already in completion). This gives continuity and compatibility across subdivisions. In a local split toroidal chart the subdivisions are toric subdivisions at that stratum; one is using the corresponding monomial order family throughout (weight ideals of boundary monomials), so that subfaces common to two cones use the same such orders.

A degree bound for angular slopes. Suppose a nonzero rational function g on F over the extended field has pole degree $\leq m$, $m \geq 1$, in the bounded embedding. It satisfies a polynomial equation over the z 's of bounded degree in g and degree $O(m)$ in the z 's. Use the equation of the image hypersurface of (z, g) in a product of projective lines. Its multidegrees are bounded by intersecting general level divisors of all but one coordinate on F on the locus of definition: individual divisor degrees are bounded by their pole degrees, so projective Bézout bounds isolated general solutions by a constant times the product bound (characteristic zero). Equivalently intersect the product of those level-divisor supports with diagonal conditions of bounded degree in a product of projective spaces, counting isolated points (the product variety has Segre degree bounded by a fixed multiple of the product of individual degrees). Write this equation as

$$\sum_{j=0}^q c_j(z)g^j = 0, \quad q \leq C, \quad \deg_z c_j \leq Cm.$$

By the Gauss property, $v(c_j)$ is a minimum of affine functions whose x -slopes have norm at most Cm . Their constant terms can involve arbitrary orders of coefficients in the trait field, but at fixed e_* these terms do not vary. At least two nonzero terms of the equation attain its least valuation. Consequently $v(g)$ is one of the finitely many quantities

$$\frac{v(c_i) - v(c_j)}{j - i} \quad (i \neq j).$$

Each is Cm -Lipschitz after enlarging the fixed constant. Along a segment the continuous, finite piecewise linear function $v(g)$ selects from this finite list; subdividing at its finitely many switches gives the same Lipschitz bound on the entire segment. Rational weights are dense, so the estimate holds also for real weights. Horizontally the argument applies on the entire cone. If the pole degree itself is < 1 the function is constant along the generic fibre. $\square$

Proof of Proposition A.5. Lemma A.6 supplies the chart, the prescribed monomial place and the integral orders. We first extend the piecewise-linearity assertion of Lemma A.7 from rational functions to the discrepancy data. Evaluations of $\mathbb{Q}$ -Cartier b-divisors are also finite piecewise linear: locally over an affine neighbourhood of the stratum write a determining Cartier multiple on a projective model as a difference of relatively generated divisors, then it uses minima on finite sets of rational functions. Generalized discrepancies differ on this cone from the reduced-log discrepancy (identically zero) by such Cartier b-evaluations. Piecewise-linearity here need not have bounded combinatorial complexity.

Two-point replacement of the nef part. Fix two rational valuations on one cone with the same e_* , and choose a common high model on which their centres and the determination of the nef part are visible. On that model let L be an ample divisor, relative to the trait, and choose a rational $\varepsilon > 0$. After twisting by a pullback from the base, take general effective representatives

$$E_+ \sim_{\mathbb{Q}} \mathbf{M} + \varepsilon L, \quad E_- \sim_{\mathbb{Q}} \varepsilon L$$

which contain neither centre. The ample perturbation is selected *after* the high model: its coefficient can therefore be made so small that its pushed horizontal degree is arbitrarily small. Add $E_+ - E_-$ to the crepant generalized boundary and remove the nef part. Since

both representative divisors have order zero at both chosen valuations, the resulting ordinary divisor data have exactly the prescribed discrepancies at these two valuations.

Let Θ be its horizontal trace on the bounded fibre F . Write degrees with respect to the bounded very ample class. The original trace B_F is effective, and the trace of the nef part has nonnegative degree. Relative triviality gives

$$\deg B_F + \deg \mathbf{M}_F = -\deg K_F.$$

Negative exceptional crepant coefficients on the high model disappear when pushed to F . Hence, if L_F denotes the pushed ample contribution, then

$$\deg \Theta^+ \leq -\deg K_F + \varepsilon \deg L_F, \quad \deg \Theta^- \leq \varepsilon \deg L_F.$$

Both degrees are uniformly bounded. This estimate uses no bound on the height of the determination or the denominator of $\mathbf{M}$. For divisible l , relative rational triviality supplies a rational l -pluriform σ for these ordinary data. Its divisor on the generic fibre is $-l\Theta$; the possible pullback term from the trait has constant value on a slice with fixed e_* . Divide by the l -power of the log form ψ from wedging our relative $d \log z_j$'s with a base log volume near this DVR. At either evaluation the discrepancy is $v(\sigma/\psi^l)/l$ up to the base term constant at fixed e_* , since ψ is a log generator at the toroidal divisors. The generic pole degree of this ratio is $O(l)$: the log volume of the z_j has controlled poles (on the regular codimension one locus of F), hence controlled zeros by the canonical degree (nonpositive here by log CY effectivity horizontally), and the horizontal boundary degree was just bounded. Lemma A.7 applied to σ/ψ^l , followed by division by l , proves the uniform metric assertion. The integer l may be arbitrarily large. Continuity extends the estimate from the two rational endpoint valuations to any two real valuations on the slice. Thus neither the nef-part index nor the vertical contact enters the constant. Horizontal traces and these canonical-degree computations on the generic fibre can use forms divided by a separating base volume (the absolute log order at a horizontal divisor is the fibre log order, by taking independent base residues). Horizontally use exactly the same replacement and log volume over the generic field. $\square$

Corollary A.8 (Horizontal variants). *The chart and metric conclusions have the following two variants.*

- (i) *The same tailorable bounded complement and homogeneous chart argument works if we are working over an algebraically closed generic field, with the relevant effective CYs on a $\mathbb{Q}$ -factorial model of Fano type, and have boundedness only up to isomorphism in codimension one of these models.*
- (ii) *A Cartier rational b -order function N with effective trace of degree $O(\xi)$ on the bounded model, b -linearly $\mathbb{Q}$ -equivalent to a difference of b -nef parts both of trace degrees $O(\xi)$, has horizontal Lipschitz constant $O(\xi)$, including when $\xi \rightarrow 0^+$.*

Proof. For (i), tailor the complement on the Fano type model using the klt perturbation in Lemma A.4, and then push the complement and the effective CY data crepantly to the bounded normal small model. Their total divisors are $\mathbb{Q}$ -linearly trivial. This supplies the bounded embedded marked family and all the degree estimates in Proposition A.5; the anticanonical degree is bounded by the same curve slicing argument.

For (ii), use the two-point construction for each of the two b -nef parts, taking the pushed ample errors to be $o(\xi)$. As above replace those parts by general effective representatives after the same arbitrarily tiny ample addition on a high determination, avoiding both centres. The residual rational principal term for N now uses functions with pole degree divided by denominator $O(\xi)$ on F ; apply Lemma A.7 (using constancy if pole degree < 1). $\square$

A.6 Integral approximation on the cones

The charts now give a bound on how discrepancy changes with the angular coordinates, even when the trait contact is large. We next approximate real chart points by integral ones. Integrality makes the resulting orders positive integral multiples of prime orders, so the small-prime hypotheses in the comparison lemmas can be used without changing their normalization.

The following approximation does not require uniform smoothness over the DVR. All norms are fixed Euclidean norms; the matrices range over a fixed finite list, so their kernel lattices have uniformly equivalent coordinate norms.

Lemma A.9 (Integral approximation). (i) *Let u_i lie in a fixed nonnegative orthant and $|u_i| \rightarrow \infty$. After passing to a subsequence there are $s_i \rightarrow 0^+$, nonzero nonnegative integral vectors n_i , and a fixed, possibly subsequence-dependent, C_* such that*

$$|n_i - s_i u_i| \leq C_* s_i, \quad s_i |u_i| \rightarrow \infty. \quad (\text{A.3})$$

(ii) *Let $M_i x_i = r_i$, where $x_i \geq 0$, r_i is integral and M_i belongs to a fixed finite list of integral matrices. If $\alpha_i > 0$ tends to zero, then after passing to a subsequence there are positive integers E_i , nonnegative integral n_i , and a fixed C_* such that*

$$M_i n_i = r_i E_i, \quad s_i = E_i \alpha_i \rightarrow 0^+, \quad |n_i - E_i x_i| \leq C_* s_i. \quad (\text{A.4})$$

(iii) *For every sufficiently small fixed $\gamma > 0$, the same matching equations admit $n_i \geq 0$ integral and $1 \leq E_i \leq J(\gamma)$ with*

$$|n_i - E_i x_i| \leq \gamma, \quad (\text{A.5})$$

where $J(\gamma)$ is uniform in i and in the contact vectors r_i .

Proof. For (i), pass diagonally so that for every integral character γ , the numbers $\langle \gamma, u_i \rangle$ either converge in $\mathbb{R}$ or diverge in absolute value. The characters with bounded values form a subgroup Γ . Their limits are additive and extend to evaluation on a vector u_* . For a fixed interval $I = [a, b] \subset (0, \infty)$, the probability distribution of $s(u_i - u_*)$ modulo the lattice, with s uniform on I , tends to Haar measure on $\Gamma^\perp$. Indeed its Fourier coefficient at a character in Γ tends to one; for a character outside Γ , integration of the exponential bounds the coefficient by

$$\frac{2}{2\pi(b-a)|\langle \gamma, u_i - u_* \rangle|} \rightarrow 0.$$

Trigonometric approximation proves weak convergence. Every neighborhood of zero in $\Gamma^\perp$ has positive Haar measure. Choose intervals tending to zero sufficiently slowly and neighborhoods of radius $o(s_i)$, while also requiring $s_i |u_i| \rightarrow \infty$. A diagonal choice gives an integral n_i with $|n_i - s_i(u_i - u_*)| = o(s_i)$, proving (A.3). The error is eventually less than one in each coordinate. Since $s_i u_i \geq 0$, an integral coordinate of n_i cannot be negative. Its norm tends to infinity, so $n_i \neq 0$.

For (ii), pass to a subsequence with $M_i = M$. Elementary integer linear algebra gives a rational $x_{0,i}$ with $M x_{0,i} = r_i$ and $l x_{0,i}$ integral for a positive integer l depending only on M . For example, solve using a fixed maximal nonzero minor and take l divisible by its determinant. Put $H_0 = \ker M$, $\Lambda = H_0 \cap \mathbb{Z}^m$, and $U_i = x_i - x_{0,i}$. On the dual lattice of Λ , pass diagonally so that every normalized distance

$$\frac{\text{dist}(\langle \gamma, l U_i \rangle, \mathbb{Z})}{l \alpha_i}$$

converges to a finite value, with a signed residual, or tends to infinity. The characters in the first case form a subgroup Γ . Their signed residual limits are additive; choose $u_* \in H_0$ whose evaluations give these limits.

Sample positive integers j for which $s = jl\alpha_i \in [a, b]$, with $[a, b] \subset (0, \infty)$ fixed. The distributions of $jl(U_i - \alpha_i u_*)$ in H_0/Λ tend to Haar measure on $\Gamma^\perp$, which may be disconnected. For $\gamma \in \Gamma$ the frequency, reduced modulo $\mathbb{Z}$, is $o(\alpha_i)$, so its Fourier coefficient tends to one. Otherwise the geometric-series bound is at most

$$\frac{C\alpha_i}{\text{dist}(\langle \gamma, l(U_i - \alpha_i u_*) \rangle, \mathbb{Z})} \rightarrow 0.$$

Use neighborhoods of zero and a slow diagonal choice as before. There are $j_i > 0$ and $k_i \in \Lambda$ such that, with $s_i = j_i l\alpha_i \rightarrow 0$,

$$|k_i - j_i l(U_i - \alpha_i u_*)| = o(s_i).$$

Set $E_i = lj_i$ and $n_i = E_i x_{0,i} + k_i$. These are integral matches and $|n_i - E_i x_i| \leq C_* s_i$. The same coordinate argument proves $n_i \geq 0$.

For (iii), apply simultaneous Dirichlet approximation to lU_i on the fixed kernel torus. It supplies a positive $j_i \leq J_0(\gamma)$ with $\text{dist}(j_i lU_i, \Lambda) \leq \gamma$. Use the same definition of n_i and $E_i = lj_i$. Matching is exact, positivity follows when $\gamma < 1$, and the bound on E_i is independent of r_i . $\square$

In a horizontal discrepancy application take $u_i = w_i/a_0(w_i)$. In a vertical application of (A.2), take $x_i = w_i/e_{*,i}$ and $\alpha_i = a_0(w_i, e_{*,i})/e_{*,i}$. The constant C_* in the vanishing-error conclusions is allowed to depend on the selected subsequence; the bound $J(\gamma)$ in (A.5) is uniform.

A.7 Proofs of the comparison statements

We apply the two preceding tools in tandem. The tailored complement places a minimizing divisor in a bounded chart; integral approximation then replaces it by a prime order of small discrepancy. This turns a failure of comparison on the base into a violation of the corresponding hypothesis upstairs.

Proof of Lemma A.1. We induct on the dimension of the total space; the birational case was settled in Subsection A.3.

Extraction and effectivity. Extract using a_0 with δ_* fixed small and take the Mori step $X_1 \rightarrow W$. Effectivity of all parameters on the extraction holds by monotonicity, or in the two-sided case by the ratio hypothesis taking $\delta_* < \min(\delta, 1)/3$, after trimming $L_i \rightarrow \infty$ more slowly. Indeed concavity using the opposite endpoint of a fixed interval controls interior upper ratios too. For example, if $0 \leq t \leq T$, then concavity at zero gives

$$a_t(T') \leq (1 + t/T)a_0(T') - (t/T)a_{-T}(T').$$

The hypothesis on all test subsequences makes the right side $(1 + o(1))a_0(T')$, uniformly on the extracted divisors. The other sign is identical. Apply this for each fixed rational T , then choose a slowly increasing interval by diagonalization. At subsequent Mori steps the discrepancy families are discriminants of the original family. They need not have a common global piecewise affine presentation. For each fixed prime P , however, resolve the original family over its generic DVR and include the fibre in the snc support. In the trimmed range, original divisorial valuations trivial on the current function field already have positive discrepancy. The discriminant is therefore the minimum of the finitely many vertical

component ratios on that resolution. Minima compose under the nested field restrictions, so the finite rational parameter transitions used below persist at each step. Generic klt over the next base can likewise be tested on a resolution of the original family over its generic field.

Generic klt over the next base. We claim generic klt over W in the Mori step for all bounded parameters eventually. Otherwise take a sign and a first horizontal lc threshold $t_* \neq 0$ with $|t_*|$ bounded, and a horizontal minimizing divisorial T . This first threshold is rational by resolution; if tested on the original family, a place giving it restricts nontrivially to the current field (by the previous generic klt), hence gives an lc place there too. Tailor the chart using a_{t_*} . Concavity at this place gives

$$a_{\pm L} \leq (1 - L/|t_*|)a_0$$

for arbitrarily large fixed rational L , sign as chosen. Let w be the chart vector representing T . Put $u = w/a_0(w)$, and bound $|t_*|$ by a fixed T_0 . If $|u|$ were bounded along a subsequence, the uniform homogeneous Lipschitz bound would bound $|a_{\pm L}(u)|$ independently of fixed L , whereas $a_{\pm L}(u) \leq 1 - L/T_0$. Choosing L large gives a contradiction. Thus, after a further subsequence, $|u| \rightarrow \infty$. Lemma A.9(i) gives $|n_i - s_i u_i| \leq C_* s_i$, $n_i \neq 0$, and hence

$$0 < a_0(n_i) \leq (1 + CC_*)s_i, \quad a_{\pm L}(n_i) \leq (1 - L/T_0 + CC_*)s_i.$$

The same C works for every effective CY discrepancy $a_{\pm L}$. Choose $L > T_0(1 + CC_*)$ after fixing the subsequence. The restriction of n_i is $k_i \text{ord}_{T_i}$, with $k_i \geq 1$. Therefore $a_0(T_i) \rightarrow 0$, while $a_{\pm L}(T_i) < 0$, contradicting the appropriate ratio hypothesis. This proves the claim; trim again to keep generic klt throughout and induce effective generalized data on W .

Transfer of the ratio hypothesis. We must pass the corresponding ratio hypothesis to $b_t(P)$ on W . Suppose there is failure with $b_0(P) \rightarrow 0$, respectively with $b_0(P) < \delta'$ where $\delta' > 0$ is fixed sufficiently small below. Concavity (also using the other sign in the two-sided case) gives a bounded parameter with $b_t \leq 0$. Move from zero along a sign to the first $b_t = b_0/2$; we may take $b_s \leq 2b_0$ up to that time: if there is a higher peak use the opposite sign, where the drop is then forced. In the one-sided case this uses monotonicity instead.

There is a rational parameter p on this segment and a simultaneous minimizer over P with

$$a_0(T)/e(T) \rightarrow 0 \quad (\text{respectively } a_0(T)/e(T) \leq 10\delta'), \quad a_p(T) \leq \tfrac{1}{2}a_0(T).$$

Indeed take a cutoff radius $\sqrt{b_0}$ or $8\delta'$. At zero all minimizers have ratio $a_0/e = b_0$. At the half-drop any minimizer below the cutoff works. If minimizers exit the cutoff first, at a transition interpolate where a minimizer below and one at/above tie, to obtain a ratio near the cutoff, where the desired discrepancy drop holds since $a_p/e = b_p \leq 2b_0$. It suffices to track the finite minimizing candidates from log resolution (or the divisorial restrictions of the finite candidates minimizing in the original family at each parameter; restriction of any minimizing original candidate gives a current minimizer by composition); use the LC-place connected interpolation above at the rational transition parameter.

Tailor the vertical chart to this T . In its notation $\alpha = \rho a_0(T)/e(T)$ and concavity gives a decrease as above by arbitrarily large factors at a fixed signed L . If $\alpha \rightarrow 0$, kernel approximation and uniform Lipschitz at fixed uniformizer value give again

$$a_0(n, E) = O(s), \quad a_{\pm L}(n, E) < 0$$

for sufficiently large fixed L , impossible. Otherwise we are in the two-sided case with $\alpha \leq C\delta'$ but bounded below after passage to a subsequence. Use the bounded Dirichlet approximation with γ so small that the Lipschitz error for a_0 is $< \delta/2$. The constants here (also ramification and J) are uniform for this Mori step, so choose δ' small enough that $E\alpha < \delta/2$. A sufficiently

large signed L fixed on the subsequence still makes the discrepancy negative by the uniform Lipschitz bound even for $a_{\pm L}$. This contradicts the strong ratio hypothesis. We have thus transferred that hypothesis with fixed $\delta' > 0$ (and the one-sided hypothesis without a fixed cutoff as required). Iterate using $\dim W < \dim X$, stopping if birational over S . This proves Lemma A.1. $\square$

Proof of Lemma A.2. Absorb the input multiple into H . Use the same Mori and discriminant steps with fixed klt CY a . For $b(P) \rightarrow 0$ at a step choose a minimizer, so in the vertical chart $\alpha = a(w, e_*)/e_* = \rho b(P) \rightarrow 0$. Here write $h = v_P(H_W)$ where H_W is the base pullback. Use the kernel argument on the augmented torus with sampling vector $l(U, \rho h)$. Write (u_*, z_*) for an extension of the signed limits divided by $l\alpha$ on the bounded-frequency subgroup Γ . The same sampling equidistributes modulo the augmented lattice after subtracting $s(u_*, z_*)$.

Every vector in the Lie space of $\Gamma^\perp$ must have last coordinate zero. Otherwise choose its sign with negative last coordinate and target a small multiple ξv with $\xi \rightarrow 0^+$ on that torus. By taking intervals $0 < s \ll \xi \rightarrow 0$ sufficiently slowly and approximating the target to $o(\xi)$ along a diagonal subsequence, the integral matched valuation has angular error $O(\xi)$, hence $a(n, E) = O(\xi)$, but signed phase $\xi v_{\text{last}} + o(\xi)$, contradicting the lower sign bound. The sign bound applies to integral multiples whenever their total discrepancy tends to zero. Indeed, for $v_i = k_i \text{ord}_{T_i}$ with $a(v_i) \rightarrow 0$, the primitive discrepancies also tend to zero. If their phase lifts are $r_i = O(a(T_i))$, $r_i \geq -\varepsilon_i a(T_i)$, with $\varepsilon_i \rightarrow 0$, then $k_i r_i$ are lifts with

$$k_i r_i = O(a(v_i)), \quad k_i r_i \geq -\varepsilon_i a(v_i).$$

They tend to zero, so no phase wrapping affects the sign. The projection of the compact subgroup $\Gamma^\perp$ to its last circle now has zero Lie algebra, hence finite image. Some $q > 0$ annihilates this image; by character duality $(0, q) \in \Gamma$. Consequently $\text{dist}(ql\rho h, \mathbb{Z}) = O(l\alpha)$. Its normalized residual limit is nonnegative: indeed $z_* \geq 0$ by the same sign test targeting zero on the torus with precision $o(s)$, angular error $O(s)$. Pass to a subsequence with l, ρ fixed.

For every test sequence $b(P) \rightarrow 0$ we thus have a subsequence and a fixed multiple there giving both the $O(b)$ bound and the lower $-o(b)$. There exists a *single* fixed multiple giving the upper size bound for all such sequences. Otherwise choose diagonal violations with

$$\text{dist}(k!h, \mathbb{Z}) > k!k b(P), \quad b(P) \rightarrow 0.$$

The subsequence property gives a fixed good multiple M on a further subsequence, with $\text{dist}(Mh, \mathbb{Z}) \leq Cb(P)$. For large k , $M \mid k!$, and thus

$$\text{dist}(k!h, \mathbb{Z}) \leq (k!/M)\text{dist}(Mh, \mathbb{Z}) \leq (k!/M)Cb(P),$$

a contradiction. Fix such a multiple M_0 . If its small lift has a negative ratio to $b(P)$ bounded away from zero along a subsequence, apply the subsequence property again and take a common multiple of M_0 and the resulting good multiple. Both small lifts, multiplied by fixed positive integers, tend to zero and represent the same phase, so eventually they are equal. Their incompatible lower signs give a contradiction. This proves the required lower $-o(b(P))$ bound for M_0 . This gives the required induction and proves Lemma A.2. $\square$

Proof of Lemma A.3. Use the Mori step extracting by $a + t_0 d$. Since a has bounded denominator and $d \geq 0$, all extracted divisors with this discrepancy tiny have $a = 0$, d tiny. Thus in the extra-data case too, effectivity on the extraction holds uniformly, by the upper bounds on $a = 0$.

Claim A.10 (Quantized chart cells). *The bounded-denominator functions a, h , when present, have only a bounded finite set of possible slopes in a horizontal chart, or in integral kernel coordinates on a vertical slice $e_* = 1$. Their vertical affine constant terms belong to a rational grid with bounded denominator, although their magnitudes need not be bounded. Their common cells are cut out by a fixed finite set of possible normal vectors, with offsets of bounded denominator. They are pointed, their vertices have bounded denominator, and they satisfy a uniform polyhedral distance bound.*

Proof of the claim. Horizontally, the functions of bounded denominator a, h have homogeneous linear pieces with slopes in a fixed bounded finite rational grid: their Lipschitz constants are bounded using values of effective CY discrepancies at finitely many fixed parameters, and their values on integral vectors lie in a prescribed rational grid, which controls the linear form on any full-dimensional open conical piece (take lattice differences there). Thus their cells can be cut by the comparisons of those finitely many linear forms. Vertically work on $e_* = 1$, writing $x = x_0 + U$ as before in an integral kernel basis. Linear parts in U again belong to a bounded finite rational grid, and the possibly unbounded constant terms at x_0 to a rational grid with bounded denominator (test lattice differences on the open *homogeneous* pieces, using integrality after multiplying x_0 by l). Coordinates forced zero on the slice can first be dropped (use x_0 for the reduced system), giving full relative dimension in $x_0 + \ker M$. More explicitly, an interior of linearity there lifts by homogeneity to an open in the linear space of matches. Take a lattice point interior to this homogeneous piece and translate a sufficiently large positive multiple of it by each of (lx_0, l) and the kernel basis vectors (with last coordinate zero). Both evaluations in each difference are on integral matches in the same linearity piece, which gives the intercept and kernel-slope quantization regardless of the size of x_0 . Closures of cells can be cut by inequalities from a finite set of normals, with offsets of bounded denominator: compare pairs of the affine forms occurring and include nonnegativity constraints. Values are affine on these comparison cells by continuity. Vertices have bounded denominator, and the polyhedra are pointed. We also use the elementary polyhedral error bound (Hoffman's bound): distance to a nonempty polyhedron with rows in a fixed finite list is bounded by a constant times the maximum violation. Here is the distance estimate explicitly. Write a nonempty polyhedron as $P = \{y : R_\nu y \leq b_\nu\}$, with rows R_ν in the fixed finite list. If y is the nearest point of P to x , express $x - y$ in its normal cone using independent active rows:

$$x - y = \sum_{\nu \in I} \lambda_\nu R_\nu^\top, \quad \lambda_\nu \geq 0, \quad \sum_{\nu \in I} \lambda_\nu \leq C|x - y|.$$

The constant is uniform because only finitely many independent row sets can occur. Taking the scalar product with $x - y$ gives

$$|x - y|^2 = \sum_{\nu \in I} \lambda_\nu (R_\nu x - b_\nu) \leq C|x - y| \max_{\nu} (R_\nu x - b_\nu)_+.$$

Cancel $|x - y|$ when it is nonzero. □

Horizontal signs and generic klt. First in the extra-data case, $h < 0$ on a horizontal a -lc place over W is impossible eventually. Tailor using a . The homogeneous comparison cells for a, h come from finitely many rational cones. On a cell containing a witness, the face $a = 0$ is rational and h is linear. Some extremal ray of that face has $h < 0$. The possible primitive integral ray generators form a bounded finite set, so there is a bounded nonzero integral witness $a = 0$, $h < 0$. But c on it is bounded by Lipschitz (again by fixed parameters), and h quantized and $\geq -\eta_i c$, impossible. Next generic klt for $a + \lambda(h + \theta c)$ holds uniformly for small $\lambda > 0$, $0 < \theta \leq \theta_0$: otherwise take $\lambda_i \rightarrow 0$ equal to a first positive lc threshold

with the corresponding fixed θ_i in each example. There is such a first threshold by common resolution, since on horizontal $a = 0$ we have $h \geq 0$, $c > 0$. Tailor using this threshold place. In the homogeneous chart distance from its vector w to $a = 0$ (including the zero vector) is bounded by $C_1 a(w)$ by the slope/polyhedral bound. On that set $h + \theta_i c \geq 0$, by the divisorial conditions and rational approximation there. Thus $h + \theta_i c \geq -C_2 a(w)$ at w , inconsistent with the threshold tending to zero ($a(w) > 0$).

Lexicographic minima and their denominators. For P over W denote the claimed lexicographic minima by a', d' (and a', h', c'). They exist: on a common resolution over the DVR use the vertical components and the formula for discrepancies by monomial orders plus the nonnegative reduced-log discrepancy. Horizontal a is nonnegative, and in the extra-data case on horizontal $a = 0$, h is nonnegative and $c > 0$. Explicitly include the fibre among the divisors on the resolution, so e is a nonnegative linear combination of vertical orders only. At a generically klt parameter a minimizer has reduced-log discrepancy zero (on the resolution) and all its positive orders in those coordinates lie on minimizing vertical components, by subtracting the minimum times e ; thus tracking these finitely many components captures departures from the faces in the argument. For each example and P , sufficiently small strictly positive parameters in lexicographic order thus have only lexicographically minimizing components, and give a simultaneous minimizer for tailoring using a generically klt parameter. On that chart the minimum a/e_* is $\rho a'$, and likewise for h on the minimizing locus if used. By linear optimization at cell vertices on the slice $e_* = 1$, both of these minima have bounded denominator. Thus so do a' and h' , uniformly in P .

Uniformity of the chambers. If minimizers of $a + td$ leave the a -minimum for some t tending to zero, take the first (strictly positive rational) transition where minimizers of both types tie. Connected interpolation gives a simultaneous minimizing place with $(a/e) - a'(P)$ arbitrarily small positive. Tailor at that place for the transition discrepancy. On the cell of its normalized weights (value $e_* = 1$), the affine function $a - \rho a'(P)$ is nonnegative. Its minimum is attained at a vertex, because the cell is pointed. Both the vertex values of a and $\rho a'(P)$ lie in a fixed rational grid. A strictly positive minimum would therefore be bounded below by a fixed positive number. Taking the interpolated excess smaller than this number forces the face $a = \rho a'(P)$ in that cell to be nonempty. Let $\varepsilon = a(w) - \rho a'(P) > 0$. The distance estimate in Claim A.10 gives w_0 on that face with $|w - w_0| \leq C_1 \varepsilon$. If C_2 is the Lipschitz constant of d , then

$$(a + td)(w_0) - (a + td)(w) \leq -\varepsilon + tC_1C_2\varepsilon < 0$$

for a uniformly small $t > 0$. This contradicts minimality; continuity permits the comparison at real weights and rational approximation recovers divisorial tests. This proves uniform exact minimization on the a -face for $a + td$. Identical reasoning with $h + \theta c$ yields uniform small λ with *all* minimizers still on the a -face for every $0 < \theta \leq \theta_0$ in the extra-data case. Fix a uniformly positive small such λ . If first positive θ -transitions away from the h -minimum within the a -minimum approach zero, now interpolate with tiny positive excess of h/e above h' ; all simultaneous minimizers stay on the a -minimum. Use joint a, h cells with the additional affine face $a = \rho a'$; the grid and distance argument within this face applied to h contradicts minimality for $h + \theta c$. Within the a -minimum the minimizer condition no longer depends on the magnitude of $\lambda > 0$, only on θ . Thus the full formulas hold uniformly on a smaller sector.

Induction on the base. For induction $a' + td'$ gives klt data for positive t by the minimum formula, and data at zero by the endpoint limit already explained. In the extra-data case on $a' = 0$, $d' \leq c'$, $h' \geq -\eta_i c'$, and testing at a d' -minimizer at fixed uniform small $\theta > 0$ gives $h' + \theta c' \leq C_3 d'$ by the exact formula. Hence $c' \leq C_4 d'$ and $h' \leq C_3 d'$ there. All hypotheses now hold at the next step. Any negative h at $a = 0$ vertical in this step marks a negative h' at $a' = 0$, by restriction. Induct to the base to obtain all assertions. $\square$

B Nonvanishing from signed phases

We seek sections of a bounded positive multiple of a rational polarization, even when its denominators are unbounded. The hypothesis will control the fractional orders of the polarization at primes whose discrepancies tend to zero. The sign of that control will determine which corrections can be removed in the proof.

For a rational Weil divisor H on a normal variety, set

$$R(H) = \sum_P \left(1 - \frac{1}{r_P}\right) P, \quad r_P = \min\{r \in \mathbb{Z}_{>0} : r \text{ coeff}_P(H) \in \mathbb{Z}\}. \quad (\text{B.1})$$

Only finitely many terms are nonzero. When H is $\mathbb{Q}$ -Cartier, its order at a higher divisor means the order of its Cartier b-pullback, rather than a rounded trace. A *signed phase* of this order is a lift of its image in $\mathbb{R}/\mathbb{Z}$; whenever this lift tends to zero, it is eventually the unique lift in $(-1/2, 1/2)$.

We use the comparison results of Section A and two boundedness theorems. Rationally connected projective varieties carrying effective generalized δ -lc Calabi–Yau data in bounded dimension are bounded up to isomorphism in codimension one [5, Theorem 1.7]. No bound on the Cartier index of the nef b-divisor occurs in that theorem. We also use effective birationality for a nef and big integral Weil divisor N on a projective δ -lc variety when $N - K$ is pseudo-effective [6, Theorem 1.1]. Our applications of the latter theorem take place on $\mathbb{Q}$ -factorial models, so that N is $\mathbb{Q}$ -Cartier, but need not be Cartier.

Lemma B.1 (Phase nonvanishing). *Let (V_i, H_i) be a sequence in bounded dimension, where V_i is projective of Fano type and H_i is a nef and big $\mathbb{Q}$ -Cartier rational Weil divisor. Suppose that, for rational $t \in [0, L_i]$, where $L_i \rightarrow \infty$, there are generalized data with discrepancies $a_i(t)$ such that*

$$B_i(t) \geq R(H_i), \quad K_{V_i} + B_i(t) + \mathbf{M}_i(t)_{V_i} \sim_{\mathbb{Q}} tH_i, \quad \mathbf{M}_i(t) \text{ is b-nef and } \mathbb{Q}\text{-Cartier}. \quad (\text{B.2})$$

Assume that $a_i(0)$ is klt and that $a_i(t)$ is nondecreasing in t at every divisorial valuation.

Assume the following condition, also after restriction to any subsequence. For every choice of rational $t_i \rightarrow \infty$ in the parameter range, one can pass to a further subsequence and choose a fixed positive integer q such that, for every sequence of primitive divisorial valuations T_i with $a_i(t_i, T_i) \rightarrow 0$, the signed phases $\theta_i(T_i)$ of $qH_i(T_i)$ satisfy

$$-o(a_i(t_i, T_i)) \leq \theta_i(T_i) \leq O(a_i(t_i, T_i)). \quad (\text{B.3})$$

Then there is an integer $M > 0$ such that for every i some $m_i \in \{1, \dots, M\}$ satisfies $|\lfloor m_i H_i \rfloor| \neq \emptyset$. Equivalently, after replacing M by a common multiple, the same positive degree works for all i .

All the constants in this lemma are sequence bounds. In particular, the phase condition is tested along arbitrary subsequences and with arbitrarily chosen divisors. This quantifier has two useful consequences. First, at a fixed growing parameter and after the allowed scaling, there are $C > 0$ and a sufficiently small fixed $\eta > 0$ such that all phases at $a_i(t_i, T) < \eta$ have their small lifts bounded in absolute value by $Ca_i(t_i, T)$, eventually. Otherwise one selects $a_i(t_i, T_i) \rightarrow 0$ with an unbounded ratio. Second, the condition also applies to integral multiples $w_i = e_i \text{ord}_{T_i}$ whenever $a_i(t_i, w_i) \rightarrow 0$. Indeed, $a_i(t_i, T_i) \rightarrow 0$, and

$$e_i \theta_i(T_i) = O(a_i(t_i, w_i)) \longrightarrow 0, \quad e_i \theta_i(T_i) \geq -o(a_i(t_i, w_i)). \quad (\text{B.4})$$

Thus multiplication introduces no wrap around the phase circle. This observation is needed each time a divisorial valuation restricts to a multiple of an angular prime.

B.1 Seifert bundles and two consequences of comparison

We use the graded-algebra description of Seifert bundles and its compactification [22, Definition 5, Theorem 7 and Section 14]. The discrepancy and phase formulas needed here are proved below.

Lemma B.2 (Bundle construction and discrepancies). *Let V be projective normal, and let H be a $\mathbb{Q}$ -Cartier rational divisor. There is a projective normal compactified Seifert bundle $p : Y \rightarrow V$ with disjoint $\mathbb{Q}$ -Cartier zero and infinity sections E_0, D and a rational fibre coordinate z such that*

$$\operatorname{div}(z) = p^*H + E_0 - D, \quad K_Y + E_0 + D = p^*(K_V + R(H)). \quad (\text{B.5})$$

*The second equality uses compatible canonical divisors. The divisor D is relatively ample; if H is nef, then D is nef, and if H is nef and big, then $D + p^*H$ is nef and big.*

In these identities, pullback of a rational Weil divisor along the equidimensional map p is defined at the generic points of its prime components, using the multiplicities of their inverse images. This agrees with Cartier pullback whenever the divisor is $\mathbb{Q}$ -Cartier. Orders of H , E_0 , and D on higher models use their Cartier b -pullbacks.

Suppose $B \geq R(H)$ and $\mathbf{M}$ is a nef $\mathbb{Q}$ -Cartier b -divisor on V . Give Y boundary $p^(B - R(H))$ and nef part $p^*\mathbf{M}$. Write a for the generalized discrepancy on V and A for that on Y . If a primitive divisorial valuation v of $k(Y)$ restricts nontrivially to w on $k(V)$, then*

$$A(v) = a(w) + |h(v)| + \operatorname{def}(v), \quad h(v) = v(E_0) - v(D) \equiv -w(H) \pmod{\mathbb{Z}}, \quad (\text{B.6})$$

where $\operatorname{def}(v)$ is a nonnegative integer. It vanishes precisely for the Gauss, or fibre-torus-invariant, valuations. If $w = 0$, the primitive valuations are the ordinary places of the generic $\mathbb{P}^1$, and have discrepancy 1. Consequently klt base data give klt data on Y without the two sections; adding both sections gives lc data whose non-klt locus is contained in the sections.

Proof. The two affine charts are the relative spectra of the algebras of rational monomials gz^j satisfying $\operatorname{div}(g) + jH \geq 0$, with $j \geq 0$ and $j \leq 0$, respectively. If $\ell H = \operatorname{div}(b)$ locally, these charts are finite cyclic quotients of the ordinary line charts obtained using $b^{1/\ell}$ and the coordinate $z/b^{1/\ell}$. They glue to the stated normal projective bundle. This description gives the two sections, their $\mathbb{Q}$ -Cartier property, and the first equality in (B.5). The restriction of D to itself is numerically H , while its restriction to E_0 is trivial; this and relative ampleness give the positivity assertions.

At a prime P of V the reduced vertical divisor has multiplicity r_P . Computing the relative canonical divisor by $d \log z$ therefore gives the second equality in (B.5), including exactly the coefficient $1 - 1/r_P$.

For the discrepancy formula, write $w = e \operatorname{ord}_T$ on a smooth base model and choose a uniformizer x at T . Wedge a base logarithmic volume form with $d \log z$. The additional logarithmic order is a nonnegative integer; it is zero exactly when the residue of the valuation-zero monomial $z^e/x^{v(z)}$ is transcendental over $\kappa(T)$. This is the Gauss condition. Removing the two sections from the boundary adds $v(E_0) + v(D) = |h(v)|$, since their supports are disjoint. Evaluating the first identity in (B.5) gives the congruence. The same calculation on a determination of the nef part proves the generalized formula. For trivial restriction the assertion is the usual discrepancy calculation on $\mathbb{P}_{k(V)}^1$. $\square$

In the setting of (B.2), the data without the sections have total logarithmic divisor

$$K_Y + p^*(B(t) - R(H)) + p^*\mathbf{M}(t)_Y \sim_{\mathbb{Q}} (t+1)p^*H - 2D. \quad (\text{B.7})$$

At $t = 0$, adding both sections instead gives effective generalized lc Calabi–Yau data.

Lemma B.3 (Clearing the small denominators). *In bounded dimension, suppose V is projective of Fano type and carries effective generalized klt Calabi–Yau data with $B \geq R(H)$. After multiplying H by a bounded positive integer, there is an ordinary klt log Fano boundary $\Theta \geq R(H)$. For this scaled divisor the Seifert bundle is of Fano type. The assertion does not require a bound on the Cartier index of the nef b -divisor.*

Proof. Take a small $\mathbb{Q}$ -factorialization. The class $-(K_V + R(H))$ is pseudo-effective, because it is the sum of $B - R(H)$ and the trace of a b -nef divisor. Moreover $(V, R(H))$ is ordinarily klt: its discrepancies dominate the given generalized discrepancies, by $B \geq R(H)$ and b -nef negativity. Run its anticanonical MMP, using Fano type to obtain a nef model. Ordinary lc comparison holds along this MMP by the negativity argument of Section A. The coefficients of $R(H)$ belong to the fixed hyperstandard set $\{1 - 1/r : r \in \mathbb{N}\}$. The bounded ordinary lc complement theorem [3, Theorems 1.7–1.8], followed by pullback through this MMP, gives an ordinary lc boundary $\Delta \geq R(H)$ with $n(K_V + \Delta) \sim 0$, where n is bounded in terms of dimension. The pullback dominates $R(H)$ because the MMP has the anticanonical sign; this is the same complement pullback used in Lemma A.4.

If $r_P > n$, the $1/n$ -quantization of Δ forces $\text{coeff}_P(\Delta) = 1$. Multiply H by the least common multiple of the integers at most the bound for n . All smaller denominators disappear. Every remaining component of $R(H)$ is therefore contained in $\lfloor \Delta \rfloor$. Let Θ_0 be any klt log Fano boundary on V . For a sufficiently small positive rational ρ , chosen separately for each variety,

$$\Theta = (1 - \rho)\Delta + \rho\Theta_0$$

is klt, dominates the new $R(H)$, and has ample negative log canonical class. No uniform lower bound for ρ is needed.

On the bundle use boundary $p^*(\Theta - R(H)) + (1 - \rho')(E_0 + D)$, with $\rho' > 0$ sufficiently small. Lemma B.2 proves klt. If $A = -(K_V + \Theta)$, its negative log canonical class is $p^*(A - \rho'H) + 2\rho'D$, which is ample for small ρ' by relative ampleness and base positivity. Thus the bundle is of Fano type. $\square$

Lemma B.4 (An unbounded-index obstruction). *Let F_i be projective of positive bounded dimension and let G_i be ample $\mathbb{Q}$ -Cartier rational divisors such that $(F_i, R(G_i))$ is klt and*

$$-(K_{F_i} + R(G_i)) \sim_{\mathbb{Q}} I_i G_i, \quad I_i > 0, \quad I_i \longrightarrow \infty.$$

Let C_i be the compactified Seifert cone obtained by contracting the zero section, and denote its infinity divisor by D_i . Then C_i is klt of Fano type, D_i is ample, and $-K_{C_i} \sim_{\mathbb{Q}} (I_i + 1)D_i$. It is impossible that

$$v_i(D_i) = o(A_{C_i}(v_i)) \tag{B.8}$$

for every sequence of primitive divisorial valuations with $A_{C_i}(v_i) \rightarrow 0$.

Proof. The contraction exists by semiampleness of the infinity divisor. One can also construct it by the finite fibre-power map to the ordinary compactified bundle for a very ample Cartier multiple of G_i , contracting its zero section and then Stein factorizing. The bundle identity gives

$$K_Y + (1 - I_i)E_0 + (1 + I_i)D \sim_{\mathbb{Q}} 0.$$

The pair with subboundary $(1 - I_i)E_0$ is sub-klt by (B.6); the discrepancy of the contracted zero section is $I_i > 0$. This proves the assertions about the cone and its canonical divisor. Near infinity the discrepancy formula reads, for $u = v(D) > 0$,

$$A_C(v) = A_{F,R(G)}(w) + u + \text{def}(v), \quad u \equiv w(G) \pmod{\mathbb{Z}}. \tag{B.9}$$

If (B.8) held, apply Lemma A.1 over a point to the nonincreasing family with discrepancies $A_C - tv(D)$, boundary tD , and nef part $(I_i + 1 - t)D$ on a range tending to infinity. The phase-free ratio hypothesis is exactly (B.8). The comparison lemma would make this family klt on every fixed bounded parameter range, whereas at $t = 1$ the prime D has discrepancy zero. This is a contradiction. $\square$

Lemma B.5 (Bounding a principalization order). *Let F_i be projective of Fano type in bounded dimension, and let $P_i \sim_{\mathbb{Q}} 0$ be $\mathbb{Q}$ -Cartier rational divisors. Suppose that F_i carries effective generalized klt Calabi–Yau data with discrepancies a_i and boundary at least $R(P_i)$. Suppose, after a fixed integral scaling if necessary, that the signed phases of P_i at every sequence with $a_i(T_i) \rightarrow 0$ lie in $[-o(a_i(T_i)), O(a_i(T_i))]$. Then the least positive integer q_i for which $q_i P_i$ is principal is bounded.*

More generally, the same cyclic-cover argument applies whenever the induced isotrivial fibration below has $u = o(A)$ at its small-discrepancy divisors over zero. This formulation permits the argument to be used after an independent estimate of the height u .

Proof. Apply Lemma B.3 to obtain, after a further bounded scaling, ordinary klt log Fano data dominating $R(P_i)$. This does not affect boundedness of the original principalization orders. A small $\mathbb{Q}$ -factorialization preserves all the data and these orders. Suppress the index and suppose for contradiction that $q \rightarrow \infty$. Write

$$\operatorname{div}(\phi) = qP, \quad \tilde{F} = \operatorname{Norm}_F(k(F)(r)), \quad r^q = \phi.$$

The degree is q by minimality of q and the Kummer criterion over a field containing all roots of unity. The ramification in codimension one is encoded by $R(P)$. In particular the log pullbacks of both the ordinary log Fano boundary and the generalized boundary are effective and klt.

Form the diagonal quotient

$$X = (\tilde{F} \times \mathbb{P}_y^1) / \mu_q \longrightarrow \mathbb{P}_{y^q}^1, \quad (\text{B.10})$$

where μ_q acts inversely on y and r , so that yr is invariant. The map from the product to X is quasi-étale: a nonidentity element has a fixed locus of codimension at least one on $\tilde{F}$ and only the two fixed points on $\mathbb{P}^1$. The generic fibre of (B.10) is connected. Log pulling the ordinary log Fano data to the product and descending proves that X is of Fano type over the curve.

Pull back the generalized data to the product and descend them. They give effective generalized klt Calabi–Yau data over the curve. The invariant nef b-divisor descends on a sufficiently high finite quotient; its $\mathbb{Q}$ -Cartier property and relative rational triviality descend by taking norms. For a primitive valuation over zero put $u = v(y^q)/q > 0$, and let w be its restriction to $k(F)$. The product discrepancy calculation gives

$$A(v) \geq a(w) + u, \quad u \equiv -w(P) \pmod{\mathbb{Z}}, \quad (\text{B.11})$$

where $a(w) = 0$ if $w = 0$. The congruence follows by evaluating the invariant function yr . These formulas can equally be computed upstairs and divided by the ramification index, using log pullback and homogeneous discrepancies.

If $A(v_i) \rightarrow 0$, trivial restriction is impossible because it would make u_i a positive integer. For nontrivial restriction, (B.4) and (B.11) give $u_i = o(A(v_i))$: all terms tend to zero, so there is no wrap, and the lower bound for the signed phase bounds the positive number u_i from above.

Apply Lemma A.1 over the curve to $A - tv(f^*[0])/q$. These are effective generalized Calabi–Yau data obtained by adding $tf^*[0]/q$ to the boundary; the nef part is unchanged.

The parameter range may tend to infinity. The reduced zero fibre has discrepancy 1 and multiplicity q , hence its initial discriminant discrepancy satisfies $0 < b_0(0) \leq 1/q$. Since the perturbation is a pure base pullback,

$$b_t(0) = b_0(0) - t/q, \quad b_1(0) \leq 0.$$

This contradicts the ratio conclusion of Lemma A.1, because $b_0(0) \rightarrow 0$. $\square$

B.2 Proof of phase nonvanishing

We now prove the main nonvanishing lemma. The first task is to subtract small positive phase corrections from the polarization and show that the remaining divisor is not big. Its subsequent MMP will either contradict the unbounded-index obstruction or produce a lower-dimensional base to which the same phase hypotheses descend.

Proof of Lemma B.1. Choosing the phase corrections. We argue by dimension induction. The assertion for a point is immediate. If the assertion fails in positive dimension, pass to a sequence and choose integers $N_i \rightarrow \infty$ such that

$$|[mH_i]| = \emptyset \quad \text{for all integers } 1 \leq m \leq N_i. \quad (\text{B.12})$$

Choose rational $k_i \rightarrow \infty$ sufficiently slowly relative to N_i and L_i . Use the phase hypothesis at k_i and Lemma B.3, pass to a subsequence, and scale H_i by fixed positive integers. Reparametrize the given families by these same factors, and decrease N_i by bounded factors. We retain (B.12), obtain the phase hypothesis without a displayed scaling at k_i , and have Fano type Seifert bundles. These bounded changes never replace the given discrepancy families by different ones.

Fix a sufficiently small constant $\eta > 0$. Extract precisely the exceptional primes T with $a(k_i, T) < \eta$ whose small signed phase f_T of $H(T)$ is positive. Include the original primes with the same property in the list of corrections. This is a finite list: monotonicity gives $a(0, T) \leq a(k_i, T) < \eta$, and the extraction and finiteness argument of Section A applies to the klt data $a(0)$. Write the resulting Fano type model as $U \rightarrow V$. For small fixed η we have, eventually,

$$0 < f_T \leq Ca(k_i, T) \leq C\eta, \quad C_U = \sum_T f_T T, \quad P_U = H_U - C_U, \quad (\text{B.13})$$

where H_U is the pullback of H . We first prove that P_U is not big, eventually.

The threshold contradiction under bigness.

Suppose instead that P_U is big. On the Seifert bundle let A be the discrepancy of the data at k_i without the sections. Choose a small fixed $0 < \delta < 1/4$, sufficiently small relative to η and the phase bound. Extract every exceptional prime with $A < \delta$, and let $\mathcal{S}$ be the reduced sum of *all* primes with $A < \delta$ on this extraction, including original primes. Lemma B.2 shows that they are Gauss valuations. Their extraction is funded by the effective zero-parameter generalized Calabi–Yau data with both sections, mixed with a sufficiently small ordinary klt log Fano contribution as in Section A. Thus the extraction is of Fano type, and the trace boundary of A is effective on it.

Put $\mathbf{H} = p^*H$, and denote by $\mathbf{D}$ the b-pullback of D . The b-divisor

$$\mathbf{J} = \mathbf{D} + \mathbf{H}$$

is nef and big and has integral Weil trace on the original bundle: the multiplicities r_P clear the coefficients of H at all vertical primes. Let τ be the pseudo-effective threshold for subtracting $\mathcal{S}$ from its trace on the extraction. We claim that

$$\mathcal{S} \neq 0, \quad k_i \tau_i \longrightarrow 0. \quad (\text{B.14})$$

To prove the claim, if it fails pass to a subsequence and choose integers $k_i, \tau_i^{-1} \ll m_i \ll N_i$; omit τ_i^{-1} when $\mathcal{S} = 0$. Round the coefficients on the extracted primes to obtain

$$Q = \sum_{T \in \mathcal{S}} q_T T, \quad 1 \leq q_T \leq 2, \quad N = m\mathbf{J}_U - Q$$

big and integral. All nonintegral coefficients of $m\mathbf{J}_U$ are on the extracted primes, so this rounding suffices. When $\mathcal{S} = 0$, use $N = mJ$. Run the N -MMP to a nef and big model. Here and in what follows a prime denotes the output trace, and a star denotes its Cartier b-pullback to a common resolution. Set

$$e_H = (H')^* - \mathbf{H}, \quad e_D = (D')^* - \mathbf{D}, \quad e_J = e_H + e_D.$$

B-nef negativity and MMP negativity give

$$e_H, e_D \geq 0, \quad me_J \leq (Q')^* - Q^*, \quad (Q')^* \leq 2(\mathcal{S}')^*. \quad (\text{B.15})$$

The total logarithmic divisor for A has trace-pull error $(k_i + 1)e_H - 2e_D$, by (B.7). Its effective trace boundary dominates $(1 - \delta)\mathcal{S}'$. Comparing generalized and ordinary discrepancies on the output therefore gives

$$\begin{aligned} A'_{\text{ord}} &\geq A + (1 - \delta)(\mathcal{S}')^* - (k_i + 1)e_H \\ &\geq A + \left(1 - \delta - \frac{2(k_i + 1)}{m}\right)(\mathcal{S}')^* \geq A + \frac{1}{2}(\mathcal{S}')^*. \end{aligned} \quad (\text{B.16})$$

The comparison also discards only a nonnegative trace-pull error of the nef part.

This gives a gap at contracted primes as well as surviving ones. Indeed, every primitive T with $A(T) < \delta$ was put on the extraction, and hence $Q^*(T) = q_T \geq 1$, even if the subsequent MMP contracts T . Equation (B.15) implies

$$(Q')^*(T) \geq Q^*(T) \geq 1, \quad (\mathcal{S}')^*(T) \geq \frac{1}{2}. \quad (\text{B.17})$$

Such a prime has $A'_{\text{ord}}(T) \geq 1/4$ by (B.16); all other primitive primes have $A'_{\text{ord}}(T) \geq A(T) \geq \delta$. Thus the output is ordinarily $\min\{\delta, 1/4\}$ -lc.

Fano type makes $-K'$ pseudo-effective, so $N' - K'$ is pseudo-effective. Integral-Weil effective birationality [6, Theorem 1.1] supplies a nonzero section of a bounded multiple of N' . MMP negativity pulls it back to the same multiple of N , and then to a bounded multiple of $m\mathbf{J}$. The fibre torus acts on the latter section space, so it contains a weight vector gz^j . A section of ℓJ satisfies

$$0 \leq j \leq \ell, \quad \text{div}(g) + (j + \ell)H \geq 0. \quad (\text{B.18})$$

Indeed the coefficients of its divisor along E_0, D are $j, \ell - j$, respectively, by (B.5). Thus it gives a section of $\lfloor (j + \ell)H \rfloor$ in positive degree at most $2\ell = O(m_i) \ll N_i$, contrary to (B.12). This proves (B.14).

Run the MMP of $J - \tau\mathcal{S}$ to its semiample nonbig model and let its Iitaka fibration be $Y' \rightarrow Z_\tau$. The threshold is rational because the extraction is of Fano type. Negativity now gives

$$0 \leq e_J \leq \tau((\mathcal{S}')^* - \mathcal{S}^*). \quad (\text{B.19})$$

Choose a small fixed c with $0 < c < 1 - \delta$. On this output use the boundary $B'_Y - c\mathcal{S}'$ and nef b-divisor

$$\mathbf{M}^\# = \mathbf{M}(k_i) + \left(2 + \frac{c}{\tau}\right)\mathbf{D} + \left(\frac{c}{\tau} - k_i - 1\right)\mathbf{H}. \quad (\text{B.20})$$

This boundary is effective. Both displayed coefficients in the nef part are eventually nonnegative by (B.14); in particular the nefness is absolute, and uses no Cartier-index bound. Its total logarithmic divisor is $\mathbb{Q}$ -linearly equivalent to $(c/\tau)(J' - \tau\mathcal{S}')$. The discrepancies are

$$A^\# = A + c \left((\mathcal{S}')^* - \frac{e_J}{\tau} \right) \geq A + c\mathcal{S}^*. \quad (\text{B.21})$$

Every primitive prime with $A < \delta$ has $\mathcal{S}^*(T) = 1$, including a prime contracted along the MMP. The remaining primes have $A \geq \delta$. Thus $A^\#$ has a uniform generalized gap $\min\{c, \delta\}$.

On a geometric generic fibre F of $Y' \rightarrow Z_\tau$ these are effective generalized Calabi–Yau data. Restriction can be computed using horizontal divisors on all higher models over the generic field, and then separable extension of constants; it is crepant and preserves the displayed gap. The fibre is of Fano type and rationally connected. Apply [5, Theorem 1.7] to $A^\#$ on F and take a bounded normal model isomorphic to it in codimension one. Normalization of the bounded models is still bounded after stratification and spreading, and does not add divisors to a codimension-one isomorphism.

The effective boundary $B'_Y - c\mathcal{S}'$ has coefficients at least $1 - \delta - c$ on $\mathcal{S}'$. Intersecting its Calabi–Yau equality with a fixed bounded polarization bounds its degree: the trace of the nef b-part is pseudo-effective, and the canonical degree is bounded in the bounded family. Consequently

$$\deg \mathcal{S}' = O(1), \quad \deg J' = O(\tau), \quad \deg \mathbf{H}_F, \deg \mathbf{D}_F = O(\tau). \quad (\text{B.22})$$

Here $J' \sim_{\mathbb{Q}} \tau\mathcal{S}'$ on the fibre, and the last two degrees are nonnegative and sum to that of J' .

We now use the assumed bigness of P_U . It gives an effective rational divisor $\Sigma \sim_{\mathbb{Q}} H$ such that for every v listed in $\mathcal{S}$, with restriction w , one has

$$w(\Sigma) > v(D) - v(E_0). \quad (\text{B.23})$$

For completeness, if $v(D) > 0$, write $w = e \operatorname{ord}_T$. The discrepancy formula gives $ea(k_i, T) < \delta$ and $v(D) < \delta$. Since $e|f_T| \leq Cea(k_i, T)$ and δ is sufficiently small, the congruence has no wrap and $v(D) = ef_T$. Thus $f_T > 0$, and this prime was corrected in P_U . An effective representative of the big class P_U may be chosen to vanish strictly at these primes and at any of the remaining finitely many projected centres. Every listed valuation has nontrivial angular restriction, since trivial restriction would give $A = 1$; its centre on U is therefore proper. Reserve a tiny ample summand and choose a sufficiently divisible member through all these centres. Adding back C_U gives (B.23). Pushing down yields an effective representative of H on V ; the representative on U is its pullback, since it differs from H_U by a rational principal divisor.

Every homogeneous normalized threshold section has divisor

$$(1 - b)D + bE_0 + p^*\Xi, \quad 0 \leq b \leq 1, \quad \Xi \sim_{\mathbb{Q}} (1 + b)H. \quad (\text{B.24})$$

If $b < 1$, replace b by $b + \alpha$ and Ξ by $\Xi + \alpha\Sigma$ for a sufficiently small positive rational α . The divisor remains effective, while every $\tau\mathcal{S}$ constraint improves strictly by (B.23). The finite list of strict improvements would increase the pseudo-effective threshold, a contradiction. Hence $b = 1$ for every such section. All these systems are torus invariant, because the listed valuations and the original J are torus invariant. Equal-degree ratios therefore lie in $k(V)$, by their weight decompositions and $b = 1$. The valuation $v_0 = \operatorname{ord}_{E_0}$ is trivial on this field, so it is horizontal over the Iitaka base Z_τ .

We obtain a contradiction on the geometric generic fibre. Tailor a bounded ordinary lc complement preserving v_0 as an lc place, using the effective zero-parameter generalized Calabi–Yau data with the two sections and the Fano type funding of Lemma A.4. This

may first be done on a small $\mathbb{Q}$ -factorialization of the fibre, using the horizontal lift of v_0 , and then transferred to the bounded model. Work on its homogeneous horizontal chart. Proposition A.5 bounds the Lipschitz constant of $A^\#$. The difference $A^\# - A$ is the order of

$$\frac{c}{\tau} (\mathbf{J} - (J' - \tau \mathcal{S}')^*).$$

The parenthesized b-divisor has effective trace $\tau \mathcal{S}'$ of degree $O(\tau)$ on this fibre and is b-nef up to a rational principal divisor. The shrinking-degree estimate of Corollary A.8 therefore gives an $O(1)$ Lipschitz bound for the difference, and hence for A .

The same estimate gives an $O(\tau)$ Lipschitz bound for $h_0(v) = v(E_0)$. Indeed E_0 has effective trace and is b-linearly equivalent to $\mathbf{D} - \mathbf{H}$, whose two nef terms have degree $O(\tau)$ by (B.22). In applying the estimate one replaces these nef terms by effective representatives with arbitrarily small error, avoiding the two centres in question. The function-degree argument on the bounded chart then applies: a rational function with pole degree strictly below 1 is constant. This is why the Lipschitz bound tends to zero, rather than merely remaining bounded.

Let w_0 be a chart vector representing v_0 . We have $h_0(w_0) = A(w_0) > 0$. The shrinking Lipschitz bound implies $|w_0/A(w_0)| \rightarrow \infty$. The horizontal approximation in Lemma A.9, applied to this ray, gives integral divisorial vectors n and numbers $s \rightarrow 0^+$ such that

$$A(n) = O(s) \rightarrow 0, \quad h_0(n) \geq s/2.$$

On the other hand, any such small-discrepancy valuation with $h_0 > 0$ has $v(D) = 0$. Formula (B.6) and the lower signed phase bound at $a(k_i)$, including (B.4), imply $h_0 = o(A)$; trivial restriction would instead make h_0 a positive integer and is impossible. This contradicts the two displayed estimates. We have proved that P_U is not big.

Running the correction MMP.

Diagonalize the preceding conclusion to choose $\eta_i \rightarrow 0^+$ sufficiently slowly so that P_U is still not big. Run its MMP. All parameters used from now on may grow more slowly, remain below k_i , and be $o(1/\eta_i)$. On the output put $\mathcal{T}'$ equal to the reduced sum of surviving correction primes. With $\mathbf{H}$ now denoting the original angular b-pullback, MMP negativity and b-nefness give

$$0 \leq C_U^* \leq E := \mathbf{H} - (P')^* \leq (C')^*, \quad C' \leq O(\eta_i) \mathcal{T}'. \quad (\text{B.25})$$

More explicitly, if $e_H = (H')^* - \mathbf{H} \geq 0$, then $E = (C')^* - e_H$, and P -negativity gives $(C')^* - C_U^* \geq e_H$.

Excluding a Mori fibre output. Suppose first that P_U is not pseudo-effective. On the geometric generic fibre F of the final anti- P' Mori contraction, put $G = -P'$, which is ample. Choose a new slow parameter $s_i \rightarrow \infty$, with $s_i \eta_i \rightarrow 0$, and obtain the phase condition at $a(s_i)$ after a fixed scaling. Scale H, P_U, C_U together. The correction coefficients remain $O(\eta_i)$, and P_U is integral at correction primes: subtracting their phases was precisely what made those coefficients integral. Thus $R(G)$ is supported away from $\mathcal{T}'$, and the pushed boundary satisfies

$$B(s_i)' \geq R(G) + (1 - o(1)) \mathcal{T}'.$$

Comparing with ordinary discrepancies on F , the only adverse total-divisor error is $s_i e_H$, which is at most $s_i (C')^*$. B-nef negativity for the moduli part and (B.25) therefore give

$$A_{F, R(G)} \geq a(s_i) + \frac{1}{2} (\mathcal{T}')^*. \quad (\text{B.26})$$

In particular $(F, R(G))$ is klt. A uniform ordinary gap is not asserted or needed in this case.

The relative Picard number of the Mori contraction is one. The classes used here, including $R(G)$, are restrictions from this model. Horizontal prime coefficients restrict unchanged over the generic field and split with unchanged multiplicities after a separable constant extension, so the same class comparisons hold on the geometric generic fibre. There we have

$$G + H' = C' \leq O(\eta_i)\mathcal{T}',$$

where H' is pseudo-effective. The zero-parameter generalized Calabi–Yau equality also shows that $-(K_F + R(G))$ dominates a fixed positive multiple of $\mathcal{T}'$ in this one-dimensional space of restricted classes: its boundary contributes $(1 - o(1))\mathcal{T}'$, and the moduli trace is pseudo-effective. Consequently

$$-(K_F + R(G)) \sim_{\mathbb{Q}} I_i G, \quad I_i \longrightarrow \infty. \quad (\text{B.27})$$

Initially this is a numerical proportionality, with $I_i \gg 1/\eta_i$. It is rational linear proportionality after clearing denominators because F is rationally connected: pull numerically trivial Cartier multiples to a smooth rationally connected resolution, where they are torsion, and descend the resulting rational equivalence.

Consider the compactified cone for (F, G) and a primitive divisorial valuation with ordinary discrepancy $A \rightarrow 0$ and infinity height $u > 0$. Its angular restriction w has $A_{F, R(G)}(w) \leq A$ and $u \leq A$ by (B.9). Moreover (B.25) and (B.26) imply

$$0 \leq E(w) \leq O(\eta_i)A_{F, R(G)}(w) \leq O(\eta_i)A. \quad (\text{B.28})$$

Divisorial tests on the geometric fibre lift to homogeneous horizontal tests on the original model. Thus the signed phase bound for H applies, and

$$u \equiv w(G) \equiv -w(H) + E(w) \pmod{\mathbb{Z}}.$$

All three quantities have small signed representatives. There is no wrap, so the lower phase bound and (B.28) give $u = o(A)$. If $w = 0$, the congruence makes u a positive integer, already impossible. Valuations with $u = 0$ satisfy the same conclusion trivially. This contradicts Lemma B.4 and (B.27).

Descent from the semiample output. We are left with a semiample nonbig P' and its contraction $g : U' \rightarrow W$. For a newly trimmed parameter range tending to infinity, with $t\eta_i \rightarrow 0$ uniformly on that range, use boundary and nef part

$$\tilde{B}(t) = B(t/2)' - tC', \quad \tilde{\mathbf{M}}(t) = \mathbf{M}(t/2) + (t/2)\mathbf{H}. \quad (\text{B.29})$$

Their total logarithmic divisor is tP' up to rational linear equivalence, and their discrepancies are exactly

$$\tilde{a}(t) = a(t/2) + tE. \quad (\text{B.30})$$

To see the equality, first pushing the old total divisor contributes $-(t/2)e_H$; adding the nef term and subtracting tC' contributes another $-(t/2)e_H + t(C')^*$. Their sum is tE .

The new boundary dominates $R(P')$. At correction primes P' is integral, the old boundary has coefficient $1 - o(1)$, and the amount subtracted is $O(t\eta_i) = o(1)$. Away from them the old denominator condition is unchanged. The nef part is b-nef; (B.30) and $E \geq 0$ show that the discrepancies are increasing and klt, with effective Calabi–Yau data at zero.

This family has the required signed phase condition for P' . Indeed, at a new growing parameter, apply the original condition to $a(t/2)$ and scale all the relevant divisors by the resulting fixed integer. If $\tilde{a}(t, T) \rightarrow 0$, then

$$a(t/2, T) \leq \tilde{a}(t, T), \quad E(T) \leq \tilde{a}(t, T)/t = o(\tilde{a}(t, T)).$$

Since $P'(T) = H(T) - E(T)$, subtracting this negligible nonnegative order preserves the lower $-o(\bar{a})$ bound and the upper $O(\bar{a})$ bound for the phase.

On the geometric generic fibre of g , the divisor P' is $\mathbb{Q}$ -linearly trivial. Its integral principalization order is bounded by Lemma B.5: along any putative unbounded-order subsequence choose one growing parameter with the just-proved phase bound, use the restricted effective generalized klt Calabi–Yau data, and clear the remaining bounded small denominators as in that lemma. These operations use only fixed multipliers. A small $\mathbb{Q}$ -factorialization of the geometric fibre preserves all the divisor orders and permits the cyclic-cover construction.

The principalization descends from the geometric generic fibre to its field of definition. Indeed being Cartier descends faithfully flatly, and the geometrically trivial line bundle is trivial by Hilbert 90; equivalently its one-dimensional space of trivializing rational functions descends. Thus for a bounded positive integer q and some $\phi \in k(U')^*$, the divisor $qP' - \text{div}(\phi)$ has no horizontal terms. There is then an exact equality

$$qP' - \text{div}(\phi) = g^*H_W \quad (\text{B.31})$$

for an ample $\mathbb{Q}$ -Cartier rational divisor H_W . To justify exactness, semiampleness already gives $P' \sim_{\mathbb{Q}} g^*A_W$ with A_W ample. After clearing denominators the difference between the left side of (B.31) and qg^*A_W is principal and vertical. A rational function with vertical divisor restricts to a nowhere vanishing regular function on the projective generic fibre, hence belongs to $k(W)^*$ because g is a contraction. Absorb this rational principal divisor into qA_W to obtain H_W .

If W is a point, (B.31) and $E \geq 0$ already give a section of a bounded multiple of H . Otherwise W has strictly smaller dimension than V . It is of Fano type. Apply the generalized canonical bundle formula to (B.29), and replace the old parameter s by $s = qt$. The induced data on W are effective generalized data with total tH_W , klt at zero and with nondecreasing discrepancies. Their nef parts are b-nef and $\mathbb{Q}$ -Cartier.

We check the denominator condition on W explicitly. For a prime P on W , choose a nonexceptional vertical component T above it with multiplicity e . Let r be the denominator of $\text{coeff}_T(P')$ and let r' be that of $\text{coeff}_T(qP' - \text{div}(\phi))$. Then $r' \mid r$. The denominator r_P of H_W divides er' , by (B.31). Since the upstairs boundary dominates $R(P')$, the discriminant discrepancy at P satisfies

$$b_t(P) \leq \frac{1}{er} \leq \frac{1}{er'} \leq \frac{1}{r_P}.$$

Thus the discriminant boundary dominates $R(H_W)$.

For each growing parameter, the pure pullback phase condition descends by Lemma A.2, after its allowed fixed multiple. Every hypothesis of Lemma B.1 is now satisfied in smaller dimension. Induction gives bounded-degree sections of H_W . They pull back through (B.31) to bounded-degree sections of P' , and the b-divisor inequality $\mathbf{H} \geq (P')^*$ in (B.25) carries them to sections of the same bounded multiples of H on V . This contradicts (B.12) and completes the induction.

The contradiction excludes every sequence of failures with $N_i \rightarrow \infty$, and hence proves boundedness for the original sequence. Any finite initial segment omitted in an eventual assertion can also be included: each of its big divisors has a section in some positive degree. Taking a common multiple of the bounded degrees gives the common degree in the statement. $\square$

C Cutoff fields, phase bounds, and section counts

We continue to work with sequences of bounded dimension, and with the subsequence convention of Section A. In particular, a statement about all valuations with an order tending to zero is required on every subsequence. All constants below may depend on the dimension bound and on the fixed constants explicitly present in the hypotheses. They do not depend on an individual member of a sequence. Unless explicitly stated otherwise, quantified divisorial tests are primitive prime orders. Homogeneous estimates extend to integral multiples; arbitrary positive rescaling is used only for homogeneous inequalities or exact zero conditions.

C.1 Polarized discrepancy data and cutoff fields

Let V be a projective variety of Fano type. Let a be the discrepancy function of an effective generalized rational lc Calabi–Yau pair over the point, and suppose that

$$a(T) \in \mathbb{Z} \quad (T \text{ a primitive prime order}).$$

Let D be an effective nef and big rational $\mathbb{Q}$ -Cartier divisor on V . Its Cartier b-pullback is denoted by $\mathbf{D}$. Assume that $b = a + \mathbf{D}$ is klt and has effective trace on V , or equivalently $b(Q) \leq 1$ for primes Q on V . This generalized pair is obtained by adding $\mathbf{D}$ to the nef part and subtracting its trace from the boundary; its total log canonical class is unchanged. We call a prime order T with $a(T) = 0$ a *floor prime*. Thus on V every positive coefficient of D lies on the floor, and every nonfloor prime has $a(Q) = 1$ and $D(Q) = 0$.

For a rational divisor A on a normal projective model, put

$$S(A) = \{0\} \cup \{f \in k(V)^* : \operatorname{div}(f) + A \geq 0\}. \quad (\text{C.1})$$

Tests on this space always mean tests on its nonzero elements. If $A \geq 0$, it contains 1, so the field it generates is unambiguous. We take the varying main scale $L \geq 1$ to be rational. Rational upper approximations within a fixed factor give the same statements for real scales, after changing constants. A $\mathbb{Q}$ -principal order is the order function of a rational principal divisor. Unless a new trace model is specified, orders of rational divisors on V mean their b-pullbacks.

We shall also use geometric generic fibres. Divisors on such a fibre spread over a finite extension of the generic field, and their closures on resolutions give horizontal divisors on the total space. Conversely, horizontal divisors restrict divisorially. Finite extensions of the field of constants in characteristic zero are separable and are étale on these generic-field models, so actual prime orders restrict to primitive horizontal orders; integral multiples restrict to integral multiples. Canonical divisors from the base have horizontal order zero, and discrepancies and Cartier b-orders therefore agree by homogeneity. Nef b-parts restrict on a determination. Relative numerical proportionality for a Picard-number-one Mori contraction remains true on a geometric generic fibre: spread any test curve over a finite extension and specialize. These fibres are of Fano type. Consequently rational numerical equivalence on them implies rational linear equivalence, by rational connectedness and the Fano type basepoint-free consequences recalled in Section A.

Apply the extraction construction of Subsection A.3, funded by $a + \mathbf{D}/2$, to obtain a projective $\mathbb{Q}$ -factorial Fano type model $U \rightarrow V$ containing every primitive prime with $a = 0$ and $\mathbf{D} \leq 1$, and extracting no other prime. The list is finite: each selected prime has discrepancy at most $1/2$ for the fixed generalized klt pair $a + \mathbf{D}/2$. On U , precisely the floor primes form $\operatorname{Supp} D_U$, and their coefficients lie in $(0, 1]$. For rational $0 < \beta \leq 1$, set

$$I_\beta = \{Q : a(Q) = 0, D(Q) \leq \beta\}, \quad P_\beta = D_U - \sum_{Q \in I_\beta} D(Q)Q. \quad (\text{C.2})$$

All the primes in this sum occur on U . Run a negative P_β -MMP and let

$$p : U' \longrightarrow W_\beta$$

be the contraction defined by the semiample output P'_β . The semiample divisor is the pullback of an ample rational divisor, allowing a point as the base. In fact there is an effective ample rational divisor D_W such that

$$P'_\beta = p^* D_W, \quad D' = p^* D_W + \sum_{Q \in I_\beta} D(Q) Q'. \quad (\text{C.3})$$

To check the first equality as divisors, rather than just as classes, P'_β is effective and has no horizontal component. A rational principal difference between it and a pullback has divisor zero on the projective generic fibre, hence is a function from the base. Absorb that function into the divisor downstairs. Here the connected-fibre property makes $k(W_\beta)$ relatively algebraically closed in $k(V)$. Writing $\mathbf{P}$ for the nef Cartier b-divisor defined by P'_β , MMP negativity gives

$$0 \leq \mathbf{P} \leq P_\beta^* \leq \mathbf{D}. \quad (\text{C.4})$$

Every prime in I_β survives the MMP. Its coefficient in P_β^* is zero, so the effective error $P_\beta^* - \mathbf{P}$ has value zero there; a prime contracted by a negative MMP would instead have strictly positive error.

Define

$$K_\beta = k(W_\beta) \subset k(V).$$

The field is generated by the spaces $S(mP_\beta)$, $m \in \mathbb{N}$, and is relatively algebraically closed. These fields decrease with β . A strict inclusion of two nested relatively algebraically closed subfields decreases transcendence degree: if their transcendence degrees were equal, the larger would be algebraic over the smaller. Thus there are at most $\dim V$ strict changes. We have $K_1 = k$, and $K_{0+} = k(V)$, since below the least coefficient in the finite support of D_U the divisor P_β equals the big divisor D_U .

The following maximal-order condition, called (C- α), will recur:

$$T(\Gamma) \leq (1 + \eta)D(T) \quad \text{for every effective } \Gamma \sim_{\mathbb{Q}} D, \quad a(T) = 0. \quad (\text{C.5})$$

The quantifier over Γ is part of the condition. When imposed on a set of valuations in a sequence, the same $\eta = \eta_i$ works for that entire set.

Lemma C.1 (Uniform cutoff models and order functions). *Suppose $\dim W_\beta > 0$. After passage to a subsequence there is a fixed $\delta > 0$ and a projective $\mathbb{Q}$ -factorial Fano type extraction $\widehat{W} \rightarrow W_\beta$ such that $\widehat{W}$ has ordinary δ -lc singularities and every positive coefficient of the pullback of D_W/β is at least δ . A bounded constant C can be chosen so that $S(CD_W/\beta)$ generates $k(W_\beta)$.*

There are algebraically independent functions $\phi_1, \dots, \phi_{\dim W_\beta}$ in this space and a b-nef $\mathbb{Q}$ -Cartier order function H on $k(W_\beta)$, pulled back to $k(V)$, with uniformly bounded denominator of its values on primitive orders, such that:

- (i) *If $a(T) = 0$, then $H(T) \leq CD(T)/\beta$. If T is nontrivial on K_β and all $T(\phi_j) \geq 0$, then $H(T) \leq -1$.*
- (ii) *If $D(Q) = 0$ and $a(Q) \leq 1$, then $H(Q) \leq 0$.*
- (iii) *If $a(T) = 0$ and (C.5) holds at T , then $H(T) \geq -C\eta D(T)/\beta$.*

In particular, if (C.5) holds on all of I_β with $\eta_i \rightarrow 0$, those primes are eventually horizontal over K_β ; that is, their restrictions to K_β are trivial.

Proof. On $U' \rightarrow W_\beta$ consider

$$d = \frac{\mathbf{D} - \mathbf{P}}{\beta} + \mathbf{P}. \quad (\text{C.6})$$

It is nonnegative and strictly positive on $a = 0$. The discrepancies $a + td$ have effective traces for $0 \leq t \leq 1$: at a surviving prime in I_β their trace increment is $tD(Q)/\beta \leq 1$, and at an unremoved floor prime it is $tD(Q) \leq 1$. Realize the family by a nef increment $t\mathbf{D}/\beta$ and total log canonical divisor equal, up to a rational principal divisor, to the pullback of $t(1/\beta - 1)D_W$. Lemma A.3 gives, on a fixed smaller interval, discriminants

$$a' + td'$$

with a' an effective generalized lc Calabi–Yau discrepancy over the point and with uniformly bounded denominator. These are klt for positive t .

We first make explicit a lifting consequence needed here. If $a'(E_i) = 0$, $d'(E_i) \rightarrow 0$, and $k_i > 0$ tends to zero, then, after a subsequence, there are primitive lifts T_i with

$$T_i|_{k(W)} = e_i \text{ord}_{E_i}, \quad a(T_i) = 0, \quad d(T_i) \rightarrow 0, \quad e_i k_i \rightarrow 0. \quad (\text{C.7})$$

At one Mori step, choose a minimizer of d on the a -minimum by computing the discriminant at a sufficiently small positive parameter. Its tailored vertical chart from Proposition A.5 has normalized values $0, \rho d'$ on the slice $e_* = 1$. Lemma A.9, with a fixed small error γ , gives an integral point with $e_* = l_j$ bounded in terms of γ , by rounding $l_j U$ in the kernel lattice. The error in a is less than the gap of its value lattice; nonnegativity and quantization therefore give $a = 0$ exactly. The slope bound in the same chart gives

$$d \leq e_* \rho d' + C\gamma.$$

First fix γ and go sufficiently far in the sequence, then let $\gamma \downarrow 0$ diagonally slowly enough that also $e_* k_i \rightarrow 0$. Primitivizing the resulting order only decreases the orders and the restriction multiplicity. Iterate through the bounded chain of Mori steps, at each step including the already accumulated multiplicity times k_i among the quantities required to be negligible. Birational steps preserve the discrepancy functions. This proves (C.7) without asserting a uniform bound for e_i .

There cannot be a sequence of primes E with, for one fixed sufficiently small $t > 0$,

$$(a' + td')(E) \rightarrow 0, \quad 0 < D_W(E)/\beta \rightarrow 0. \quad (\text{C.8})$$

The bounded denominator gives $a'(E) = 0$ eventually. Apply (C.7) with $k_i = D_W(E)/\beta$. It gives

$$a(T) = 0, \quad \frac{D(T)}{\beta} \leq d(T) + \frac{eD_W(E)}{\beta} \rightarrow 0.$$

Thus $T \in I_\beta$, where $\mathbf{P}(T) = 0$, whereas $\mathbf{P}(T) = eD_W(E) > 0$, a contradiction. Nor can an original prime E of W have $0 < D_W(E)/\beta \rightarrow 0$. A prime Q above it on U' satisfies

$$D(Q) = \mathbf{P}(Q) = eD_W(E) > 0, \quad a(Q) = 0, \quad a'(E) = 0, \quad d'(E) \leq D_W(E).$$

It would again give (C.8).

Choose fixed rational $0 < t \leq 1/4$ in the chamber. Extract every exceptional prime of $A = a' + td'$ below a sufficiently small fixed $\delta_* < 1/4$. This finite extraction is funded by mixing a' slightly with an ordinary klt Calabi–Yau pair, as in Subsection A.3; the extracted primes have $a' = 0$ and positive boundary coefficient. The preceding exclusions imply a uniform positive lower bound on all nonzero coefficients of

$$E_0 = (D_W/\beta)|_{\widehat{W}}.$$

Here and below the notation means the rational pullback. The ordinary discrepancies on the extraction dominate those of A , by b-nef negativity and $\mathbb{Q}$ -factoriality. Every remaining exceptional primitive prime has $A \geq \delta_*$, and every extracted prime has ordinary discrepancy one. This proves a uniform ordinary gap on $\widehat{W}$.

For completeness, we verify the additional ordinary gap needed to apply polarized birationality to an integral Weil divisor. Let B_A be the trace boundary of A on $\widehat{W}$. On every component of E_0 its coefficient is at least $1/2$. For an original component the covering-prime calculation above gives $a' = 0$ and $d' \leq D_W(\cdot) \leq 1$; extracted components have $A < \delta_*$. Choose a fixed integer $m > 4/\delta$ and an integral divisor

$$N = mE_0 - Q_0,$$

where the coefficients of Q_0 belong to $[1, 2]$ on $\text{Supp } E_0$ and are zero elsewhere. Such rounding is possible coefficient by coefficient. Since $mE_0 - Q_0$ dominates a positive multiple of E_0 , N is big. Run an N -MMP to a nef and big integral Weil divisor N^+ on a $\mathbb{Q}$ -factorial model. With stars denoting pullbacks to a common determination, b-nef negativity and MMP negativity give

$$0 \leq e_E = (E_0^+)^* - E_0^*, \quad me_E \leq (Q_0^+)^* - Q_0^*. \quad (\text{C.9})$$

The total generalized log canonical divisor of A is $t(1 - \beta)E_0$ up to a rational principal divisor. Comparing its nef part before and after the MMP, ordinary discrepancies on the output are at least

$$A + (B_A^+)^* - t(1 - \beta)e_E \geq A + \left(\frac{1}{4} - \frac{t}{m}\right)(Q_0^+)^*. \quad (\text{C.10})$$

Indeed $B_A^+ \geq Q_0^+/4$ and $e_E \leq (Q_0^+)^*/m$ by (C.9). If a primitive prime with $A < \delta_*$ is in $\text{Supp } E_0$, it has $(Q_0^+)^* \geq Q_0^* \geq 1$. If it is not in this support, then it is already a prime on $\widehat{W}$ and cannot be contracted by the N -MMP: N has coefficient zero there, the output pullback is effective, and a contraction would require strictly positive MMP error. Its ordinary discrepancy on the output is one. All other primitive orders have $A \geq \delta_*$. Formula (C.10) therefore gives a uniform ordinary gap on the output. Also $N^+ - K^+$ is pseudo-effective, since the output is of Fano type. Birkar's integral-Weil polarized birationality theorem [6, Theorem 1.1] now gives birational sections of a bounded multiple of N^+ . Negativity identifies these with sections of that multiple of N , hence with functions in $S(CD_W/\beta)$ for bounded C . This proves the field-generation assertion.

Choose algebraically independent ϕ_j in this space. If $\omega = \bigwedge_j d\phi_j$, define on the base

$$H(E) = a'(E) - \nu_{\log, E}(\omega), \quad \nu_{\log, E}(\omega) = 1 + \text{ord}_E(\omega). \quad (\text{C.11})$$

The Calabi–Yau identity shows that H is the moduli b-divisor after a rational principal adjustment. It is thus b-nef and $\mathbb{Q}$ -Cartier, and its values have bounded denominator because a' does and the log orders are integral. If $T|_{k(W)} = e \text{ord}_E$ and $a(T) = 0$, the discriminant minimum gives $a'(E) = 0$. Pole bounds for the ϕ_j on every model give

$$\nu_{\log, E}(\omega) \geq -CD_W(E)/\beta.$$

If all $E(\phi_j) \geq 0$, the form is regular at that discrete valuation ring and its logarithmic order is at least one. These prove (i). For (ii), if restriction is nontrivial, then $D_W(E) = 0$ by $\mathbf{P} \leq \mathbf{D}$, so the functions are regular and

$$H(Q) = ea'(E) - e\nu_{\log, E}(\omega) \leq a(Q) - e \leq 0.$$

The trivial restriction gives $H(Q) = 0$ directly.

On W itself we also have $H_W \leq CD_W/\beta$: use $a' \leq 1$, the lower coefficient bound, and the same log-order estimate. For any arbitrarily small positive rational λ , choose on a determining model an effective

$$\Sigma \sim_{\mathbb{Q}} H + \lambda \mathbf{D}_W/\beta;$$

this is possible because the displayed class is nef and big. The rational principal order

$$\chi = \Sigma - H - \lambda \mathbf{D}_W/\beta$$

has poles at most CD_W/β on W , and therefore at most CD/β upstairs. Condition (C.5), applied to $D + (\beta/C)\chi$, gives $\chi(T) \leq C\eta D(T)/\beta$. Since $\Sigma \geq 0$,

$$H(T) \geq -C\eta D(T)/\beta - \lambda \mathbf{P}(T)/\beta.$$

Let $\lambda \downarrow 0$ to prove (iii). Finally, for $T \in I_\beta$, $\mathbf{P}(T) = 0$, so each ϕ_j has nonnegative order there. If the restriction were nontrivial, (i) and (iii), together with $D(T) \leq \beta$, would give $-1 \geq H(T) \geq -C\eta_i$, which is impossible eventually. $\square$

C.2 A phase estimate at a varying scale

The cutoff construction identifies the part of the function field visible at a prescribed pole cost. We next use it to control the denominators of rational principal orders at floor primes of very small polarization order. This is the phase estimate needed to regularize cone characters in Section D.

Lemma C.2 (Scaled phase bound). *Assume (C.5) with $\eta_i \rightarrow 0$ whenever $a(T) = 0$ and $LD(T) \rightarrow 0$, along arbitrary subsequences. Let ℓ be a rational principal order. Suppose that for every nonfloor prime Q on V there is an integer p_Q with*

$$|p_Q - \ell(Q)| \leq G(Q), \quad G \geq 0, \quad G \sim_{\mathbb{Q}} c_g LD, \quad (\text{C.12})$$

where $c_g \geq 0$ is rational and uniformly bounded. There is a fixed positive integer q such that

$$\text{dist}(q\ell(T), \mathbb{Z}) = O(LD(T)) \quad \text{if } a(T) = 0 \text{ and } LD(T) \rightarrow 0. \quad (\text{C.13})$$

Proof. The hypothesis on nonfloor primes also holds on U , since every additional extracted prime is on the floor. If $c_g > 0$, condition (C.5) applies to $G/(c_g L)$. If $c_g = 0$, the effective rationally trivial divisor G is zero. Consequently, at every floor order to which the smallness hypothesis applies,

$$G(T) \leq c_g L(1 + \eta_i)D(T). \quad (\text{C.14})$$

Suppose the conclusion fails. Factorial diagonalization gives a subsequence and primitive targets T with

$$a(T) = 0, \quad \beta = D(T), \quad L\beta \rightarrow 0, \quad \frac{\text{dist}(q\ell(T), \mathbb{Z})}{L\beta} \rightarrow \infty \quad \text{for each fixed } q \in \mathbb{N}_{>0}. \quad (\text{C.15})$$

To justify the simultaneous last assertion, at stage i choose a failure for the multiplier $i!$ with ratio tending sufficiently quickly to infinity. For $q \mid i!$, $\text{dist}(i!x, \mathbb{Z}) \leq (i!/q) \text{dist}(qx, \mathbb{Z})$; choosing the failure ratio larger than all these factors gives the claimed diagonal sequence. Among such sequences maximize the eventually constant $\text{trdeg } K_\beta$. Lemma C.1 says that T is horizontal over this field. Let τ be the first positive cutoff for which $K_\tau = K_\beta$. It exists because there are finitely many coefficient cutoffs and the field $K_{0+} = k(V)$ cannot have a nonzero horizontal divisorial valuation. In particular,

$$0 < \tau \leq \beta, \quad L\tau \rightarrow 0. \quad (\text{C.16})$$

After multiplying ℓ and G by one fixed integer, maximality implies a uniform bound

$$\text{dist}(\ell(Q), \mathbb{Z}) \leq CLD(Q) \quad (a(Q) = 0, D(Q) < \tau). \quad (\text{C.17})$$

Indeed, a failure of this assertion for every fixed multiple would, by the same factorial diagonalization, produce a sequence at a cutoff strictly before τ , hence at strictly larger transcendence degree.

The first threshold. We first prove (C.17) with $D(Q) \leq \tau$, after another fixed multiplication. If it fails, choose a primitive S with $D(S) = \tau$ such that every fixed multiple has phase mismatch much larger than $L\tau$. The prime S lies on U . Fix rational $0 < \gamma < 1/4$ and put

$$\mathcal{S} = S + \sum_{\substack{Q \text{ on } U, \\ a(Q)=0 \\ Q \neq S}} \gamma \min\{1, D(Q)/\tau\} Q. \quad (\text{C.18})$$

Let t be the pseudo-effective threshold of $D_U - t\mathcal{S}$. It is rational, since U is of Fano type. The divisor $D_U - \tau\mathcal{S}$ is effective, while an effective representative of $D_U - t\mathcal{S}$ would give an effective divisor rationally equivalent to D whose order at S is at least t . Rational pseudo-effectivity on a Fano type model is realized by such an effective representative. Hence (C.5) at S gives

$$\tau \leq t \leq (1 + \eta_i)\tau, \quad \eta_i \rightarrow 0. \quad (\text{C.19})$$

Run the threshold MMP and take the semiample nonbig fibration. On its generic fibre,

$$D' \sim_{\mathbb{Q}} t\mathcal{S}', \quad \sum_{Q \neq S} D(Q)Q' \lesssim 2\eta_i\tau\mathcal{S}'. \quad (\text{C.20})$$

Here $A \lesssim B$ means that $B - A$ is rationally linearly equivalent to an effective rational divisor. To see the second inequality, the coefficient of every $Q' \neq S'$ in $D' - t\mathcal{S}'$ is at least $D(Q)/2$ for large i , whereas the coefficient of S' is at least $-\eta_i\tau$. Restriction of the rationally trivial threshold class proves the claim. Enlarge η_i rationally if necessary. The restriction of D' is big, so S' is horizontal and big on this generic fibre. For the threshold MMP and every subsequent MMP over its base, the pullback errors satisfy

$$0 \leq e_D = (D'')^* - \mathbf{D} \leq t((\mathcal{S}'')^* - \mathcal{S}^*). \quad (\text{C.21})$$

The first comparison uses that $\mathbf{D}$ is nef; the second is MMP negativity for $D - t\mathcal{S}$, followed by its crepancy over the threshold base.

We need a signed integral multiplier, allowed to grow slowly, for which

$$m = m_i \neq 0, \quad \sigma = \{m\ell(S)\} \rightarrow 0^+, \quad \eta_i + |m|L\tau = o(\sigma). \quad (\text{C.22})$$

Pass first to a limiting phase. If it is irrational, choose fixed signed powers whose limiting residues approach zero from above, and then let these powers vary sufficiently slowly. For a rational limit, first take a fixed multiple clearing that limit. Its signed residual $\zeta_i \rightarrow 0$ satisfies $|\zeta_i|/(L\tau) \rightarrow \infty$ by the assumed failure. Choose its sign positive and multiply it by an integer so that the resulting residual still tends to zero but dominates η_i . This is possible with no wrap: choose a target residual $s_i \rightarrow 0$ with $s_i \gg \eta_i + |\zeta_i|$, and a multiplier of size $s_i/|\zeta_i|$. Its product with $L\tau$, divided by the new residual, is $O(L\tau/|\zeta_i|) \rightarrow 0$. This proves (C.22).

On the threshold model choose a rational correction n so that $N = m\ell' + n$ is an integral Weil divisor, as follows:

$$\begin{aligned} n(S') &= -\sigma, & (\text{C.23}) \\ |n(Q')| &\leq C|m|LD(Q) & (Q \neq S, a(Q) = 0, D(Q) < \tau), \\ 0 \leq n(Q') &< 1 & (Q \neq S, a(Q) = 0, D(Q) \geq \tau), \\ |n(Q')| &\leq |m|G'(Q') & (a(Q) > 0). \end{aligned}$$

Use nearest integers in the second line, upward rounding in the third, and mp_Q in the last; outside the finite relevant support the correction is zero. Since ℓ' is rational principal, (C.20) bounds the positive errors in class by

$$C|m|LD' + |m|G' + \tau^{-1} \sum_{Q \neq S} D(Q)Q' \lesssim O(|m|L\tau + \eta_i)S'.$$

Thus

$$N \lesssim -(\sigma - o(\sigma))S',$$

and N is not relatively pseudo-effective. Run its Mori fibre MMP over the threshold base. On the geometric generic fibre F of the last Picard-number-one contraction,

$$H_1 = -N'' \text{ is integral and ample,} \quad H_1 \lesssim (\sigma + o(\sigma))S''. \quad (\text{C.24})$$

The upper comparison uses the lower bounds in (C.23); all the generic class comparisons survive pushing and restricting. The variety F is of Fano type, and $-K_F$ dominates $S''|_F$ numerically. In fact the pushed crepant boundary of a is effective and contains S'' with coefficient one, while the moduli trace on F is pseudo-effective. For this last assertion restrict its nef determination first, add arbitrarily small ample errors and a pullback of an ample divisor if needed, and push effective approximations to F . Letting the errors tend to zero proves pseudo-effectivity. Picard-number-one proportionality on the total contraction, followed by its geometric generic restriction and rational linear equivalence on F , therefore gives

$$-K_F \sim_{\mathbb{Q}} I_i H_1, \quad I_i \longrightarrow +\infty. \quad (\text{C.25})$$

We verify the small-discrepancy hypothesis that contradicts Lemma B.4. Let w be an integral divisorial angular evaluation on F , allowing a nonprimitive one, with ordinary discrepancy $A_F(w) \rightarrow 0$. Generalized-to-ordinary comparison gives

$$A_F(w) \geq a(w) + w((S'')^*). \quad (\text{C.26})$$

Indeed the pushed boundary has coefficient one at every surviving floor prime, dominates S'' , and b-nef negativity controls the moduli trace. Hence integrality gives $a(w) = 0$ eventually and, by (C.21), $w(S^*) \leq w((S'')^*) \rightarrow 0$. The formula on U gives $\mathbf{D} \leq CS^*$, with fixed C because $D(Q) \leq 1$ and $\tau \leq 1$. Thus the primitive restriction of w is eventually one of the primes extracted on U . Since $w(S^*) \rightarrow 0$, that prime is neither S nor a floor prime of coefficient at least τ : their $\mathcal{S}$ -coefficients are respectively 1 and γ . The remaining coefficient is $\gamma D(Q)/\tau$, so

$$D(w) \leq C\tau A_F(w). \quad (\text{C.27})$$

The solved estimate (C.17), extended by integral homogeneity, and (C.22) imply that $m\ell(w)$ has a residual $o(A_F(w))$ modulo $\mathbb{Z}$. Equations (C.14) and (C.21) also give

$$|m|w((G'')^*) = |m|(G(w) + c_g Le_D(w)) = o(A_F(w)). \quad (\text{C.28})$$

The equality holds because $G - c_g LD$ is rational principal and its pullback is unchanged. In the estimate, $e_D(w) \leq tw((S'')^*) = O(\tau A_F(w))$.

The signs of the rounding corrections now matter. Their negative part is at most

$$\sigma(S'')^* + C|m|L(D'')^* + |m|(G'')^*;$$

its value is $o(A_F(w))$ by the preceding bounds. The positive unit-size errors are supported on floor components, whose reduced sum has order at most $A_F(w)$ by boundary comparison. Thus

$$-o(A_F(w)) \leq w((n'')^*) \leq O(A_F(w)). \quad (\text{C.29})$$

All these are inequalities between effective divisors pulled back from $\mathbb{Q}$ -factorial models before restriction to F .

Consider the compact Seifert cone associated to the integral polarization H_1 . For a divisorial evaluation over infinity with positive height u and ordinary discrepancy $A \rightarrow 0$, Lemma B.2 gives $A \geq u + A_F(w)$ and

$$u \equiv -m\ell(w) - w((n'')^*) \pmod{\mathbb{Z}}.$$

The trivial angular restriction is impossible when $A \rightarrow 0$, since then u is a positive integer. Otherwise all terms tend to zero, so there is no wrap in this congruence. The residual of $m\ell(w)$ is $o(A_F(w))$ and the negative part in (C.29) is $o(A_F(w))$; positivity of u therefore forces $u = o(A)$. Orders with centre outside infinity have $v(D) = 0$, so the same small-height conclusion holds for every small-discrepancy prime. This contradicts Lemma B.4 and (C.25). We have proved the phase estimate also at $D(S) = \tau$, after a fixed multiplication.

Rational triviality on the cutoff fibre. Return to the cutoff model $U^\tau \rightarrow W_\tau$. On its generic fibre,

$$D' = \sum_{Q \in I_\tau} D(Q)Q',$$

and no other floor prime on this model is horizontal. With the fixed multiplications already made, choose an integral Weil divisor $N = \ell' + n$ whose horizontal trace satisfies

$$|n| \leq CLD' + G'. \quad (\text{C.30})$$

If either N or $-N$ is not relatively pseudo-effective, run its Mori fibre MMP. On the final geometric generic Mori fibre, put $R = \sum_{Q \in I_\tau} Q''$. The same pullback and class comparisons give

$$H_1 = \mp N'' \lesssim O(L\tau)R, \quad 0 \leq \mathbf{D} \leq (D'')^* \leq \tau R^*. \quad (\text{C.31})$$

Again $-K$ dominates R numerically, so the integral ample divisor H_1 has anticanonical index tending to infinity. If $A_F(w) \rightarrow 0$, ordinary comparison gives $a(w) = 0$, $w(R^*) \leq A_F(w)$, and hence

$$D(w) = O(\tau A_F(w)), \quad w((G'')^*) = O(L\tau A_F(w)).$$

For the latter estimate use (C.14) at the primitive restriction and the positive trace-pullback error bounded by $(D'')^*$. That primitive restriction is eventually in I_τ , since $D(w) \rightarrow 0$, and all such primes survive on U^τ . Its solved phase gives a residual $o(A_F(w))$ for $\ell(w)$. Equation (C.30), pulled back after the MMP, gives $w((n'')^*) = o(A_F(w))$. The Seifert congruence consequently forces $u = o(A)$ as before, contradicting Lemma B.4.

Both N and $-N$ are therefore relatively pseudo-effective. On a Fano type generic fibre they each have effective rationally equivalent representatives, by semiample termination. Their sum is an effective rationally trivial divisor and is zero, so N is rationally linearly trivial on that fibre.

A bounded principalization multiple. Run an MMP over W_τ making $R = \sum_{I_\tau} Q''$ nef. On the geometric generic fibre the integral divisor N'' remains rationally principal, and its pullback agrees with that of N , since a rationally trivial divisor is crepant through these modifications. Fix $0 < c < 1$. The discrepancies

$$a_* = a + cR^* \quad (\text{C.32})$$

are an effective generalized klt Calabi–Yau pair. The trace is effective because R is reduced and lies on the floor; klt follows from $\mathbf{D} \leq \tau R^*$ and positivity of $\mathbf{D}$ on $a = 0$. The additional b-part is nef. The same small-order argument shows

$$\text{dist}(N''(w), \mathbb{Z}) = o(a_*(w)) \quad \text{whenever } a_*(w) \rightarrow 0.$$

We record why this bounds its torsion order. If its minimal principalization order l were unbounded, write $lN'' = \text{div}(g)$ and take the normalization in a root of g . The cover is connected by minimality of l and is unramified in codimension one because N'' is integral. Thus ordinary Fano type data pull back without an angular ramification boundary. Form the product with $\mathbf{P}^1$ and take the finite diagonal cyclic quotient, as in Lemma B.5. The product-to-quotient map is quasi-etale; over zero the fibration has multiplicity l . The log pullback of (C.32), product with the line factor, and descent provide effective generalized klt Calabi–Yau data over the curve.

For a prime over zero, put $u = \text{ord}(\text{base coordinate})/l > 0$. Its discrepancy satisfies $A \geq u + a_*(w)$, and the cyclic congruence is $u \equiv -w(N'') \pmod{\mathbb{Z}}$. A trivial angular restriction makes u a positive integer and hence cannot occur when $A \rightarrow 0$. For a nontrivial restriction the preceding phase estimate and the absence of wrap give $u = o(A)$. Add the rational divisor $f^*[0]/l$ with varying coefficient, extending the height by zero away from the origin. The discrepancies $A - su$ remain an effective generalized Calabi–Yau family over the curve for arbitrarily long nonnegative parameter intervals. Lemma A.1 applies because every $A \rightarrow 0$ has $u = o(A)$. But the base discrepancy at zero is at most $1/l \rightarrow 0$ and is nonpositive after the coefficient $s = 1$ is added. This contradicts its conclusion. Thus l is bounded.

The bounded principalization descends to the original generic field. Cartierness descends under the algebraically closed field extension for the corresponding Weil divisor, and a geometrically trivial line bundle and its inverse each have a nonzero section over the original field by flat base change. Their product is a nonzero constant, so those sections give a trivialization. This also explains explicitly why no unbounded field extension is absorbed into the asserted bounded multiple.

The final two-sided comparison. There is now a fixed positive integer q and a rational function χ on the generic fibre, hence in $k(V)$, such that the rational principal order

$$h = q\ell - \text{div}(\chi)$$

has horizontal trace on U^τ bounded by

$$|h'| \leq CLD' + qG'. \quad (\text{C.33})$$

For example, take $\text{div}(\chi) = qN$ on that fibre, so $h' = -qn$. The original target T is horizontal over $K_\tau = K_\beta$, and subtracting an integral principal order does not remove its phase failure:

$$\frac{|h(T)|}{LD(T)} \rightarrow \infty. \quad (\text{C.34})$$

On the geometric generic fibre consider

$$a_0 = a + c\mathbf{D}/\tau, \quad a_s = a_0 + \frac{sh - q|s|\mathbf{G}}{J_i L \tau}. \quad (\text{C.35})$$

Choose $J_i \rightarrow \infty$ sufficiently slowly compared with the ratio in (C.34), and then a two-sided parameter range $|s| \leq R_i \rightarrow \infty$ with $R_i/J_i \rightarrow 0$. These are concave, finite piecewise affine effective generalized Calabi–Yau data. Indeed (C.33) and $D' \leq \tau \sum_{I_\tau} Q'$ give trace effectivity

for this range; off the floor the term $-q|s|\mathbf{G}$ absorbs the possible positive trace of sh . Up to rational principal terms the added nef part has $\mathbf{D}$ -coefficient $c/\tau - q|s|c_g/(J_i\tau) > 0$. Thus b-nefness also holds.

The central member is klt. If a primitive geometric generic order satisfies $a_0 < \min\{1, c\}/2$, then $a = 0$ and $D < \tau/2$. Its primitive horizontal restriction is therefore a surviving prime of I_τ on U^τ . Equations (C.14) and (C.33) give

$$|h| + q\mathbf{G} = O(LD)$$

there, uniformly for all such orders: these are primes already on the model where the horizontal trace bound was proved. Thus the fixed discrepancy cutoff controls every order required by the two-sided comparison lemma. For every fixed signed s the relative difference $(a_s - a_0)/a_0$ is consequently $O(|s|/J_i) \rightarrow 0$. This is the strong two-sided hypothesis of Lemma A.1, with a fixed positive discrepancy cutoff. Its conclusion over a point says that a_s is klt for fixed s eventually. At the lift of T , however, choose $s = 1$ or $s = -1$ so that $sh(T) = -|h(T)|$. Equations (C.35) and (C.34), with the choice of J_i , give $a_s(T) < 0$. This contradiction proves the lemma. $\square$

C.3 Exact saturation of neutral fields

For the pinning argument, a small error on a fixed list of functions is not enough: later steps use rational expressions of unrestricted degree in those functions. The next lemma puts every bounded-cost function in one relatively algebraically closed cutoff field and makes the relevant valuations exactly trivial on that field. Its count estimates use one normalization for both the upper and lower bounds.

Lemma C.3 (Saturation and section counts). *Let $f_1, \dots, f_s$ be a finite list of genuine rational functions on V , whose length may vary with the sequence, and let*

$$J(T) = \max\{0, \max_j (-T(f_j))\}, \quad 0 \leq J \leq CL_0\mathbf{D}, \quad 1 \leq L_0 = O(L). \quad (\text{C.36})$$

Thus J is an integral basepoint-free Cartier b -divisor on a high model. Suppose (C.5) holds uniformly with $\eta_i \rightarrow 0$ on

$$\mathcal{L} = \{T : a(T) = 0, J(T) = 0\}.$$

In addition, on every subsequence on which $L_0 \rightarrow \infty$, assume that all valuations in $\mathcal{L}$ eventually restrict trivially to the field generated by $S(L_0D)$.

After passage to a subsequence, there are cutoff fields K_ of fixed transcendence degree p , a fixed constant B , and positive real numbers v_i with the following properties.*

- (i) *For every fixed $C_1 > 0$, eventually $S(C_1LD) \subset K_*$.*
- (ii) *Eventually $S(BLD)$ generates K_* . If $T \in \mathcal{L}$ is nonnegative on this space, its restriction to K_* is trivial.*
- (iii) *There is $C' > 0$, independent of the positive integer n , such that for every fixed n , eventually*

$$\dim S(nLD) \leq C'v_in^p, \quad \dim \text{Span}\{g_1 \cdots g_n : g_j \in S(BLD)\} \geq v_in^p. \quad (\text{C.37})$$

The eventual index in (i) may depend on C_1 , and that in (iii) may depend on n ; B and C' are fixed before n . All assertions concern divisorial evaluations. Exact triviality also holds on rational monomial subcones cut out by the indicated conditions.

Proof of the field assertions. Because J is integral and $L_0 = O(L)$, choose a fixed small $c > 0$ so that

$$a(T) = 0, \quad LD(T) \leq c \implies J(T) = 0.$$

There is, after a subsequence, a single field plateau containing two cutoffs

$$0 < \beta_0 \leq \beta_1 \leq c/L, \quad L\beta_0 \longrightarrow 0, \quad L\beta_1 \geq c' > 0, \quad K_{\beta_0} = K_{\beta_1} = K_*. \quad (\text{C.38})$$

Here is a precise reason for this choice. Pass to limits of the finitely many change locations multiplied by L , adding dummy endpoints if the number of changes varies. The last change tending to zero lies below a cutoff β_0 with $L\beta_0 \rightarrow 0$; choose it decreasing slowly enough to lie beyond that change. The next change either has a positive limiting scaled location or lies beyond c/L . Choose β_1 strictly before it, with $L\beta_1$ fixed positive and at most c . This gives (C.38). Passing again fixes $p = \text{trdeg } K_*$.

All I_{β_1} belong to $\mathcal{L}$, so Lemma C.1 makes them horizontal over K_* . It follows, for *every* rational $m > 0$, that

$$S(mP_{\beta_0}) = S(mP_{\beta_1}). \quad (\text{C.39})$$

Indeed a function in the left side belongs to K_* : enlarge m to an integer and use the section-field description. Its order at every removed prime in I_{β_1} is zero by horizontality. Removing those pole allowances therefore leaves it in the right side. The reverse inclusion follows from $P_{\beta_1} \leq P_{\beta_0}$. For fixed C_1 , a function in $S(C_1LD)$ has no pole at I_{β_0} eventually, because its order is integral and $C_1L\beta_0 < 1$. Hence

$$S(C_1LD) \subset S(C_1LP_{\beta_0}) = S(C_1LP_{\beta_1}) \subset K_*.$$

This proves (i). The cutoff generation statement at β_1 , and $\beta_1 \geq c'/L$, place a birational field-generating space for K_* inside $S(B_0LD)$ for fixed B_0 . Enlarging B will therefore make $S(BLD)$ generate K_* ; a basis contains p algebraically independent elements, by greedily choosing a transcendence basis from any field-generating set.

We prove that some fixed B forces the asserted exact triviality. Otherwise choose $B_i \rightarrow \infty$ slowly and $T_i \in \mathcal{L}$ that are nonnegative on $S(B_iLD)$ but nontrivial on K_* . Since $K_1 = k$, there is a change at $\beta \geq \beta_1$ where T is trivial on K_β but nontrivial on the preceding field K_{β_-} . Take β_- sufficiently close to β from below that $\beta_- > \beta/2$. Use H, ϕ_j from Lemma C.1 at β_- . Their pole bounds are $C\mathbf{D}/\beta$, and $1/\beta = O(L)$; thus $T(\phi_j) \geq 0$ eventually. Its nontrivial restriction gives

$$H(T) \leq -1. \quad (\text{C.40})$$

If $L_0\beta \rightarrow \infty$, then $L_0 \rightarrow \infty$ because $\beta \leq 1$. The same bounded-cost field-generating space for K_{β_-} is contained in $S(L_0D)$ eventually. The hypothesis of triviality on the *field* generated by the latter space contradicts the choice of T . It remains to treat a subsequence on which $L_0\beta$ is bounded.

Restrict to the geometric generic fibre of the cutoff contraction $U^\beta \rightarrow W_\beta$. There T is horizontal and

$$D' \leq \beta \sum_{I_\beta} Q'.$$

Apply the extra-data part of Lemma A.3, over a point, with

$$\tilde{a} = a + \kappa J, \quad d = c_* = \mathbf{D}/\beta, \quad h = H, \quad (\text{C.41})$$

where $\kappa > 0$ is fixed rational and sufficiently small; c_* denotes the additional nonnegative direction of that lemma. We verify all its hypotheses. The orders $\tilde{a}$ have bounded denominator because a and J are integral. The order H has bounded denominator by the cutoff lemma.

Both J and $\mathbf{D}/\beta$ are b-nef, and H is pulled back from a b-nef divisor on K_{β_-} ; after a birational determination it restricts to a b-nef divisor on this fibre. Their Calabi–Yau realizations hold the total class rationally trivial. At floor primes on this fibre,

$$H \leq CD/\beta, \quad J \leq CD/\beta,$$

where the second bound uses boundedness of $L_0\beta$. At other primes on the model, $a = 1$, $D = 0$, $J = 0$, and $H \leq 0$. These inequalities give effective traces on uniform small positive ranges in (C.41), including the family with $\lambda H + sc_*$ for $0 < s \leq \lambda\theta_0$. The family $\tilde{a} + td$ is klt for $t > 0$ because D is positive on $a = 0$. Finally,

$$\tilde{a} = 0 \iff a = J = 0, \quad -C\eta_i c_* \leq H \leq Cc_* \quad \text{on this zero set.}$$

The lower bound is part (iii) of the cutoff lemma, using the assumed uniform (C.5). Thus the exact sign conclusion of Lemma A.3 over a point prohibits $H < 0$ at the lift of T , contradicting (C.40). This proves (ii).

The proof of (iii), including its common normalization v_i , is given below after establishing the adjoint lattice-point statement. For the extension to rational monomial subcones, take any finite list of tested functions, resolve their divisors, and subdivide the cone so their orders are linear on each piece. Divisorial rational rays are dense in each rational piece. The exact zero assertions extend by continuity; applying this separately to each function gives triviality on the whole field. $\square$

C.4 Adjoint lattice points for integral Weil polarizations

We prove the precise lattice-point supply needed for the lower count. The polarization will be nef and big and may be an integral Weil divisor with unbounded Cartier index. The rounding and pushdown steps below keep those hypotheses throughout.

The weighted-blowup and adjoint-lifting construction below adapts the “Adjoint lattice points” and “Orbifold vanishing” lemmas in [26]. Here the argument is developed for a nef and big integral Weil polarization with a klt Calabi–Yau boundary, including the rounding and global regularity checks given below; the earlier theorem is not used as a black box.

Lemma C.4 (Vanishing on a smooth orbifold). *Let $\mathcal{X}$ be a smooth proper Deligne–Mumford stack over an algebraically closed field of characteristic zero, with trivial generic stabilizer and projective coarse space. Let Θ be an effective rational divisor with simple normal crossing support and coefficients less than one. If D is an integral Cartier divisor and $D - (K_{\mathcal{X}} + \Theta)$ is nef and big, then $H^q(\mathcal{X}, \mathcal{O}_{\mathcal{X}}(D)) = 0$ for $q > 0$.*

Proof. Write $q : \mathcal{X} \rightarrow X$ for the coarse morphism. For each prime B of X , let P be its reduced inverse-image divisor and e_B its generic stabilizer order. Thus $q^*B = e_BP$. Rational divisors on $\mathcal{X}$ descend uniquely to rational divisors on X ; use a subscript c for the descended divisor. The local normal-direction equation $t = z^{e_B}$ gives

$$K_{\mathcal{X}} = q^*(K_X + R), \quad R = \sum_B (1 - e_B^{-1})B.$$

Choose compatible canonical divisors using one rational top form. Set $\Delta = R + \Theta_c - \{D_c\}$. If the coefficients of D and Θ at P are $d \in \mathbb{Z}$ and $\theta \in [0, 1)$, write $d = e_B n + s$ with $0 \leq s < e_B$, also when d is negative. Then

$$\text{coeff}_B \Delta = \frac{e_B - 1 + \theta - s}{e_B} \in [0, 1).$$

In particular Δ is effective.

The pair (X, Δ) is klt, including at higher-codimensional quotient singularities. Indeed, étale locally on X there is a finite quotient presentation $[U/G]$, with U smooth and $f : U \rightarrow U/G$ finite. Its log pullback boundary is $\Theta_U - f^*\{D_c\}$. This possibly negative boundary is bounded above by the simple normal crossing boundary Θ_U . Thus for every divisorial valuation F over U ,

$$A(F; U, \Theta_U - f^*\{D_c\}) = A(F; U, \Theta_U) + F(f^*\{D_c\}) > 0.$$

The finite-map discrepancy formula multiplies downstairs log discrepancies by positive ramification indices, so the downstairs pair is klt. This argument requires only local quotient charts.

The coarse pushforward is

$$q_* \mathcal{O}_X(D) = \mathcal{O}_X(\lfloor D_c \rfloor).$$

To check it, a rational section h is regular along P precisely when $e_B \operatorname{ord}_B(h) + d \geq 0$, equivalently $\operatorname{ord}_B(h) + \lfloor d/e_B \rfloor \geq 0$. Both sides extend these codimension-one conditions across codimension at least two: on a smooth finite quotient chart this follows by normal extension and then taking invariants. The stack is tame in characteristic zero, so coarse pushforward is exact and identifies the cohomology groups.

Quotient singularities are locally $\mathbb{Q}$ -factorial. Consequently $M = \lfloor D_c \rfloor$ is an integral $\mathbb{Q}$ -Cartier Weil divisor, and

$$M - (K_X + \Delta) = D_c - (K_X + R + \Theta_c)$$

is nef and big. Kawamata–Viehweg vanishing for a klt pair and an integral $\mathbb{Q}$ -Cartier Weil divisor gives the result. We use [13, Theorem 3.2.4, manuscript version of 9 October 2023]; the additional log-bigness condition on log canonical centres is vacuous for a klt pair. The characteristic-zero statement also follows from its complex form by descending the finite-type data to a finitely generated field and using invariance of nefness, bigness and cohomology under algebraically closed field extension. $\square$

Lemma C.5 (Adjoint Taylor exponents). *Let X be a projective $\mathbb{Q}$ -factorial variety of Fano type of dimension $p > 0$, and let N be a nef and big integral Weil divisor. Suppose that $H^0(X, \mathcal{O}_X(N))$ defines a birational map. Choose an effective rational boundary Δ with (X, Δ) klt and $K_X + \Delta \sim_{\mathbb{Q}} 0$. At a general smooth point outside the supports of N and Δ , fix regular parameters and take total-degree-then-lexicographic Taylor initial exponents, using the compatible local trivializations of the powers of N . Let*

$$K = \overline{\operatorname{conv}} \left\{ \frac{\nu(s)}{j} : j \in \mathbb{Z}_{>0}, \quad 0 \neq s \in H^0(X, \mathcal{O}_X(jN)) \right\}.$$

If t is a positive integer and $\alpha \in \mathbb{Z}_{\geq 0}^p$ satisfies $\alpha + \mathbf{1} \in \operatorname{int}(tK)$, then some section of tN has initial exponent α .

Proof. The shift by $\mathbf{1}$ supplies the adjoint correction in this argument. A weighted blowup will convert it into the canonical discrepancy, and the same shift will place the required monomial in a multiplier ideal on the exceptional divisor. Vanishing then lifts that monomial to a global section.

Preparing sections with weighted initials. Resolve (X, Δ) by $f : Y \rightarrow X$, isomorphically over a neighborhood of the chosen point x . Put $N_Y = f^*N$ and define the log pullback Δ_Y by

$$K_Y + \Delta_Y = f^*(K_X + \Delta).$$

The exceptional coefficients of Δ_Y may be negative. This will cause no difficulty: our vanishing boundary will be a fractional rounding correction, not Δ_Y . Both N_Y and Δ_Y have zero divisor support near x after compatible local trivialization. Choose canonical divisors with no support near x as well.

Let C be the closed cone generated by the actual pairs $(j, \nu(s))$. It has slice K at degree one. Choose rational $0 < t_0 < t$ with $(t_0, \alpha + 1) \in \text{int } C$. There is a cone generated by finitely many actual pairs that already contains this point in its interior: approximate the vertices of a small simplex around the point by finite nonnegative combinations of actual pairs. Sufficiently close approximations still surround the point.

Choose positive primitive integral weights $u_1, \dots, u_p$ that replace the Taylor order by scalar weight on this finite list, each listed section having its specified initial as a single weight initial. Also require

$$\gamma \text{ precedes } \alpha \implies u \cdot \gamma < u \cdot \alpha.$$

Here is the finite-order justification. Take an integer d at least the total degree of every listed initial and of α . Weights sufficiently close to $(1, \dots, 1)$ preserve every strict total-degree comparison through degree d , and satisfy $(d+1) \min u_i > d \max u_i$, so all higher-degree monomials are larger than every relevant initial. Successive sufficiently small rational perturbations realize lexicographic comparisons within the finitely many monomials of degree at most d . Clearing denominators and dividing by the common divisor preserves all strict comparisons. This also proves the assertion for formal Taylor expansions.

Put $p_0 = u \cdot \alpha$ and $U = u \cdot (\alpha + 1)$. The finite cone meets the plane with degree t_0 and weight U in a neighborhood of $(t_0, \alpha + 1)$ relative to that entire plane. Choose finitely many rational points surrounding it there. Write these points as nonnegative rational combinations of the cone generators and clear denominators. Taking products, and then common powers as necessary, gives a sufficiently divisible positive integer l and sections of the Cartier divisor $lt_0 N_Y$ with single weight initials $z^{\beta^{(1)}}, \dots, z^{\beta^{(r)}}$ such that

$$u \cdot \beta^{(a)} = lU, \quad \alpha + 1 \in \text{relint conv}\{\beta^{(1)}/l, \dots, \beta^{(r)}/l\}.$$

The relative interior contains a neighborhood in the whole plane $u \cdot v = U$. Pulling these sections to Y is legitimate after clearing denominators: the effective divisor of each product is $\mathbb{Q}$ -Cartier and its rational pullback is effective.

The weighted modification and the vanishing divisor. Let $\pi : \mathcal{Z} \rightarrow Y$ be the smooth stack weighted blowup at x with weights u , and let E be its exceptional Cartier divisor. On the coordinate model it is

$$\left[(\mathbb{A}_{x_1, \dots, x_p, \tau}^{p+1} \setminus \{x_1 = \dots = x_p = 0\}) / \mathbb{G}_m \right], \quad \rho \cdot (x_i, \tau) = (\rho^{u_i} x_i, \rho^{-1} \tau), \quad z_i = \tau^{u_i} x_i.$$

Its exceptional divisor is the weighted projective stack $\mathbb{P}(u)$. This description pulls back along an étale parameter map near x and glues to the identity away from x . Equivalently use the weighted ideal filtration generated by monomials of weight at least k ; each ideal is supported on a finite infinitesimal neighborhood of x and extends as the unit ideal outside x . The finitely generated weighted Rees algebra makes the coarse modification projective. The stack is smooth with finite stabilizers and trivial generic stabilizer. On the displayed atlas the Jacobian gives

$$K_{\mathcal{Z}} = \pi^* K_Y + (\sum_i u_i - 1)E.$$

For $p = 1$ this construction is the identity modification and $E = x$; the same formulas and restriction sequence apply.

Write $\Delta_{\mathcal{Z}} = \pi^* \Delta_Y$, supported away from E , and define

$$P_0 = \pi^*(tN_Y) - p_0 E, \quad H_0 = \pi^*(lt_0 N_Y) - lUE.$$

The divisor P_0 is rational globally but integral Cartier near E ; H_0 is Cartier globally. The selected sections, after their common weighted order is removed, are sections of H_0 . Their base ideal $\mathfrak{b}$ does not vanish identically on E , and its restriction to E is generated by their monomial initials. Take a projective log resolution $\sigma : \mathcal{Y} \rightarrow \mathcal{Z}$ of these data and all rational divisor supports used below, with strict transform $\tilde{E}$. Arrange that the total support is simple normal crossing, that σ is an isomorphism at the generic point of E , and that $\tau : \tilde{E} \rightarrow E$ resolves $\mathfrak{b}|_E$. First resolve the additional divisor supports, then principalize the ideal with the resulting simple normal crossing boundary, including the transform of E , on smooth charts [35, Theorems 2.4.1–2.4.3]. Smooth functoriality identifies these constructions on chart overlaps, so they descend to the required smooth stack; the sequence of blowups has projective coarse space. Write

$$\mathfrak{b}\mathcal{O}_{\mathcal{Y}} = \mathcal{O}_{\mathcal{Y}}(-G).$$

Then $G \geq 0$ does not contain $\tilde{E}$, and $\sigma^*H_0 - G$ is globally generated.

Set

$$Q = P_0 - K_{\mathcal{Z}} - \Delta_{\mathcal{Z}} - E, \quad R = \sigma^*Q - G/l, \quad A = K_{\mathcal{Y}} + [R], \quad \Theta = [R] - R.$$

Thus A is an integral Cartier divisor, and Θ is effective with simple normal crossing support and coefficients in $[0, 1)$. The weighted canonical identity, $U - p_0 = \sum u_i$, and $K_Y + \Delta_Y \sim_{\mathbb{Q}} 0$ give

$$Q - H_0/l \sim_{\mathbb{Q}} (t - t_0)\pi^*N_Y.$$

Consequently

$$A - (K_{\mathcal{Y}} + \Theta) = R \sim_{\mathbb{Q}} (t - t_0)\sigma^*\pi^*N_Y + (\sigma^*H_0 - G)/l.$$

The first summand is nef and big and the second is nef. Lemma C.4 therefore gives $H^1(\mathcal{Y}, \mathcal{O}(A)) = 0$ and the restriction surjection

$$H^0(\mathcal{Y}, A + \tilde{E}) \longrightarrow H^0(\tilde{E}, (A + \tilde{E})|_{\tilde{E}}).$$

The boundary monomial. We now exhibit the required section on the exceptional divisor. Near E the original boundary vanishes and P_0 is integral. Thus, on a neighborhood of $\tilde{E}$,

$$A = K_{\mathcal{Y}} + \sigma^*(P_0 - K_{\mathcal{Z}} - E) - \lfloor G/l \rfloor.$$

Adjunction identifies the restricted bundle with

$$\tau^*(P_0|_E) + K_{\tilde{E}} - \tau^*K_E - \lfloor (G|_{\tilde{E}})/l \rfloor.$$

Round-down commutes with this restriction: simple normal crossings prevent two components of G from restricting to a common prime divisor on $\tilde{E}$, and each disconnected intersection retains the same coefficient. This is the log-resolution expression for $P_0|_E \otimes \mathcal{J}_E((\mathfrak{b}|_E)^{1/l})$.

The weighted-homogeneous monomial z^α is a global section of $P_0|_E = \mathcal{O}_E(p_0)$, after the fixed local frame. On the chart $x_k \neq 0$, use the smooth finite atlas setting $x_k = 1$, with residual group μ_{u_k} . The ideal becomes the ordinary monomial ideal generated by $x_{\text{rem}}^{\beta_{\text{rem}}^{(a)}}$ and the proposed section becomes $x_{\text{rem}}^{\alpha_{\text{rem}}}$. Dropping coordinate k is an affine isomorphism from the plane $u \cdot v = U$ to $\mathbb{R}^{p-1}$, because $u_k > 0$. Hence $\alpha_{\text{rem}} + \mathbf{1}$ lies in the ordinary interior of the convex hull of the projected vectors $\beta_{\text{rem}}^{(a)}/l$, and therefore in the interior of the scaled Newton polyhedron. Howald's monomial multiplier-ideal formula [15, Main Theorem] gives the required membership on this smooth atlas. It is equivariant for μ_{u_k} , so descends on every chart. No fractional coordinate exponents are introduced: the root needed to set $x_k = 1$ is already part of the atlas.

Lifting and pushing down. The restriction surjection now lifts this monomial. Near E its lifting bundle differs from σ^*P_0 by

$$K_Y - \sigma^*K_Z - \sigma^*E + \tilde{E} - \lfloor G/l \rfloor.$$

The part preceding the last term is exceptional over Z ; indeed it is effective by the smooth blowup discrepancy calculation for the smooth pair (Z, E) . At every nonexceptional prime the displayed correction is nonpositive. Therefore the lifted rational section pushes down to a regular section of P_0 near E . Its restriction to E is z^α , initially generically and then everywhere by regularity. Pushing further to the smooth neighborhood of x gives a regular section with weight initial exactly z^α .

It remains to check global regularity downstairs, since N_Y was rational and Δ_Y could have negative coefficients. At an original prime B of X away from x , write $n_B = \text{coeff}_B(tN) \in \mathbb{Z}$, $\delta_B = \text{coeff}_B \Delta \geq 0$, and let $g_B \geq 0$ be the corresponding coefficient of G . Canonical coefficients cancel inside the ceiling, and the allowed divisor coefficient of the pushed lifting bundle is

$$n_B + \lceil -\delta_B - g_B/l \rceil \leq n_B.$$

Thus no forbidden pole appears at any original prime. The point x was already handled by the local weighted calculation (also when $p = 1$). Normality now gives a section of $\mathcal{O}_X(tN)$. Its weight initial excludes every monomial preceding α in the original Taylor order, by the choice of u , and the coefficient of z^α is nonzero. Its Taylor initial is therefore α . $\square$

Lemma C.6 (Uniform count from the jet body). *In the setting of Lemma C.5, K is bounded and contains the standard simplex $\Sigma = \text{conv}(0, e_1, \dots, e_p)$. For every integer $n \geq 1$,*

$$h^0(X, nN) \leq (n + p)^p \text{vol}(K).$$

Fix any integer $t > 1 + 2p$. The space spanned by products of n sections of tN has dimension at least $n^p \text{vol}(K)$.

Proof. On a nonempty open set, the map given by $|N|$ is an immersion and has no base point, because it is birational in characteristic zero. At the chosen general point its values and first derivatives thus span all first jets. Suitable linear combinations give initial exponents $0, e_1, \dots, e_p$, proving $\Sigma \subset K$. The order of vanishing at this point of a section of jN is at most Cj : intersect its effective divisor with a general complete intersection curve through the point, chosen not to lie in its support. Its degree is $jN \cdot H^{p-1}$ for a fixed sufficiently ample H , and bounds the local intersection multiplicity. The constant may depend on this individual example, which suffices to make K bounded.

For a finite-dimensional space of germs, the number of distinct leading exponents is its dimension, by successive elimination of leading coefficients; alternatively use the filtration by the well-ordered monomials, whose nonzero graded pieces are one-dimensional. The exponent set at degree n is contained in $nK \cap \mathbb{Z}^p$. The disjoint half-open unit boxes based at these lattice points lie in

$$nK + [0, 1]^p \subset nK + p\Sigma \subset (n + p)K.$$

Their total volume is their number, proving the upper bound.

For the lower bound let $y = nx \in nK$. Write $x_i = a_i + f_i$ with $a_i \in \mathbb{Z}_{\geq 0}$ and $0 \leq f_i < 1$. Choose n integers in coordinate i , namely $a_i + 1$ in exactly $\lfloor nf_i \rfloor$ of the positions and a_i in the others. Doing this independently in every coordinate gives $\alpha^{(1)}, \dots, \alpha^{(n)}$ with

$$\sum_{r=1}^n \alpha^{(r)} = \lfloor y \rfloor, \quad 0 < \alpha_i^{(r)} + 1 - x_i < 2.$$

When $f_i = 0$ all entries equal a_i , so the strict inequalities still hold. Therefore $\alpha^{(r)} + \mathbf{1} - x \in \text{int}((t-1)\Sigma)$: all coordinates are positive and their sum is less than $2p < t-1$. Since $\Sigma \subset K$ and $x \in K$, a neighborhood of $\alpha^{(r)} + \mathbf{1}$ lies in $K + (t-1)K = tK$. Lemma C.5 supplies a degree- t section with each exponent $\alpha^{(r)}$. Their product has exponent $\lfloor y \rfloor$.

Consequently all lattice points $\lfloor y \rfloor$, for $y \in nK$, occur among product initials. Their unit boxes cover nK , so their number is at least $\text{vol}(nK) = n^p \text{vol}(K)$. Sections with distinct initials are linearly independent, proving the claimed lower bound. The case $p = 0$ is immediate separately. $\square$

C.5 Completion of the saturation count

The field and exact-triviality assertions are already established. We finish by placing both section counts on one nef integral Weil polarization. The preceding two lemmas then give the same jet-body volume as the normalization in the upper and lower estimates.

Proof of part (iii) of Lemma C.3. If $p = 0$, then $K_* = k$. By part (i), each fixed-cost section space is eventually the one-dimensional constant space. Take $v_i = 1$. Assume $p > 0$ and use the extraction $\widehat{W}$ from Lemma C.1 at β_1 of (C.38). Let D_* be the pullback of D_W to $\widehat{W}$. For every rational $m > 0$, the cutoff spaces identify exactly as function spaces:

$$S(mP_{\beta_1}) = S(mD_*). \quad (\text{C.42})$$

The negative-MMP error is effective and exceptional, so pushing and pulling the divisor inequality for an individual rational function identifies the space with the one on the semiample output. A function in that space belongs to $k(W)$; the equality $P'_{\beta_1} = p^*D_W$ then identifies its inequalities with those downstairs. Pullback to $\widehat{W}$ does not change them. This argument works for each rational m , with no requirement that mP_{β_1} be Cartier.

Every positive coefficient of LD_* is at least $c'\delta$, because $L\beta_1 \geq c'$ and the coefficients of D_*/β_1 are at least δ . Choose a fixed sufficiently large integer M and put

$$N = \lfloor MLD_* \rfloor. \quad (\text{C.43})$$

The coefficient lower bound gives $N \geq LD_*$ once M is large enough. Increase M so that $N \geq CD_*/\beta_1$, with the constant from the field-generation construction. Then $S(N)$ defines a birational map and N is big and integral. Run its MMP to a nef and big integral Weil divisor N' on a $\mathbb{Q}$ -factorial projective Fano type variety X' . For every positive integer j , negativity gives

$$S(jN) = S(jN'), \quad S(jN') \subset S(jMLD). \quad (\text{C.44})$$

Indeed the first equality follows by pushing an effective divisor and, in the reverse direction, adding the effective exceptional MMP error; the second follows from $N \leq MLD_*$ and (C.42). For every fixed positive integer n , part (i) and the exact plateau identities give eventually

$$S(nLD) \subset S(nLD_*) \subset S(nN) = S(nN'). \quad (\text{C.45})$$

Choose an effective ordinary rational klt Calabi–Yau boundary Δ on X' . Such a boundary is obtained from a klt Fano type boundary by adding a general sufficiently divisible rational member of its ample anti-log-canonical class. At a general smooth point outside the boundary and divisor supports, let K_i be the jet body of N' in Lemma C.5. Its volume is positive and finite by Lemma C.6. Set

$$v_i = \text{vol}(K_i).$$

Equations (C.45) and Lemma C.6 imply

$$\dim S(nLD) \leq (n+p)^p v_i \leq (p+1)^p v_i n^p.$$

Fix once and for all an integer $t > 1 + 2p$. The same lemma supplies a space of products of n sections of tN' of dimension at least $v_i n^p$. By (C.44), $S(tN') \subset S(tMLD)$. Enlarge the constant B in part (ii) to at least tM . Those products then lie in the product space in (C.37), proving the lower bound with the same v_i . Both t and M , and hence B , were fixed before n . This completes the lemma. $\square$

Corollary C.7 (Fixed generators and neutral algebraic extensions). *In Lemma C.3, let $g_{i,1}, \dots, g_{i,r_i}$ be any finite list contained in $S(C_1 L_i D_i)$ for one fixed C_1 . For all sufficiently large i , the entire field $F_i = k(g_{i,1}, \dots, g_{i,r_i})$, and its relative algebraic closure inside $k(V_i)$, lie in $K_{*,i}$. Thus every test from part (ii) is exactly zero on every nonzero element of these fields at that same index, without a degree-dependent eventual qualification. More generally, a valuation trivial on a subfield remains trivial on any algebraic extension of it inside its valued overfield.*

Proof. Part (i) of the lemma includes the whole space $S(C_1 L_i D_i)$ at a single eventual index; it does not supply separate indices for its individual elements. Field closure gives $F_i \subset K_{*,i}$. The cutoff field is relatively algebraically closed, so it contains the stated relative algebraic closure as well. To see the last assertion directly, let $x \neq 0$ be algebraic and choose a polynomial relation for it with nonzero constant term and coefficients in the trivially valued subfield. If the value of x were positive, the constant term would have uniquely least value; if it were negative, the highest nonzero term would have uniquely least value. Neither is possible in a vanishing sum. Hence its value is zero. This applies to every rational expression and every polynomial degree once the finite generators have been included, rather than by a further limit in that degree. $\square$

D Characters and regularization on affine cones

We translate the projective results of Sections B and C into statements about homogeneous functions on affine cones. The distinction between a primitive prime order and a rescaled valuation is essential in this translation.

D.1 Angular fields and the discrepancy normalization

Let $\check{R}$ be a finitely generated normal domain over k , equipped with a positive torus grading and a commuting diagonal action of a finite cyclic group G . Positivity means that there is a linear functional strictly positive on every nonzero occurring weight and that the degree-zero ring is k . Set $R = \check{R}^G$ and $X = \text{Spec } R$, and assume that X is $\mathbb{Q}$ -Gorenstein and klt. We use the effective torus acting on R , with character lattice M of rank $r_T > 0$. Write

$$C = \text{cone}\{m : R_m \neq 0\} \subset M_{\mathbb{R}}, \quad \sigma = C^\vee \subset \text{Hom}(M, \mathbb{R}), \quad \mathcal{K} = \text{Frac}(R)^\mathbb{T}.$$

Thus C is a full-dimensional pointed rational polyhedral cone. Choose a multiplicative splitting $m \mapsto z^m$ of the homogeneous rational functions. Such a splitting exists because M is free abelian, and it gives

$$\text{Frac}(R) = \mathcal{K}(z^M). \tag{D.1}$$

Let $h \in \check{R}$ be a nonzero homogeneous function of G -character χ and torus weight $\mathbf{u} \in M_{\mathbb{Q}}$. If r_G is the order of G , set $H = r_G^{-1} \text{div}(h^{r_G})$ on X . Assume that there is a torus-invariant rational top form θ on X such that

$$K_X(\theta) + H = 0, \quad (X, H) \text{ is lc.} \tag{D.2}$$

Evaluation of a semi-invariant function always means evaluation of an invariant power, divided by its exponent. In particular, writing $h^{r_G} = f_h z^{r_G \mathbf{u}}$ gives

$$(P, y)(h) = \gamma(P) + \langle y, \mathbf{u} \rangle, \quad \gamma = \frac{1}{r_G} \operatorname{div}(f_h). \quad (\text{D.3})$$

For a primitive divisorial valuation P of $\mathcal{K}$, define

$$\Delta_P = \{y \in \operatorname{Hom}(M, \mathbb{R}) : P(f) + \langle y, m \rangle \geq 0 \text{ whenever } f z^m \in R\}. \quad (\text{D.4})$$

The inequalities for finitely many homogeneous algebra generators suffice. The polyhedron is nonempty: adding a sufficiently large multiple of a strictly positive grading satisfies all these inequalities. For rational $y \in \Delta_P$, the Gauss rule

$$(P, y) \left(\sum_m f_m z^m \right) = \min_{f_m \neq 0} (P(f_m) + \langle y, m \rangle)$$

defines a valuation nonnegative on R . After a positive rational rescaling it is a primitive divisorial valuation. For trivial restriction to $\mathcal{K}$, the corresponding Gauss valuations are the nonzero rational pure weights $y \in \sigma$, with value $\langle y, m \rangle$ on $f z^m$.

Choose a basis $z_1, \dots, z_{r_T}$ of the weight monomials. Torus invariance gives a rational top form θ_0 on $\mathcal{K}$ such that

$$\theta = \theta_0 \wedge d \log z_1 \wedge \dots \wedge d \log z_{r_T}.$$

Here θ_0 is a nonzero constant when $\mathcal{K} = k$. For a primitive P put $a(P) = \nu_{\log, P}(\theta_0)$, where log order is the order of a top form plus one. Extend a homogeneously to positive multiples of prime orders.

Proposition D.1 (Cone dictionary). *In the above setting the following statements hold.*

- (i) *The discrepancy of (X, H) at (P, y) is $a(P)$, and its discrepancy at a pure weight is zero. Thus $a(P)$ is a nonnegative integer for every primitive P , and $\mathbf{u} \in \operatorname{int} C$.*
- (ii) *For rational $\mu \in \operatorname{int} C$, the ray quotient V_μ is normal and projective, with function field $\mathcal{K}$. The order function*

$$\mathbf{L}(\mu)(P) = \min_{y \in \Delta_P} \langle y, \mu \rangle \quad (\text{D.5})$$

is the b -pullback of an ample rational divisor on V_μ . Consequently

$$\mathbf{D}(P) = \min_{y \in \Delta_P} (P, y)(h) = \gamma(P) + \mathbf{L}(\mathbf{u})(P) \quad (\text{D.6})$$

is the b -pullback of an effective ample rational divisor D on $V_{\mathbf{u}}$.

- (iii) *At every actual prime P of V_μ , a minimizer y of (D.5) can be chosen so that $e(P, y)$ is a primitive prime order on X , for some positive integer e . For this choice,*

$$a(P) + (P, y)(h) = \frac{1}{e}, \quad ey \in \operatorname{Hom}(M, \mathbb{Z}). \quad (\text{D.7})$$

On each V_μ , a is the discrepancy function of an effective ordinary lc Calabi–Yau pair, and $a + \mathbf{D}$ is that of an effective generalized klt Calabi–Yau pair with nef part $\mathbf{D}$. Each V_μ is of Fano type. At an actual prime outside the floor $\{a = 0\}$, the same minimizer satisfies

$$a(P) = 1, \quad e = 1, \quad (P, y)(h) = 0. \quad (\text{D.8})$$

Proof. Take a uniformizer t at P and separating residue coordinates $s_1, \dots, s_{p-1}$, where $p = \text{trdeg}_k \mathcal{K}$. Up to a unit the base form is $t^{a(P)} d \log t \wedge ds_1 \wedge \dots \wedge ds_{p-1}$. For a rational Gauss extension, integral powers of the z_j can be adjusted by powers of t to obtain angular units with algebraically independent residues. The resulting logarithmic wedge has log order zero. Together with (D.2), this proves the discrepancy formula. The calculation with trivial restriction to $\mathcal{K}$ is the same, using a basis of the kernel of the pure weight. Nonnegativity follows from lc. Integrality is the ordinary integral log order of θ_0 . At a nonzero rational pure weight y , klt of X gives $A_X(y) = y(h) = \langle y, \mathbf{u} \rangle > 0$. Positivity on the nonzero rays of σ is equivalent to $\mathbf{u} \in \text{int } C$.

Choose a positive integer d with $d\mu \in M$ and set

$$V_\mu = \text{Proj} \bigoplus_{j \geq 0} R_{jd\mu}.$$

The ray ring is finitely generated and normal: it is the appropriate diagonalizable invariant subring, followed, if necessary, by a Veronese subring. Its degree-zero part is k , so its Proj is projective. An invariant rational function is a ratio of two homogeneous elements of the same weight. Multiplying their common weight by a suitable occurring weight moves both elements onto a sufficiently divisible multiple of the interior ray. To see the latter assertion directly, express μ as a strictly positive rational combination of all the finitely many generating weights and clear denominators; sufficiently high multiples leave room to subtract any prescribed occurring weight. This proves $k(V_\mu) = \mathcal{K}$.

Write the algebra generators as $f_j z^{m_j}$. Linear programming duality gives

$$\min_{y \in \Delta_P} \langle y, \mu \rangle = \max \left\{ - \sum_j \lambda_j P(f_j) : \lambda_j \geq 0, \sum_j \lambda_j m_j = \mu \right\}. \quad (\text{D.9})$$

The polytope on the right is compact because the grading is positive. Its rational vertices depend on the weights and μ , not on P . After clearing their denominators and taking a Veronese generated in degree one, these vertices give actual monomials on the ray. Conversely, a sum of monomials has no worse pole order than its worst summand. Therefore (D.9) is exactly the maximum allowed pole order in the homogeneous linear system, expressed using $z^{d\mu}$ as rational frame. On the Proj this system is basepoint-free in sufficiently divisible degree. Its maximum-pole formula on every higher model is the pullback of its ample tautological divisor. This proves (ii), including the b-divisor assertion. Formula (D.3) identifies $\mathbf{D}$ with a rational principal adjustment of this polarization. Since h is regular upstairs, its invariant powers have nonnegative value at every Gauss extension, so $\mathbf{D} \geq 0$.

For (iii), choose a homogeneous section s on the defining ray whose nonvanishing chart contains the generic point of P . The map $\text{Spec } R_s \rightarrow \text{Spec}(R_s)_0$ is the torus quotient, and the latter is the corresponding chart of V_μ . Localize the base at P . Its uniformizer remains a nonunit upstairs: the invariant ring is a direct summand, and an inverse upstairs would give an inverse downstairs by taking its invariant part. A component of its zero divisor therefore has height one and dominates P , by the principal ideal theorem. Such a component is torus invariant, since a connected torus cannot permute its finitely many irreducible components nontrivially.

The primitive order of a torus-invariant prime is Gauss. Indeed its valuation ideals are torus invariant, hence are direct sums of weight spaces; there is no cancellation between different weights, and on each weight space restriction to $\mathcal{K}$ supplies the indicated base order. Write the resulting valuation as $e(P, y)$. Its restriction to $\mathcal{K}$ is an integral multiple eP , and its values on z^M are integral. Invertibility of s implies that y minimizes the ray polarization. A prime on a normal variety has ordinary log discrepancy one. Thus $1 = e(a(P) + (P, y)(h))$,

which proves (D.7). Notice that this argument concerns actual primes on V_μ ; it asserts no such identity at an arbitrary exceptional prime.

On V_μ set $\Omega = -K_{V_\mu}(\theta_0)$. The log discrepancies of (V_μ, Ω) are a , and $K_{V_\mu}(\theta_0) + \Omega = 0$. At actual primes (D.7) gives $0 \leq a \leq 1$; hence Ω is effective. Moreover $a(P) + \mathbf{D}(P) \leq 1$ there. At every divisorial P , a rational minimizer for (D.6) gives a Gauss extension, so klt of X implies $a(P) + \mathbf{D}(P) > 0$. Subtracting the trace of $\mathbf{D}$ from Ω and taking $\mathbf{D}$ as nef part gives the claimed generalized klt Calabi–Yau data.

For completeness these data imply Fano type without any uniform index assumption. On a common resolution on which $\mathbf{D}$ is semiample, replace it by a general divided member of a sufficiently divisible free system. Added to the generalized boundary this gives an ordinary klt sub-pair; its pushdown has effective boundary and is crepant, with total log canonical divisor rationally trivial. The added divisor is big. Choose an effective ample divisor A on V_μ . Bigness permits a second effective representative of $\mathbf{D}$ containing a small positive multiple of the pullback of A . Mixing this representative with the general free representative with sufficiently small positive weight preserves klt. The resulting effective boundary downstairs contains εA for some $\varepsilon > 0$. Subtracting that summand leaves an effective klt boundary whose negative log canonical class is $\mathbb{Q}$ -linearly equivalent to εA , hence ample. Finally, at a non-floor prime the integer $a(P) > 0$ in (D.7) forces (D.8). $\square$

D.2 Which divisorial lc tests occur

Lemma D.2 (All divisorial lc tests are Gauss). *Every nontrivial divisorial valuation nonnegative on R and having zero discrepancy for (X, H) is Gauss. Its restriction to $\mathcal{K}$ is either eP with $a(P) = 0$, or trivial, in which case it is a nonzero pure weight in σ . Conversely every rational Gauss extension with $a(P) = 0$, and every nonzero rational pure weight, has zero discrepancy and a center whose closure contains the contracting vertex.*

If $\text{Spec } \check{R} \rightarrow X$ is quasi-étale and the pairs are related by log pullback, every divisorial h-lc test upstairs restricts to one of these tests, up to scaling. Conversely all the rational tests just listed lift to divisorial h-lc tests upstairs whose center closures contain the upstairs vertex. These statements suffice in particular for testing invariant rational functions.

Proof. Let v be a primitive divisorial valuation as in the first assertion. By the divisorial restriction statement in Section A, a nontrivial restriction to $\mathcal{K}$ has the form eP , with P primitive and e a positive integer. Use the uniformizer t and separating residue coordinates from the proof of Proposition D.1. Every remaining logarithmic differential has nonnegative log order at v . Thus

$$A_{X,H}(v) \geq ea(P). \quad (\text{D.10})$$

More explicitly, pass to the finite-index sublattice of M on which $v(z^m)$ is divisible by e , and replace each of its basis monomials by the unit $z^{mt-v(z^m)/e}$. If their residues are algebraically dependent over the residue field of P , their leading differential wedge vanishes, so the inequality in (D.10) is strict. Zero discrepancy therefore forces $a(P) = 0$ and independence of these angular unit residues. For a finite sum of tied terms, divide by a term of least value. All differences of its exponents lie in that finite-index lattice; independence of the adjusted residues prevents cancellation. Thus v is exactly the Gauss valuation determined by its restrictions to $\mathcal{K}$ and z^M .

If the restriction to $\mathcal{K}$ is trivial, use a separating transcendence basis of $\mathcal{K}$ in the same wedge computation. If all angular weights were zero, the unit residues could have transcendence degree at most $r_T - 1$ over $\mathcal{K}$, and the leading top wedge would vanish. Zero log discrepancy therefore forces a nonzero angular weight. It further forces algebraic independence, over $\mathcal{K}$, of the residues of a basis of the weight-zero monomials. The same tied-sum argument now proves that v is a pure Gauss weight. Nonnegativity on R places that weight in σ .

The converse discrepancy assertion was computed in Proposition D.1. The center of a Gauss valuation is torus invariant. A positive grading contracts its closure to the vertex, so the vertex lies in that closure; the valuation need not itself be centered at the closed vertex.

For a finite quasi-étale map with log-pulled pairs, the ramification formula makes discrepancies homogeneous under restriction. Hence an upstairs lc valuation restricts to a downstairs lc valuation, and every divisorial extension of a downstairs lc valuation is lc upstairs. Such extensions exist by taking the normalization of a model carrying the downstairs prime. The finite image of an upstairs center closure is the downstairs center closure and contains the vertex. There is only one point upstairs over that vertex: every positive homogeneous semi-invariant generator has an invariant positive power, which vanishes there. Therefore the upstairs center closure contains the upstairs vertex as claimed. $\square$

D.3 Sections and a uniform sandwich argument

Let $\xi = fz^k \in \text{Frac}(R)$ be a nonzero rational eigenfunction, invariant under the finite group, and let $m \in \mathbb{Z}$ satisfy $\mu = k + m\mathbf{u} \in \text{int } C$. Define

$$\mathbf{J}_m(P) = \min_{y \in \Delta_P} (P, y)(\xi h^m) = \text{div}(f)(P) + m\gamma(P) + \mathbf{L}(k + m\mathbf{u})(P). \quad (\text{D.11})$$

This is the b-pullback of an ample rational divisor J_m on V_μ ; the divisor itself may be signed. At a non-floor prime of V_μ it has integral coefficient, by (D.8) and the integral value of the finite-group invariant function ξ at a primitive cone prime.

For a rational b-divisor $\mathbf{A}$ that descends to a model, write

$$S(\mathbf{A}) = \{0\} \cup \{g \in \mathcal{K}^* : \text{div}(g) + \mathbf{A} \geq 0\}.$$

Testing on its determining model is equivalent to testing all divisorial orders. This notation agrees with that of Section C and also applies to signed divisors.

Lemma D.3 (Sections as regular symbols). *For every positive integer q and every $g \in \mathcal{K}$,*

$$g \in S(q\mathbf{J}_m) \iff g\xi^q h^{qm} \in \check{R}. \quad (\text{D.12})$$

In particular the neutral functions of cost $b \in \mathbb{N}$ are precisely

$$\{g \in \mathcal{K} : gh^b \in \check{R}\} = S(b\mathbf{D}). \quad (\text{D.13})$$

Proof. Regularity implies all Gauss inequalities, hence the implication to the left. Conversely take a sufficiently divisible power so that the semi-invariant becomes invariant, its weight is on the integral ray, and its degree is large enough for the ray ring to agree with the section ring of the tautological polarization in that degree. The divisor inequalities and (D.9) then put that power in R . The original function belongs to $\text{Frac}(\check{R})$ and is integral over $\check{R}$, so normality puts it in $\check{R}$. The second formula is the case $\xi = 1$. When $\mathcal{K} = k$, the same argument says that an interior weight becomes effective in a divisible degree and then uses normality. $\square$

All statements that follow are sequence statements in bounded dimension. Constants implicit in $O(\cdot)$ are uniform in the sequence, and hypotheses about small divisorial orders are imposed along every subsequence, as in Sections A–C.

Proposition D.4 (Sandwich regularization). *Let $L \geq 1$ vary with the sequence. Choose integers $m > m_0$ with $m - m_0 \geq cL$ for a fixed $c > 0$, and suppose both $k + m_0\mathbf{u}$ and $k + m\mathbf{u}$*

are interior. Assume that whenever primitive divisorial valuations P satisfy $a(P) = 0$ and $\mathbf{LD}(P) \rightarrow 0$, there are integers j_P such that

$$j_P \leq \mathbf{J}_{m_0}(P) \leq \mathbf{J}_m(P) \leq j_P + O(\mathbf{LD}(P)). \quad (\text{D.14})$$

Then there is a bounded positive integer q and a nonzero $g \in S(q\mathbf{J}_m)$. Equivalently, $g\xi^q h^{qm}$ is regular on $\text{Spec } \check{R}$.

Proof. For $\mathcal{K} = k$ the preceding lemma gives the assertion with $q = 1$. Otherwise work first on V_μ , where $\mu = k + m\mathbf{u}$. Use the generalized klt Calabi–Yau discrepancy $a + \mathbf{D}$ to extract onto a $\mathbb{Q}$ -factorial Fano type model U all exceptional primes below a fixed small positive discrepancy cutoff, and no other exceptional primes. The extraction statement is supplied by Section A. All extracted primes are floor primes: a is integral and nonnegative. On U , the positive coefficients of the trace D_U are precisely the floor primes, and they are at most one. This holds for surviving primes by (D.7), and for extracted primes by their small $a + \mathbf{D}$ discrepancy.

For each floor prime Q on U put $\rho_Q = \min\{1, \mathbf{LD}(Q)\}$. Choose a small fixed rational $c_1 > 0$, with $2c_1 \leq c$, and an effective correction R_c , supported on these primes, such that

$$c_1\rho_Q \leq R_c(Q) \leq 2c_1\rho_Q, \quad \text{den}(J_m(Q) - R_c(Q)) \leq C_1/\rho_Q. \quad (\text{D.15})$$

Here den is the reduced positive denominator. Such a choice is obtained by taking a rational grid of mesh at most $c_1\rho_Q$ and denominator at most C_1/ρ_Q in the interval $[J_m(Q) - 2c_1\rho_Q, J_m(Q) - c_1\rho_Q]$. There is no correction outside the floor: no such primes were extracted, and J_m is integral there.

Minima of sums give the b-divisor inequality

$$\mathbf{J}_m - \mathbf{J}_{m_0} \geq (m - m_0)\mathbf{D}. \quad (\text{D.16})$$

Since $R_c \leq 2c_1\mathbf{LD}_U$, the divisor $P_U = (J_m)_U - R_c$ dominates the trace of $\mathbf{J}_{m_0}$ and is big. Run its MMP on U to a nef big semiample divisor P' on a $\mathbb{Q}$ -factorial Fano type model U' . On a common higher model, denote pullback from a displayed model by a star. Then

$$\mathbf{J}_m \geq (P')^* \geq \mathbf{J}_{m_0}, \quad R_c^* \leq \mathbf{E} := \mathbf{J}_m - (P')^* \leq (R'_c)^*. \quad (\text{D.17})$$

Here is justification for both sides, including the comparison with the possibly different ray m_0 . The divisor J_m descends on U , and MMP negativity gives $(P')^* \leq P_U^* = \mathbf{J}_m - R_c^*$. For the middle inequality, every divisible section of $\mathbf{J}_{m_0}$ is a section of P_U , hence of P' by the MMP. The maximum-pole description of semiample divisors, applied on a common determination, gives $(P')^* \geq \mathbf{J}_{m_0}$. Finally b-nef negativity gives $\mathbf{J}_m \leq (J'_m)^*$; since $J'_m = P' + R'_c$, this is the last inequality in (D.17). Also $\rho_Q \geq D_U(Q)$, so

$$\mathbf{E} \geq R_c^* \geq c_1(D_U)^* \geq c_1\mathbf{D}.$$

The final comparison is again b-nef negativity for $\mathbf{D}$.

We apply Lemma B.1 to the polarization P' . Choose fixed rational $\delta_2 > 0$ small enough for the denominator bound below, and then $0 < \delta_1 \leq \delta_2 c_1$. For $0 \leq t \leq \delta_2$ use the discrepancy family and nef part

$$\begin{aligned} a_t &= a + (1 - t/\delta_2)\delta_1\mathbf{D} + t\mathbf{E}, \\ \mathbf{M}_t &= (1 - t/\delta_2)\delta_1\mathbf{D} + t\mathbf{J}_m. \end{aligned}$$

Their total log canonical divisor is $\mathbb{Q}$ -linearly equivalent to tP' . For $t \geq \delta_2$ keep $a_t = a + \delta_2\mathbf{E}$ and add $(t - \delta_2)(P')^*$ to the nef part. This gives families on arbitrarily long parameter intervals. They are nondecreasing, klt, and have nef $\mathbb{Q}$ -Cartier b-parts.

We check effectivity and the denominator trace requirement of that lemma. Every prime of U' survives from U . At a non-floor prime, $a = 1$, $\mathbf{D} = \mathbf{E} = 0$, and P' has integral coefficient. At a surviving floor prime Q , $\mathbf{E}(Q) = R_c(Q) \leq 2c_1\rho_Q$, while

$$\frac{1}{\text{den}(P'(Q))} \geq \frac{\rho_Q}{C_1}.$$

Take $2\delta_2 c_1 C_1 \leq 1$. Monotonicity gives $0 < a_t(Q) \leq \delta_2 R_c(Q)$ throughout the family, so $a_t(Q) \leq 1/\text{den}(P'(Q))$. Equivalently, its effective boundary dominates the denominator boundary $R(P')$ used in Section B.

It remains to check the signed phase hypothesis. At a growing parameter, small discrepancies mean $a(P) = 0$ and $\mathbf{E}(P) \rightarrow 0$. Because $\mathbf{E} \geq c_1 \mathbf{D}$, these valuations eventually belong to the fixed-cutoff list already extracted on U . On that list, (D.15) and (D.17) give

$$\mathbf{E}(P) \geq c_1 \min\{1, L\mathbf{D}(P)\}.$$

Thus $L\mathbf{D}(P) \rightarrow 0$ and $L\mathbf{D}(P) = O(\mathbf{E}(P))$. Combining (D.14) with (D.17) yields

$$j_P \leq (P')^*(P) \leq j_P + O(\mathbf{E}(P)).$$

The phase of $(P')^*(P)$ therefore has a nonnegative lift of size $O(a_t(P))$, which is stronger than the signed phase hypothesis. Lemma B.1 supplies a nonzero section in a bounded positive degree of P' . By (D.17) it lies in $S(q\mathbf{J}_m)$, and Lemma D.3 finishes the proof. $\square$

D.4 Bounded multiples, character counts, and tilted heights

The sandwich proposition turns a phase interval into a regular symbol. We use it in three forms: to supply symbols for bounded-width lattice characters, to count symbols with one character and weight, and to supply a symbol whose cost reflects a small tilted height. The last form will be used on newly optimized cones before any further rank refinement.

Corollary D.5 (Bounded-multiple character regularization). *Suppose that the order condition (C.5) on $V_{\mathbf{u}}$, with error $\eta_i \rightarrow 0$, holds whenever $a(P) = 0$ and $L\mathbf{D}(P) \rightarrow 0$. Fix $C_0 > 0$. If $k \in M$ satisfies*

$$|\langle y, k \rangle| \leq C_0 L \langle y, \mathbf{u} \rangle \quad (y \in \sigma), \quad (\text{D.18})$$

there are a bounded positive integer q , a nonzero invariant rational eigenfunction $Y = fz^{qk}$, and a positive integer $b \leq C_2 L$ such that $h^b Y \in \check{R}$. The constants are uniform in k . Both signs of k may be treated with the same bounded multiplier, and their costs may be increased to a common cost $O(L)$.

The same bounds hold uniformly if L is replaced by any $L' \geq L$ and (D.18) is imposed at scale L' .

Proof. Choose an integer $m_1 \asymp L$ with $m_1 > C_0 L$, so that both $\pm k + m_1 \mathbf{u}$ lie in $\text{int } C$. Bigness of their ray polarizations supplies rational-power functions g_+, g_- on $\mathcal{K}$ such that

$$(P, y)(z^{\pm k} h^{m_1}) + P(g_{\pm}) \geq 0 \quad \text{for every } P \text{ and } y \in \Delta_P.$$

Here a rational-power function means a positive integral root of an element of $\mathcal{K}^*$ used only through its $\mathbb{Q}$ -principal order; it need not be an element of $\mathcal{K}$. Put

$$\ell = -\text{div}(g_+), \quad G = 2m_1 D + \text{div}(g_+ g_-).$$

Adding the two Gauss inequalities and taking the height minimum shows that $G \geq 0$ on $V_{\mathbf{u}}$, and $G \sim_{\mathbb{Q}} 2m_1 D$. At a non-floor prime Q of this model, choose the integral minimizing y in (D.8). Since its height is zero, the two inequalities give

$$0 \leq \langle y, k \rangle - \ell(Q) \leq G(Q), \quad \langle y, k \rangle \in \mathbb{Z}.$$

Lemma C.2 therefore gives a fixed positive integer q_0 and a fixed $A > 0$ such that, on all tiny sequences in the statement, one can choose integers j_P with

$$|q_0 \ell(P) - j_P| \leq ALD(P).$$

Apply Proposition D.4 to $\xi = z^{q_0 k}$. For any sufficiently large integer $m_0 = O(L)$ and $m = 2m_0$, the first Gauss inequality gives

$$\mathbf{J}_{m_0} \geq q_0 \ell + (m_0 - q_0 m_1) \mathbf{D}.$$

At a height-minimizing angular vector the second inequality gives

$$\mathbf{J}_m \leq q_0 \ell + (m - q_0 m_1) \mathbf{D} + q_0 \mathbf{G} \leq q_0 \ell + (m + q_0 m_1) \mathbf{D} + q_0 \mathbf{G}.$$

The order condition (C.5), applied to $G/(2m_1) \sim_{\mathbb{Q}} D$, bounds $\mathbf{G}(P) \leq 2m_1(1 + \eta_i) \mathbf{D}(P)$ on the tiny sequences. Choose $m_0 - q_0 m_1 \geq AL$. These estimates give exactly (D.14). A bounded further multiplier q_1 then supplies $gz^{q_1 q_0 k} h^{q_1 m} \in \check{R}$. This is the desired regularization.

Apply the argument to $-k$ as well, raise the two resulting functions to bounded powers to align their multipliers, and multiply by powers of h to align their costs. Uniformity in the weight follows by applying the sequence argument to any proposed sequence of exceptions. For an enlarged scale $L' \geq L$, its tiny sequences satisfy $L\mathbf{D}(P) \rightarrow 0$ as well. All the preceding hypotheses therefore hold with L' in place of L . A sequence of exceptional pairs (L', k) would contradict the same proof; this proves the stated uniformity, including for growing L'/L . $\square$

Corollary D.6 (Count for a single character and weight). *Assume the hypotheses of Lemma C.3 on $V_{\mathbf{u}}$, and use its notation p and v_i . For a fixed $C_0 > 0$, let $W_{t,k}$ be the space of regular symbols in $\check{R}$ of finite-group character χ^t and torus weight $t\mathbf{u} + k$. For every fixed positive integer n , eventually*

$$\dim_k W_{t,k} \leq C_3 v_i n^p \tag{D.19}$$

simultaneously for all integers $0 \leq t \leq C_0 nL$ and all weights satisfying

$$|\langle y, k \rangle| \leq C_0 nL \langle y, \mathbf{u} \rangle \quad (y \in \sigma).$$

The constant C_3 is independent of n, t, k .

Proof. If $W_{t,k} \neq 0$, divide any nonzero member by h^t . This is a finite-group invariant rational eigenfunction, so $k \in M$. Let $\mathbf{J}^{\text{test}}$ denote the integral basepoint-free test b-divisor called J in Lemma C.3. That lemma has $0 \leq \mathbf{J}^{\text{test}} \leq CL_0 \mathbf{D}$ with $L_0 = O(L)$. At a primitive P with $a(P) = 0$ and $L\mathbf{D}(P) \rightarrow 0$, its nonnegative integral value is eventually zero. Such P belong to that lemma's test set $\mathcal{L}$ and hence satisfy (C.5). We may therefore apply Corollary D.5 at the enlarged scale nL .

Choose an opposite invariant eigenfunction Y of weight $-qk$, with q bounded and $h^b Y$ regular for $b \leq C_4 nL$. For $F \in W_{t,k}$ the neutral function

$$\Phi(F) = \frac{F^q Y}{h^{qt}}$$

satisfies $h^{qt+b}\Phi(F) = F^q h^b Y \in \check{R}$. By (D.13), it belongs to $S((qt+b)\mathbf{D}) \subseteq S(C_5 n \mathbf{LD})$. Although Φ need not be linear, it is a homogeneous polynomial map of degree q , nonzero away from the origin, and induces an injective morphism

$$\mathbb{P}(W_{t,k}) \longrightarrow \mathbb{P}(S(C_5 n \mathbf{LD})).$$

Indeed proportional images imply that $(F_1/F_2)^q \in k^*$; algebraic closedness of k then implies $F_1/F_2 \in k^*$. An injective morphism of projective varieties preserves the dimension of its source in its image, so $\dim W_{t,k} \leq \dim S(C_5 n \mathbf{LD})$. The upper count in Lemma C.3, used at an integer at most $(C_5 + 1)n$, proves (D.19); p is bounded by the ambient dimension. The uniform enlarged-scale assertion in Corollary D.5 makes q, C_4, C_5 independent of n, t, k . The eventual assertion retains the quantifier that n is fixed before passing sufficiently far along the sequence. $\square$

Corollary D.7 (Characters for a tilted height). *Suppose there are boundedly many invariant rational eigenfunctions $Y_j^\pm$ of weights $\pm k_j$, with*

$$h^{m_j} Y_j^\pm \in \check{R}, \quad 0 < m_j \leq AL_j, \quad 1 \leq L_j \leq L,$$

where A is fixed. Fix bounded real numbers u_j and $\delta > 0$. Assume that, at every nonzero pure weight and at all Gauss extensions over primitive P with $a(P) = 0$, $\mathbf{LD}(P) \rightarrow 0$, the evaluations satisfy

$$y_j = v(Y_j^+) = -v(Y_j^-), \quad \sum_j \frac{u_j y_j}{L_j} \leq \delta v(h). \quad (\text{D.20})$$

It suffices to check these inequalities on rational vectors. For fixed M_0 sufficiently large compared with $1/\delta$, choose integers

$$r_j = M_0 u_j L / L_j + O(1),$$

with uniformly bounded rounding errors. Then there is a regular nonzero symbol of finite-group character χ^{qm} and torus weight

$$q \left(m\mathbf{u} - \sum_j r_j k_j \right), \quad 0 < m \leq C_6 M_0 \delta L, \quad (\text{D.21})$$

for a bounded positive integer q . The constant C_6 depends only on the fixed bounds for the number of functions, their costs, the u_j , and rounding errors; it is independent of δ and M_0 . The bound for q may depend on the fixed choices δ, M_0 . No order condition (C.5) is assumed here.

Proof. Set $\xi = \prod_j (Y_j^+)^{-r_j}$, which has weight $-\sum_j r_j k_j$. Regularity and opposite evaluations imply $|y_j| \leq m_j v(h)$ at every test under consideration. Hence, with a fixed constant C_7 controlling the rounding errors,

$$\sum_j r_j y_j \leq (M_0 \delta + C_7) L v(h). \quad (\text{D.22})$$

Choose $M_0 \delta \geq 2C_7 + 2$ and an integer $m_0 \asymp M_0 \delta L$ strictly larger than $(M_0 \delta + C_7)L$, and set $m = 2m_0$. The constants in these choices are independent of δ, M_0 . At pure weights (D.22) implies strict positivity of the weights of ξh^{m_0} and ξh^m on $\sigma \setminus \{0\}$; both rays are therefore interior. At the angular tests it gives $\mathbf{J}_{m_0}(P) \geq 0$.

At a height-minimizing angular vector, the elementary estimate

$$\left| \sum_j r_j y_j \right| \leq \sum_j |r_j| m_j \mathbf{D}(P) \leq C_8(M_0 + 1) L \mathbf{D}(P)$$

also holds. Consequently $0 \leq \mathbf{J}_{m_0}(P) \leq \mathbf{J}_m(P) \leq C_9(M_0, \delta) L \mathbf{D}(P)$ on the tiny sequences. Proposition D.4 applies with $j_P = 0$ and produces $g\xi^q h^{qm} \in \check{R}$ for bounded q . Its character, weight, and cost are exactly those in (D.21). $\square$

E Optimization on common log canonical slices

Throughout this section, $x \in S$ is a fixed normal complex affine germ of positive dimension n , S is $\mathbb{Q}$ -Gorenstein and klt, and $R = \mathcal{O}_{S,x}$, $\mathfrak{m} = \mathfrak{m}_x$, and $A = A_S$. A valuation *over* x is nonnegative on R ; its centre may be a proper generization of x . A valuation *centred at* x has $v(\mathfrak{m}) > 0$. Unless otherwise stated, valuations are nontrivial quasi-monomial real valuations. Discrepancy is extended linearly in the weights on a log-smooth presentation. For every valuation v over x , put $v(0) = +\infty$ and, for $t \geq 0$, set

$$\mathfrak{a}_t(v) = \{f \in R : v(f) \geq t\}, \quad \mathfrak{a}_{>t}(v) = \{f \in R : v(f) > t\}.$$

The associated graded algebra and initial form are

$$\mathrm{gr}_v R = \bigoplus_{t \geq 0} \mathfrak{a}_t(v) / \mathfrak{a}_{>t}(v), \quad \mathrm{in}_v(f) = [f]_{v(f)} \quad (f \neq 0).$$

Degrees outside the value semigroup have zero summand. These conventions also apply when the centre is a proper generization of x . For a valuation centred at x , set

$$\mathrm{vol}(v) = \lim_{t \rightarrow \infty} \frac{n! \, \mathrm{length}(R / \mathfrak{a}_t(v))}{t^n}.$$

The limit exists by Cutkosky's theorem for graded families of $\mathfrak{m}$ -primary ideals [11, Theorem 1.1 (=5.5)]. Its hypothesis on the nilradical of the completion holds because R is excellent and normal, hence has reduced completion. The theorem applies first to integral thresholds; squeezing between adjacent integral thresholds gives the displayed real-parameter limit. For a klt pair (S, Δ) and a centred valuation of finite log discrepancy, the normalized volume is

$$\widehat{\mathrm{vol}}_{S, \Delta}(v) = A_{S, \Delta}(v)^n \mathrm{vol}(v).$$

Its n th root is $A_{S, \Delta}(v) \mathrm{vol}(v)^{1/n}$, with the same minimizers. Here and throughout the volume formulas, $n = \dim S$.

Fix a nonzero $h \in R$ such that (S, H) is lc, where $H = \mathrm{div}(h)$, and finitely many nonzero $F_j \in R$ and $m_j \in \mathbb{Z}_{>0}$. The *common lc slice* is

$$\mathcal{Q} = \{v : A(v) = v(h) = 1, \quad v(F_j) \geq m_j \text{ for every } j\}. \quad (\text{E.1})$$

Assume that it contains a centred valuation. Its cone of rays is defined by $A(v) = v(h) > 0$ and $v(F_j) \geq m_j v(h)$. An allowed height is

$$b(v) = b_0 v(h) - \sum_{\ell} b_{\ell} v(G_{\ell}), \quad b_{\ell} \geq 0, \quad G_{\ell} \in R \setminus \{0\}, \quad b|_{\mathcal{Q}} > 0. \quad (\text{E.2})$$

On centred valuations for which $b(v) > 0$, the degree-one homogeneous height b similarly gives the scale-invariant quantity $b(v)^n \mathrm{vol}(v)$ and its n th root $b(v) \mathrm{vol}(v)^{1/n}$, with the same

minimizers. All constants in this section may depend on this individual germ and its finite data. The constants in the strict volume gap below depend only on dimension and the stated separation.

Our goal is to minimize this functional on the common lc rays and, for rational heights, realize its minimizer as an ordinary normalized-volume minimizer of an effective rational klt pair. The realization must be exact: finite generation will be applied to that pair and its actual minimizer. We first control retraction to the lc locus and volume under interpolation; these estimates will allow one rational perturbation to exclude every competing valuation outside the common slice.

E.1 Retractions, jets, and the topology of the slice

Let $P \rightarrow S$ be a smooth model, F a reduced snc divisor on P , and v a valuation with centre on P . Write $r_F(v)$ for its monomial retraction using the components of F through the centre, with their v -weights. It is the trivial valuation if there are no such components. Retraction decreases the value of every function regular on the chart and preserves the values of its boundary monomials; see [17, Lemma 4.7, Corollary 4.8, and Proposition 5.1]. The following quantitative strengthening is needed here.

Lemma E.1 (Logarithmic jet estimate). *Fix finitely many regular affine functions on S . On a fixed resolution P , with F reduced snc, there is a constant C such that, for every polynomial $g \neq 0$ of degree at most D in those functions and every quasi-monomial v over x ,*

$$r_F(v)(g) \leq v(g) \leq r_F(v)(g) + C(D+1)A_{P,F}(v). \quad (\text{E.3})$$

The same constant works when the centre is not closed and when some weights disappear on passing to a face of the monomial cone.

Proof. Cover the relevant part of P by finitely many smooth etale coordinate charts. At the centre of v , write the coordinates of the components of F through that centre as $z_1, \dots, z_s$ and the remaining coordinates as $y_1, \dots, y_{n-s}$. Thus $v(z_i) > 0$. Choose a closed specialization of the centre in this chart, avoiding the other components of F , and translate the y -coordinates to vanish there. Both v and its retraction are nonnegative on this closed-point local ring: the original centre generalizes this point. A unit in that ring consequently has value zero for both valuations. Put $a_i = v(z_i)$ and expand g in the completed local ring. The least z -weight is

$$d_0 = \min\{\langle a, \alpha \rangle : [z^\alpha y^\beta]g \neq 0 \text{ for some } \beta\}.$$

We first prove a degree bound independent of a .

The function field of the chart is a finite extension of $\mathbb{C}(z, y)$. In a fixed basis, multiplication by each of the finitely many affine functions has a matrix with fixed rational-function entries. A polynomial of degree at most D in these matrices has a common denominator of degree $O(D+1)$ and numerator entries of the same degree. Its characteristic equation, after clearing denominators, gives

$$\sum_{j=0}^q p_j(z, y)g^j = 0, \quad \deg p_j \leq C_0(D+1), \quad (\text{E.4})$$

where q is fixed and the coefficients are not all zero. Translations of coordinates do not change this degree bound. Give *all* coordinates strictly positive rationally independent weights. Let γ be the leading exponent of g , and let δ_j be the leading exponent of a nonzero p_j . At least two distinct powers, say j and k , have the least weight in (E.4). Independence of the weights gives

$$\delta_j + j\gamma = \delta_k + k\gamma.$$

Consequently $|\gamma| \leq C_1(D+1)$, after enlarging the constant. This conclusion holds for every such choice of positive weights.

Now give the z -coordinates their fixed positive weights and give the y -coordinates weight zero. There are finitely many z -exponents of weight d_0 , and only finitely many in a sufficiently small fixed neighbourhood of this weight. Every vertex of the convex hull of the tied z -exponents is exposed by a small generic perturbation of the z -weights. Give the transverse coordinates sufficiently small positive generic weights afterwards. The leading exponent projects to that vertex: the positive gap to the other z -weights prevents an exponent outside the least face from entering. The preceding bound therefore bounds every vertex, and convexity bounds *every* tied z -exponent by $C_1(D+1)$. The same perturbation supplies at least one pair (α, β) on the least face with $|\alpha| + |\beta| \leq C_1(D+1)$.

For this choice of α , apply the product, over the z -coordinates, of the operators

$$\prod_{\substack{0 \leq l \leq C_1(D+1) \\ l \neq \alpha_i}} (z_i \partial_{z_i} - l).$$

It kills every other tied z -exponent and multiplies the chosen coefficient by a nonzero constant. Apply a transverse derivative ∂_y^β for a bounded exponent occurring in that coefficient. Its new constant term is nonzero. These operators have total order $O(D+1)$, and do not change any z -exponent. Their result g' satisfies

$$g' = uz^\alpha \pmod{I_{>d_0}}, \quad u \text{ a local unit}, \quad (\text{E.5})$$

where $I_{>d_0}$ is the monomial ideal generated by the z -monomials of weight greater than d_0 . This ideal is finitely generated. The assertion descends from completion by faithful flatness. More explicitly, faithful flatness first gives $g' = uz^\alpha + q$ with u regular and $q \in I_{>d_0}$. In the completion, u differs from the formal unit coefficient by an element of $(I_{>d_0} : z^\alpha)$. This colon ideal is proper, because z^α has weight d_0 , and hence lies in the local maximal ideal. Thus u is a unit. Each monomial generator of $I_{>d_0}$ has value greater than d_0 for both v and $r_F(v)$, and its regular coefficients have nonnegative value. Thus

$$v(g') = r_F(v)(g') = d_0 = r_F(v)(g).$$

This also explains why a nonclosed centre causes no problem.

For completeness, a regular logarithmic vector field L satisfies

$$v(Lf) \geq v(f) - A_{P,F}(v). \quad (\text{E.6})$$

For a prime-order valuation, take a local logarithmic volume form η for (P, F) and use

$$(Lf)\eta = f d\log(f) \wedge \iota_L \eta.$$

At the prime on a higher smooth model, logarithmic coordinate forms and $d\log(f)$ have nonnegative logarithmic order. Comparing the two sides gives (E.6). Homogeneity gives it for rational monomial weights, and rational approximation gives it for arbitrary quasi-monomial weights. In this approximation the signs on a fixed chart ring can be retained: the values of its finitely many generators are rational piecewise linear functions on the monomial weight cone, so one approximates in their common sign cells. Discrepancy and the finitely many function values in question are continuous there. The Euler operators and transverse derivatives used above are logarithmic vector fields, and subtracting a scalar does not worsen the estimate. Iterating (E.6) and using (E.5) proves the upper bound in (E.3). The lower bound is the retraction inequality. $\square$

Suppose more generally that Ψ is an effective lc rational sum of Cartier divisors. Resolve its support and any further finite list of ideals and divisors, and write

$$K_P + B_P = \pi^*(K_S + \Psi), \quad F = B_P^{-1}, \quad e = A_{S,\Psi}(v), \quad r = r_F(v).$$

If E is the reduced union of the support of B_P , the exceptional locus, and the additional resolved divisors, then

$$e = A_{P,E}(v) + \sum_{D \subseteq E} (1 - \text{coeff}_D B_P) v(D).$$

All summands are nonnegative. The positive numbers $1 - \text{coeff}_D B_P$ for $D \not\subseteq F$ have a positive minimum on this fixed resolution. Therefore

$$A_{P,F}(v) \leq Ce, \quad |A_S(v) - A_S(r)| \leq Ce, \quad (\text{E.7})$$

and the evaluations of each resolved ideal or divisor differ at v and r by at most Ce . These statements follow also when B_P has negative coefficients: the displayed coefficients $1 - \text{coeff}_D B_P$ are still positive away from F . In particular, if $e = 0$, then Lemma E.1 gives equality on every polynomial in the fixed affine generators. Equality extends to the local ring and its fraction field by localization. Hence $v = r$.

Lemma E.2 (Compactness and uniform centring). *The slice $\mathcal{Q}$ is a compact finite polyhedral complex of monomial valuations on any log resolution adapted to h and the F_j . Evaluations of fixed functions are continuous on it. For centred valuations with $A(v) = 1$, there are constants $c, C > 0$ depending on the germ such that*

$$\text{vol}(v) \geq c, \quad \text{vol}(v) \geq \frac{c}{v(\mathfrak{m})}. \quad (\text{E.8})$$

Volume is continuous on the centred part of $\mathcal{Q}$.

Proof. Apply (E.7) to $\Psi = H$. Every zero-discrepancy valuation is monomial on the coefficient-one part of the fixed resolution. Conversely those monomial valuations have zero discrepancy for (S, H) . Retain the strata whose closed images contain x . This condition is preserved when a stratum is replaced by a larger stratum, so the retained cones include their faces. On a cone with primitive generators E_i the equation $A(v) = \sum_i a_i A(E_i) = 1$ cuts out a compact simplex, because $A(E_i) > 0$. The additional inequalities are linear on an adapted resolution. There are finitely many cones. Function evaluations are monomial minima on them and agree on faces by localization to the larger stratum, proving compactness and continuity.

We use Li's Izumi estimate [24, Theorem 3.1]:

$$v(f) \leq C_I A(v) \text{ord}_{\mathfrak{m}}(f) \quad (f \in R \setminus \{0\}, v \text{ centred}). \quad (\text{E.9})$$

Thus, when $A(v) = 1$, the valuation ideal $\mathfrak{a}_t(v)$ is contained in $\mathfrak{m}^{\lceil t/C_I \rceil}$. Hilbert–Samuel growth gives the first bound in (E.8). Choose fixed generators $z_1, \dots, z_q$ of $\mathfrak{m}$, and let $z = z_i$ have $v(z) = \delta = v(\mathfrak{m})$. Since R is a domain,

$$(\mathfrak{a}_t(v) : z^j) = \mathfrak{a}_{t-j\delta}(v).$$

Filtering $R/\mathfrak{a}_t(v)$ by the images of $z^j R$ gives

$$\text{length}(R/\mathfrak{a}_t(v)) = \sum_{0 \leq j < t/\delta} \text{length}(R/(\mathfrak{a}_{t-j\delta}(v) + (z))).$$

For $j \leq t/(2\delta)$, Izumi bounds this summand below by $\text{length}(R/(\mathfrak{m}^{\lceil t/(2C_I) \rceil} + (z)))$. The latter grows as a positive constant times t^{n-1} ; this includes $n = 1$, when it is a positive constant. Since z belongs to a fixed finite list, the lower constant can be chosen uniformly. Divide by $t^n/n!$ to obtain the second bound in (E.8).

To prove continuity, consider valuations converging on one monomial cone to a centred limit. Drop the components whose weights tend to zero, and call the resulting retractions r_i . The retained positive weights converge multiplicatively. On regular functions, the r_i are therefore multiplicatively $1 + o(1)$ equivalent to the limit. Their maximal-ideal values, and those of the original valuations, are bounded below. The reduced-log discrepancy measuring the dropped weights tends to zero. Lemma E.1 upgrades this fact to a multiplicative comparison on *all* regular functions as follows.

Write $q = \text{ord}_{\mathfrak{m}}(f) \geq 1$. With $c_0 > 0$ a common lower bound for maximal-ideal values and $C_0 q$ the common Izumi upper bound for the values of f , fix L with $Lc_0 > 2C_0$. Represent f modulo $\mathfrak{m}^N$, $N = \lceil Lq \rceil$, by a polynomial p of degree at most N in fixed generators of $\mathfrak{m}$. Such a representative exists also for localized regular functions by expanding their unit denominators modulo $\mathfrak{m}^N$. The error has value greater than both values of f , so $v_i(p) = v_i(f)$ and $r_i(p) = r_i(f)$. The jet estimate bounds their difference by $o(1)q$, uniformly in f . Since $r_i(f) \geq c_0 q$, it follows that $r_i \leq v_i \leq (1 + o(1))r_i$ on all regular functions. Units have value zero. Inclusion of valuation ideals now proves volume continuity. Every convergent sequence has a subsequence in one of the finitely many cones, which proves the assertion on the whole centred part. $\square$

We record explicitly the uniform version of the truncation argument.

Corollary E.3 (Simultaneous approximation). *In the situation of (E.7), suppose $A_S(v)$ and $A_S(r)$ are bounded above and $v(\mathfrak{m}), r(\mathfrak{m}) \geq c_0 > 0$. Then a single constant C works for every such v and every $f \in R$:*

$$r(f) \leq v(f) \leq (1 + Ce)r(f). \quad (\text{E.10})$$

In particular, $\text{vol}(v)^{1/n} \geq (1 + Ce)^{-1} \text{vol}(r)^{1/n}$.

Proof. Use $N = \lceil L \text{ord}_{\mathfrak{m}}(f) \rceil$ with the single constant L chosen in the preceding proof. Equations (E.3) and (E.7) give $v(f) - r(f) \leq Ce \text{ord}_{\mathfrak{m}}(f) \leq Ce r(f)/c_0$. Enlarge C . The ideal inclusions give the volume inequality. $\square$

E.2 Convex domination and strict volume comparison

Lemma E.4 (Convex domination). *For $v_1, \dots, v_k \in \mathcal{Q}$ and $p_s \geq 0$ with $\sum_s p_s = 1$, there is $u \in \mathcal{Q}$ such that*

$$u(f) \geq \sum_s p_s v_s(f) \quad (f \in R). \quad (\text{E.11})$$

If a positive-weight term is centred, then u is centred. On the cone of common lc rays, the same statement holds with $A(u) = \sum_s p_s A(v_s)$, without requiring $A(v_s) = 1$. For an allowed height, $b(u) \leq \sum_s p_s b(v_s)$.

Proof. Fix finitely many functions g_i whose averaged target values $s_i = \sum_s p_s v_s(g_i)$ are positive, and choose positive rational $s'_i \leq s_i$. Include h with the *exact* target 1 and each F_j with target m_j . Choose N so that every N/s'_i is integral, and put $\mathfrak{a} = \sum_i (g_i^{N/s'_i})$. We claim that $(S, \mathfrak{a}^{1/N})$ is not klt at x . The multiplier-ideal summation formula on normal $\mathbb{Q}$ -Gorenstein complex varieties gives

$$\mathcal{J}(\mathfrak{a}^{1/N}) = \sum_{\substack{t_i \geq 0 \\ \sum_i s'_i t_i = 1}} \mathcal{J}\left(\prod_i (g_i)^{t_i}\right). \quad (\text{E.12})$$

This is the mixed-ideal formula of [34, Theorem 3.2] and [18, Corollary 2 and Theorem 3.6]; there is no smoothness or Jacobian correction in this particular formula. If the left side were the unit ideal at x , some summand would be the unit ideal in the local ring. The resulting mixed pair would be klt, so for every positive-weight v_s , $\sum_i t_i v_s(g_i) < A(v_s) = 1$. Averaging contradicts

$$\sum_i t_i \sum_s p_s v_s(g_i) = \sum_i t_i s_i \geq \sum_i t_i s'_i = 1.$$

The failure of klt is witnessed by a prime E whose centre contains x , and $A(E) \leq \min_i \{\text{ord}_E(g_i)/s'_i\}$. Thus $u = \text{ord}_E/A(E)$ meets all the lower targets. In particular $u(h) \geq 1$, whereas lc of (S, H) gives $u(h) \leq A(u) = 1$. It is already in $\mathcal{Q}$.

Here is the geometric mechanism behind (E.12). Resolve the summand ideals and their sum, with fixed divisors D_i, D , and subdivide $\{p_i \geq 0 : \sum p_i = 1/N\}$ by the rounding walls of $\mathcal{O}_P(\lceil K_{P/S} - \sum_i p_i D_i \rceil)$. The cellular complex indexed by chains of nonempty cells, with the usual alternating face maps and inclusions on removing the first cell, resolves $\mathcal{O}_P(\lceil K_{P/S} - D/N \rceil)$. In completed snc coordinates its exactness is coefficientwise: the cells admitting a given Laurent monomial form a convex relatively open subset of the exponent simplex, and their barycentric order complex is contractible when nonempty. This subset is nonempty exactly for monomials in the target. Indeed, locally one summand divided by the invertible sum ideal is a unit, so its fixed divisor there equals D . Local vanishing for each mixed multiplier-ideal term preserves the surjective augmentation under pushforward. This gives the stated sum formula in precisely the singular setting used above.

Let the rational lower targets increase to their averaged values. Compactness from Lemma E.2 gives a point meeting the finite list exactly as lower bounds. The closed sets imposing these lower bounds, indexed by all regular functions, have the finite intersection property. Compactness supplies a single u satisfying (E.11) for every function. Positivity on generators of $\mathfrak{m}$ proves the centring assertion. For unnormalized inputs, put $a = \sum_s p_s A(v_s)$, apply the normalized assertion with weights $p_s A(v_s)/a$ to $v_s/A(v_s)$, and multiply the result by a . The formula for b and the equality for $u(h)$ prove its height bound. $\square$

Corollary E.5 (Opposite ratios). *Suppose rational functions q, q' have regular numerators and power-of- h denominators, and $v(q') = -v(q)$ on $\mathcal{Q}$. The interpolant in Lemma E.4 has $u(q) = \sum_s p_s v_s(q)$ and $u(q') = \sum_s p_s v_s(q')$. The assertion holds simultaneously for any number of such pairs.*

Proof. Apply domination to both numerators and use the exact value of h . The two lower inequalities have opposite sides and so must be equalities. $\square$

Lemma E.6 (Volume interpolation and a strict gap). *Let w be centred, let v be over x , and suppose $u \geq tw + (1-t)v$ on R for $0 < t < 1$. For $f \neq 0$ with $w(f) > 0$,*

$$\text{vol}(u) \leq t^{-n} \frac{tw(f)}{tw(f) + (1-t)v(f)} \text{vol}(w). \quad (\text{E.13})$$

If v is also centred, $t = 1/2$, and $v(f) \geq (1+\gamma)w(f) > 0$ for $0 < \gamma \leq 1$, then

$$\text{vol}(w) + \text{vol}(v) - 2\text{vol}(u) \geq c(n, \gamma) \text{vol}(w), \quad (\text{E.14})$$

where $c(n, \gamma) > 0$ depends only on n, γ . Consequently valuations whose volumes are within relative $o(1)$ of the minimum on the same $A = 1$ common slice are multiplicatively $1 + o(1)$ equal on all regular functions. The conclusion is uniform when the dimension is bounded. A sufficiently small fixed relative error similarly gives any prescribed fixed multiplicative closeness to one.

Proof. For a centred filtration, write its Laplace sum as the sum over its jumps a of $\dim_{\mathbb{C}}(\mathfrak{a}_a/\mathfrak{a}_{>a})e^{-a/N}$. Integration of the cumulative dimension function gives

$$\lim_{N \rightarrow \infty} N^{-n} \sum_a \dim_{\mathbb{C}}(\mathfrak{a}_a/\mathfrak{a}_{>a})e^{-a/N} = \text{vol}(v). \quad (\text{E.15})$$

There are finitely many jumps below each bound, and polynomial growth controls the tail.

Choose a homogeneous basis of $\text{gr}_w R/(\text{in}_w f)$, lift it homogeneously, and multiply the lifts by all nonnegative powers of $\text{in}_w f$. This is a basis of $\text{gr}_w R$: its independence follows from the domain property, and successive division lowers the degree by $w(f) > 0$, proving spanning. Lift this basis to products $a_i f^j$ in R . They span modulo arbitrary high w -ideals, hence modulo high u -ideals since $u \geq tw$. The u -value of $a_i f^j$ is at least $tw(a_i) + j(tw(f) + (1-t)v(f))$. Consequently its Laplace sum is at most the w -Laplace sum at N/t , multiplied by

$$\frac{1 - e^{-tw(f)/N}}{1 - e^{-(tw(f) + (1-t)v(f))/N}}.$$

Use (E.15) to obtain (E.13).

For the strict gap, fix $T > 3$ and work in $V_N = R/\mathfrak{m}^{\lceil CTN \rceil}$, where C is large enough for this fixed triple of centred valuations. The indicated power is contained in all their threshold- TN ideals. Cap both the w and v flags at TN and choose a basis simultaneously adapted to these two flags; denote its capped values by W_i, V_i . Lifts have u -value at least $(W_i + V_i)/2$. The identity

$$e^{-W_i/N} + e^{-V_i/N} - 2e^{-(W_i+V_i)/(2N)} = (e^{-W_i/(2N)} - e^{-V_i/(2N)})^2$$

therefore bounds twice the capped u -Laplace sum above by the sum of the other two, less the sum of these squares.

Choose a vector-space complement to $\mathfrak{a}_{\gamma N/4}(w)$ in R and multiply it by $f^{\lfloor N/w(f) \rfloor}$. Every nonzero resulting element has w -value less than $(1 + \gamma/4)N$ and, for large N , has v -value at least $(1 + 2\gamma/3)N$. Multiplication is injective, and this subspace remains injective in V_N by its upper w -value bound. Its dimension is asymptotic to

$$\frac{\text{vol}(w)}{n!} (\gamma N/4)^n.$$

Its projection from the high- v subspace modulo the high- w subspace has this rank. Hence at least this many entries of a simultaneously adapted basis satisfy $W_i < (1 + \gamma/4)N$ and $V_i \geq (1 + 2\gamma/3)N$. Each contributes at least

$$\left(e^{-(1+\gamma/4)/2} - e^{-(1+2\gamma/3)/2} \right)^2$$

to the preceding sum of squares. Divide by N^n , first let $N \rightarrow \infty$, and then let $T \rightarrow \infty$. The capped tail errors are bounded by polynomial functions of T times e^{-T} and disappear. This proves (E.14), for example with any smaller positive constant than

$$\frac{(\gamma/4)^n}{n!} \left(e^{-(1+\gamma/4)/2} - e^{-(1+2\gamma/3)/2} \right)^2.$$

Finally interpolate two near-minimizers by Lemma E.4. Their interpolant has at least the minimum volume. If their values on some regular function differed by a fixed factor greater than one, apply (E.14), interchanging the valuations if necessary. It gives a fixed relative loss, contrary to near-minimality. This proves the simultaneous all-functions assertion and its fixed-error version. $\square$

E.3 Exact realization by a rational klt pair

We have established compactness, a uniform comparison with monomial retraction, and strict volume improvement when two valuations differ. The remaining task is to encode the finite order constraints in an lc boundary and then choose one rational perturbation for which every ordinary minimizer lies exactly on its zero-discrepancy locus. The parameter must work for all competing valuations at once.

Lemma E.7 (A boundary encoding the common slice). *Choose positive rational numbers θ_j with $\sum_j \theta_j < 1$. For sufficiently general scalars a_j , put $E_j = \text{div}(h^{m_j} + a_j F_j)$ and*

$$\Psi = (1 - \sum_j \theta_j)H + \sum_j \frac{\theta_j}{m_j} E_j. \quad (\text{E.16})$$

Then Ψ is effective and lc, and its nonzero zero-discrepancy rays over the germ are exactly the common lc rays. On every such ray,

$$v(h^{m_j} + a_j F_j) = m_j v(h). \quad (\text{E.17})$$

In particular, the coefficients θ_j may be chosen arbitrarily small while retaining these conclusions.

Proof. Set $\mathfrak{b}_j = (h^{m_j}, F_j)$. For any valuation,

$$\begin{aligned} A(v) - (1 - \sum_j \theta_j)v(h) - \sum_j \frac{\theta_j}{m_j} v(\mathfrak{b}_j) \\ = A(v) - v(h) + \sum_j \theta_j (v(h) - \min\{v(h), v(F_j)/m_j\}) \geq 0. \end{aligned}$$

Equality holds exactly on the common lc rays, together with the trivial valuation. Resolve the ideals and choose the E_j as general members of their pencils. On the resolution the residual free members meet the existing boundary with snc support and contain none of its strata. Their coefficients θ_j/m_j are less than one. They therefore introduce no new zero-discrepancy strata, proving the assertion about Ψ . On the coefficient-one monomial skeleton, each free residual member has value zero because it contains no stratum. The value of its pencil member is consequently the fixed-ideal value $\min\{m_j v(h), v(F_j)\} = m_j v(h)$, proving (E.17). $\square$

Theorem E.8 (Exact common-slice minimizer). *For every allowed height b , the homogeneous functional $b(v) \text{vol}(v)^{1/n}$ on centred common lc rays has a unique minimizing ray. Its $A = 1$ representative depends continuously on the coefficients of b while the height stays strictly positive on the fixed slice. It is fixed by every automorphism preserving the slice and the height.*

If the coefficients of b are rational, its minimizer w is the normalized-volume minimizer of an effective rational klt pair on $x \in S$. More precisely, one can choose an effective rational lc Cartier boundary Ψ with exactly the common lc rays as its zero-discrepancy locus, and one positive rational number λ , fixed independently of the competing valuation, such that

$$\Psi_\lambda = \Psi - \lambda b_0 H + \lambda \sum_\ell b_\ell \text{div}(G_\ell) \quad (\text{E.18})$$

is effective and klt, and w minimizes $\widehat{\text{vol}}_{S, \Psi_\lambda}$ among all centred valuations. On the common lc rays its discrepancy is λb .

For the same rational height, the algebra $\text{gr}_w R$ is finitely generated and normal. Its spectrum S^0 is a klt $\mathbb{Q}$ -Gorenstein cone. Write h^0, F_j^0, G_ℓ^0 for initial forms, w^0 for the grading valuation, and $\mathcal{Q}^0, b^0$ for the analogous central slice and height. Then $(S^0, \text{div}(h^0))$ is lc, $A_{S^0}(w^0) = w^0(h^0) = 1$, $w^0 \in \mathcal{Q}^0$, and b^0 is strictly positive on $\mathcal{Q}^0$. The valuation w^0 minimizes $b^0 \text{vol}^{1/n}$ on its centred part.

Proof. Attainment, uniqueness, and continuity. Compactness gives $\min_{\mathcal{Q}} b > 0$. A minimizing sequence has bounded volume, so (E.8) bounds its maximal-ideal values away from zero. A subsequence converges in $\mathcal{Q}$ to a centred valuation. Lemma E.2 gives volume continuity and hence attainment. If two minimizing rays existed, normalize each by $b = 1$. Their midpoint has a dominating common lc interpolant u with $b(u) \leq 1$. If the minimum value of $b \text{vol}^{1/n}$ is M , then $\text{vol}(u) \geq M^n / b(u)^n \geq M^n$. The strict gap (E.14) contradicts this unless the two height-normalized valuations agree on every regular function. Thus they are the same ray. For heights whose coefficients converge to those of a positive height, the minimum heights stay bounded below and one fixed centred competitor bounds the minimum values above. The same compactness argument shows that every subsequential limit of minimizers is the unique limiting minimizer. This proves continuity. Automorphisms preserving the data preserve the functional, so uniqueness proves equivariance.

Perturbing the boundary. Assume from now on that the height is rational. Choose Ψ by Lemma E.7, and resolve additionally all G_ℓ and $\mathfrak{m}$. For small rational $\lambda > 0$, (E.18) is effective, since the coefficient of H in (E.16) is positive. It is klt near x : on each coefficient-one component its new discrepancy is $\lambda b > 0$, while the other finitely many resolved discrepancies remain positive for sufficiently small λ . The log-smooth criterion then tests every valuation.

Uniform control of every relevant competitor. Let v be a centred quasi-monomial valuation with $A(v) = 1$ and

$$A_{S, \Psi_\lambda}(v) \text{vol}(v)^{1/n} \leq \lambda M_0, \quad (\text{E.19})$$

where M_0 is the value of $b \text{vol}^{1/n}$ at one fixed centred point of $\mathcal{Q}$. This includes every normalized-volume minimizer of the klt pair. Existence and quasi-monomiality for effective rational klt pairs are supplied by [9, Section 7] and [36, Theorem 1.2]. The following constants are uniform over *all* valuations in (E.19) and all sufficiently small λ .

Put $e = A_{S, \Psi}(v) \geq 0$ and retract v to the coefficient-one part of the fixed resolution, obtaining r . Izumi bounds every fixed function value at v when $A(v) = 1$; thus $|b(v)| \leq C$. The first bound in (E.8) and (E.19) give

$$e + \lambda b(v) \leq C\lambda, \quad e \leq C\lambda.$$

Equation (E.7) yields

$$A_S(r) = 1 + O(e), \quad b(v) = b(r) + O(e). \quad (\text{E.20})$$

For small λ , r is nontrivial and its normalized ray lies in $\mathcal{Q}$. Strict positivity of b on the compact slice therefore gives uniform constants $\beta, B > 0$ with

$$\beta \leq b(r) \leq B, \quad b(v) \geq \beta/2.$$

The sublevel inequality now bounds $\text{vol}(v)$ above. The second bound in (E.8) bounds $v(\mathfrak{m})$ below; since $\mathfrak{m}$ was resolved, its value at r differs by $O(e)$. Thus $r(\mathfrak{m})$ is also bounded below and r is an admissible centred competitor. Corollary E.3 supplies

$$v(f) \leq (1 + Ce)r(f) \quad \text{for every } f \in R, \quad (\text{E.21})$$

with a single C independent of f , v , and small λ .

One fixed parameter forces exactness. Use (E.20) and (E.21) to obtain

$$\begin{aligned} & (e + \lambda b(v)) \operatorname{vol}(v)^{1/n} - \lambda b(r) \operatorname{vol}(r)^{1/n} \\ & \geq \frac{e [1 - \lambda C' - \lambda C b(r)]}{1 + C e} \operatorname{vol}(r)^{1/n}. \end{aligned} \quad (\text{E.22})$$

Choose *one* positive rational λ small enough for all preceding bounds and for $\lambda(C' + CB) < 1$. This choice depends only on the fixed germ, boundary, and height; it is independent of every competitor in (E.19) and of every regular function. If such a competitor has $e > 0$, (E.22) is strictly positive, so retraction strictly improves its normalized-volume root functional. An actual klt minimizer consequently has $e = 0$, lies exactly on the common lc slice, and is w by uniqueness. Every competitor outside (E.19) already has larger value than the fixed slice competitor. This proves exact realization for this single rational parameter; it is not merely a limiting assertion as $\lambda \rightarrow 0$. The argument applies for every smaller positive rational parameter satisfying the same bounds.

Stable degeneration of this actual minimizer. We now apply stable degeneration to the effective rational klt pair (S, Ψ_λ) and its actual minimizer w . By [38, Theorems 1.1–1.2], its associated graded is finitely generated and defines a normal klt log Fano cone pair $(S^0, \Psi_\lambda^0; w^0)$. The grading minimizes normalized volume on the central pair; ordinary volume and pair discrepancy are preserved, as in [25, Lemma 4.10 and Theorem 3.5, arXiv version]. The general uniqueness theorem for the klt-pair minimizer is [37, Theorem 1.1]; the constrained uniqueness needed here was proved before invoking it.

Each component used in (E.18) is the divisor of a regular function. For a principal ideal (f) and a valuation filtration, multiplicativity gives $\operatorname{in}_w((f)) = (\operatorname{in}_w f)$. Equivalently its Rees homogenization, using its exact order, cuts a Cartier divisor without a fibre component. Hence Ψ_λ^0 is an effective rational sum of principal Cartier divisors. Since $K_{S^0} + \Psi_\lambda^0$ is $\mathbb{Q}$ -Cartier, so is K_{S^0} , and removing the effective boundary shows that S^0 is klt. Preservation of pair discrepancy and of all equation orders gives

$$A_{S^0}(w^0) = A_S(w) = 1, \quad w^0(h^0) = 1, \quad w^0(F_j^0) = w(F_j) \geq m_j.$$

The choice of the θ_j can be made arbitrarily small. For each such choice the exact minimizer is still the unique w , and thus the graded algebra remains the same. Choose the corresponding λ smaller as necessary. The central effective klt boundaries then contain $(1 - o(1)) \operatorname{div}(h^0)$. For every prime over S^0 , the resulting discrepancy inequalities imply, on taking the coefficient limit, $A_{S^0}(E) \geq \operatorname{ord}_E(h^0)$. This proves that $(S^0, \operatorname{div}(h^0))$ is lc.

Finally take $v' \in \mathcal{Q}^0$. By (E.17), the initial equation of $h^{m_j} + a_j F_j$ is either $(h^0)^{m_j}$, or $(h^0)^{m_j} + a_j F_j^0$ when $w(F_j) = m_j$. In either case its v' -value is at least $m_j v'(h^0)$. Therefore

$$0 < A_{S^0, \Psi_\lambda^0}(v') \leq \lambda b^0(v'), \quad (\text{E.23})$$

with equality on the right at w^0 . The left inequality proves positivity of b^0 on the entire common slice, including its noncentred faces. For centred v' , central normalized-volume minimality and (E.23) give

$$\lambda b^0(v') \operatorname{vol}(v')^{1/n} \geq A_{S^0, \Psi_\lambda^0}(v') \operatorname{vol}(v')^{1/n} \geq \lambda b^0(w^0) \operatorname{vol}(w^0)^{1/n}.$$

This is the required constrained minimality on the central slice. $\square$

Corollary E.9 (Near-minimizers for a height). *Fix an allowed height b and normalize its common rays by $b = 1$. If w_i, v_i have functional values within relative $o(1)$ of the minimum, then $w_i = (1 + o(1))v_i$ on all regular functions, with an error independent of the function. The same conclusion, after $A = 1$ normalization, holds for minimizers of heights whose coefficients converge to those of b .*

Proof. For the height-normalized pair, the midpoint interpolant has height at most one. Its volume is therefore at least the minimum of volume on the height-one slice. The proof using (E.14) applies verbatim: any fixed separation on one function would force a fixed relative volume loss. For converging heights b_i , positivity on the compact slice gives $b_i/b = 1 + o(1)$ uniformly. Their minimizing rays, normalized by $b = 1$, are thus near-minimizers for b , and the preceding assertion applies. Evaluation on h , which equals A on the common rays, shows that the factors needed to normalize by $A = 1$ converge as well. $\square$

Corollary E.10 (Domination by the minimizer). *For the minimizing ray w and every $v \in \mathcal{Q}$,*

$$\frac{v(f)}{b(v)} \leq 2^n \frac{w(f)}{b(w)} \quad (f \in R). \quad (\text{E.24})$$

For rational heights the same statement holds on the central slice with all symbols carrying a superscript 0.

Proof. Normalize both rays by $b = 1$ and interpolate them with weights $1/2$. The resulting centred u has $b(u) \leq 1$, so minimality gives $\text{vol}(u) \geq \text{vol}(w)$. For $w(f) > 0$, (E.13) gives

$$1 \leq 2^n \frac{w(f)}{w(f) + v(f)}.$$

Thus $v(f) \leq (2^n - 1)w(f)$, which implies the stated bound. If f is a unit both values are zero. The central assertion follows from the same argument and Theorem E.8. $\square$

Lemma E.11 (One-sided height tilt). *Let $G \in R \setminus \{0\}$, $m \in \mathbb{Z}_{>0}$, and $g(v) = v(G)/m - v(h)$. Suppose $g(v)/v(h) \geq 2c$ on some common lc ray, where $c > 0$ is rational. Set*

$$M = \max_{\mathcal{Q}} g \geq 2c, \quad \mu_* = (M - c)^{-1}, \quad b_\mu(v) = v(h) + \mu(cv(h) - g(v)).$$

For $0 \leq \mu < \mu_$ the height is positive. There is a rational μ in this interval whose $A = 1$ minimizing valuation w satisfies*

$$g(w) > 0, \quad b_\mu(w) \geq 1.$$

Moreover $b_\mu \leq 2 + 1/c$ on $\mathcal{Q}$, and the corresponding central height obeys the same bound on $\mathcal{Q}^0$.

Proof. As $\mu \uparrow \mu_*$, the minimum of $b_\mu \text{vol}^{1/n}$ tends to zero. To prove this even when an endpoint zero occurs at a noncentred valuation v , interpolate v with one fixed centred point $w_1 \in \mathcal{Q}$, giving w_1 weight t . For $0 < \mu_* - \mu \leq t$, the interpolant u has $b_\mu(u) = O(t)$. Since $v(h) = w_1(h) = 1$, the improved volume estimate (E.13), applied with $f = h$, gives $\text{vol}(u) \leq t^{1-n} \text{vol}(w_1)$. Hence $b_\mu(u) \text{vol}(u)^{1/n} = O(t^{1/n})$. The lower volume bound in (E.8) then forces the minimizing height to tend to zero. In particular its minimizing valuation eventually has $g(w) > c$.

If the minimizer at $\mu = 0$ has $g(w) > 0$, take $\mu = 0$. Otherwise continuous dependence in Theorem E.8 gives a nonempty open interval of parameters for which $0 < g(w) < c$. Choose a rational parameter in that interval. Then $b_\mu(w) = 1 + \mu(c - g(w)) \geq 1$. Finally $v(G) \geq 0$ gives $g(v) \geq -1$ on the normalized slice, while $\mu < \mu_* \leq 1/c$. Thus $b_\mu(v) \leq 1 + (c + 1)/c = 2 + 1/c$. The same calculation applies centrally because G^0 is regular and $v'(h^0) = 1$ on $\mathcal{Q}^0$. $\square$

F Degeneration, exact sign transfer, and rank refinement

We work with a complex klt germ $x \in S$ of dimension n , a nonzero regular function h such that (S, H) is lc, where $H = \text{div}(h)$, and a rational top form ω with $\text{div}_S(\omega) = 0$. Thus the reflexive canonical sheaf is trivialized by ω . A finite cyclic group G fixes x and acts quasi-étale on S ; we replace it by its effective image. Suppose that h and ω have the same character χ . In particular, ω/h descends to the quotient. Fix invariant ratios

$$f_j = F_j/h^{m_j}, \quad F_j \in \mathcal{O}_{S,x} \setminus \{0\}, \quad m_j \in \mathbb{N}_{>0}.$$

The common slice consists of the h -lc valuations satisfying $q(f_j) \geq 0$, with the convention on centres from Section E: the closed image of the centre contains x . Unless another normalization is displayed, $q(h) = A_S(q) = 1$.

Let w be a centred G -invariant quasi-monomial valuation for which

$$\check{R} = \text{gr}_w \mathcal{O}_{S,x}$$

is finitely generated, normal, and klt. These are hypotheses here; they are supplied in our applications by the exact optimizer of Section E. Write $S^0 = \text{Spec } \check{R}$, with vertex o , and denote initial symbols by a bar. The fine grading is by the finitely generated value group $\Lambda_w \subset \mathbb{R}$, so evaluation at w is injective on this lattice. Its torus will be denoted by $\mathbb{T}_w$. Taking invariants in each filtered exact sequence gives

$$\text{gr}_w(\mathcal{O}_{S,x}^G) = \check{R}^G. \quad (\text{F.1})$$

Indeed, the Reynolds operator makes taking invariants exact in characteristic zero. The weight lattices of $\check{R}$ and $\check{R}^G$ have finite index in one another, and hence the same rank.

F.1 Presentations and rational approximations

Lemma F.1 (Weighted presentations). *Choose $x_1, \dots, x_N \in \mathfrak{m}_x$ whose symbols generate $\check{R}$, and put $\alpha_i = w(x_i) > 0$. Any prescribed finite list of functions of positive w -value can be included in this list. Then*

$$\hat{\mathcal{O}}_{S,x} = \mathbb{C}[[X_1, \dots, X_N]]/I, \quad \check{R} = \mathbb{C}[X_1, \dots, X_N]/\text{in}_\alpha I, \quad (\text{F.2})$$

where initial forms have least weight. The valuation ideals are exactly the images of the ideals generated by monomials of weight at least the prescribed threshold. The same description holds before completion by intersection.

For every sufficiently nearby linear functional $L : \Lambda_w \otimes \mathbb{R} \rightarrow \mathbb{R}$, positive on the listed weights, the vector $\alpha_L = (L(\alpha_i))$ gives the same initial ideal, with its grading composed with L . Its filtration defines a centred quasi-monomial valuation w_L . The functionals can be chosen rational, and the resulting valuations can be computed monomially on a fixed smooth chart for w .

Proof. Subduction subtracts a polynomial in the x_i with the prescribed initial symbol, then repeats at a strictly larger value. There are only finitely many values in a bounded interval: each is a nonnegative integral combination of the finitely many positive α_i . The Izumi upper bound for the fixed valuation w implies that remainders whose w -values tend to infinity tend to zero in the $\mathfrak{m}_x$ -adic topology. In particular $\mathfrak{m}_x \subset (x_1, \dots, x_N) + \mathfrak{m}_x^2$, and Nakayama's lemma shows that the x_i generate $\mathfrak{m}_x$. Completed subduction proves both assertions in (F.2) and the assertion about threshold ideals. Monomial threshold ideals are finitely generated. Their completed extensions intersect the local ring in the original ideals, by faithful flatness of completion.

We record why the dimension and primeness arguments used below are valid also for other positive weight vectors. For any such vector γ , its monomial filtration is bounded above and below by fixed linear reindexings of the $\mathfrak{m}_x$ -adic filtration. Consequently its cumulative quotient lengths grow with exponent n . The associated graded algebra is $\mathbb{C}[X]/\text{in}_\gamma I$; comparison of its positive weighted filtration with ordinary polynomial degree shows that its growth exponent is its Krull dimension. Thus this dimension is n . If $\text{in}_\gamma I$ is prime, the filtration is multiplicative and separated, and defines a valuation. If its value group has rational rank r , the degree-zero homogeneous fraction field is its residue field and has transcendence degree $n - r$. This proves Abhyankar equality. Local monomialization of Abhyankar valuations in characteristic zero [17, Proposition 3.7] then makes the valuation quasi-monomial.

Take finitely many series in I whose α -initials generate $\text{in}_\alpha I$. Their leading faces persist after composition with L : ties are preserved because their fine weights are equal, and positivity leaves only finitely many other terms that could compete in a fixed neighbourhood of α . Therefore

$$\text{in}_\alpha I \subset \text{in}_{\alpha_L} I.$$

The first ideal is prime, and both quotients have dimension n . A nonzero ideal in a finitely generated domain lowers its dimension, so this inclusion is equality.

For the final assertion, use a smooth monomial chart for w at its stratum and expand the finitely many x_i over the residue coefficient field. Their leading faces are finite polynomials in the stratum parameters. The weights of these parameters belong to Λ_w . Composing them with nearby rational L preserves the same leading polynomials. The resulting monomial valuation dominates the presentation filtration. The induced map from the presentation graded algebra to the graded ring of this monomial valuation is injective: the polynomial relations among these leading polynomials are exactly the old relations, unchanged by regrading. Hence the two valuations agree on every function. $\square$

F.2 The canonical calculation before lc equality

Xu–Zhuang study specialization of non-degenerate reflexive differential forms under stable degeneration of klt singularities [39, Theorem 1.1 and Section 3]. The calculation below tracks a divisor-free canonical generator, including its finite-group character and any additional grading, under the finitely generated degeneration used here. We give this calculation before imposing lc equality because it will also be used to prove that equality after two gradings are compressed.

Proposition F.2 (Canonical forms under degeneration). *Under the hypotheses preceding Lemma F.1, S^0 has a divisor-free reflexive canonical generator ω^0 . It has the G -character of ω . It is homogeneous for $\mathbb{T}_w$, and its weight κ satisfies*

$$\langle w, \kappa \rangle = A_S(w). \quad (\text{F.3})$$

If an additional torus acts on the original germ, preserves w , and makes ω homogeneous, its character is preserved separately. None of these assertions assumes $A_S(w) = w(h)$. Moreover the action of G on S^0 is quasi-étale.

Proof. First track discrepancy under the rational approximations of Lemma F.1. On a fixed monomial chart with stratum parameters $z_1, \dots, z_r$ and separating residue coordinates $y_1, \dots, y_{n-r}$, write

$$\omega = \psi \frac{dz_1}{z_1} \wedge \dots \wedge \frac{dz_r}{z_r} \wedge dy_1 \wedge \dots \wedge dy_{n-r}.$$

The quasi-monomial discrepancy formula gives $A_S(w) = w(\psi) \in \Lambda_w$. Include numerators and denominators of the rational function ψ among the finite functions whose leading faces are tracked. The same formula on the same chart then gives the exact identity

$$A_S(w_L) = L(A_S(w)). \quad (\text{F.4})$$

In particular this is an identity of linear weights, rather than only a limiting assertion about discrepancies.

Choose rational L and multiply it by a positive integer so that the presentation weights are positive integers. In this paragraph write v for this integral scaling of w_L , and $a_i = v(x_i)$. After shrinking to a G -stable affine neighbourhood, the x_i have common zero set $\{x\}$. Define the extended Rees algebra

$$\mathcal{R} = \mathcal{O}(S)[t, x_1 t^{-a_1}, \dots, x_N t^{-a_N}] \subset \mathcal{O}(S)[t, t^{-1}]. \quad (\text{F.5})$$

The threshold-ideal description proves that its special fibre is the graded algebra in (F.2); positive-level quotients are supported at x , so the affine and local constructions have the same special fibre. This algebra is of finite type and torsion free over $\mathbb{C}[t]$, hence flat. Let $\mathcal{S} = \text{Spec } \mathcal{R}$ and $E = (t = 0)$. The divisor E is normal, integral, reduced, and Cartier, and $\mathcal{S} \setminus E = S \times \mathbb{G}_m$.

The total space is normal. Away from E this follows from the normality of S . At points on E , the nonzerodivisor t and the S_2 property of $\mathcal{R}/(t)$ imply S_2 for $\mathcal{R}$. At the generic point of E , the quotient by t is a field, so the total local ring is a regular discrete valuation ring. These observations give R_1 as well.

The order along E restricts to v on $\mathbb{C}(S)$ and takes value one on t . Indeed, the residues of $f t^{-v(f)}$ are the nonzero graded symbols of f , and the direct grading prevents cancellation between distinct such leading terms. It is the Gauss extension of v . The monomial log-form computation gives

$$\text{div}_{\mathcal{S}} \left(\omega \wedge \frac{dt}{t} \right) = (A_S(v) - 1)E. \quad (\text{F.6})$$

For example, on a model where the rational valuation v is ord_P , use a uniformizer of P and separating residue coordinates. The extension has weights $(c, 1)$ on that uniformizer and t ; the wedge of their logarithmic differentials has log order zero, while the remaining coefficient has value $c A_S(\text{ord}_P) = A_S(v)$. This proves the coefficient in (F.6). There are no other divisor coefficients, since off E the form is the product of the divisor-free ω and dt/t .

Since K_S is Cartier and v has integral values, $A_S(v)$ is an integer. Therefore

$$\eta = t^{-A_S(v)} \omega \wedge \frac{dt}{t} \quad \text{satisfies} \quad \text{div}_{\mathcal{S}}(\eta) = -E.$$

This establishes that the ordinary canonical divisor of the total space is Cartier. At every codimension-one point of E , the fibre is regular and the flat morphism to the line is smooth. The residue of η is consequently a rational top form on E with neither zeros nor poles at any codimension-one point. It trivializes the reflexive canonical sheaf of E . In particular this establishes ordinary canonical Cartierness, not just Cartierness after adding a boundary.

Give t weight -1 and $x_i t^{-a_i}$ weight a_i . The residue has weight $A_S(v)$ and the same G -character as ω . If an old torus is present, choose homogeneous generators; t is invariant for that torus, so its canonical character is unchanged. In the cone case such homogeneous generators have common zero only at the vertex: their common zero set is invariant and is isolated there, whereas the closure of every positive-grading orbit contains the vertex.

Units of a positively graded affine domain with degree-zero part $\mathbb{C}$ are constant. Hence any two divisor-free canonical generators on E differ by a scalar. The full fine grading torus

therefore acts on the generator by a character κ , independent of the rational regrading used to construct it. For all nearby rational L , its evaluation is $L(A_S(w))$ by (F.4) and the residue computation. Equality on this open set of rational functionals identifies the two lattice weights and proves (F.3), including its equivariant version.

Finally suppose a nonidentity element of G acquired a fixed locus of codimension at most one on E . At a general point of that locus the fibre, and hence the morphism $\mathcal{S} \rightarrow \mathbb{A}^1$, is smooth. The fixed locus of the cyclic subgroup generated by that element is smooth there. Its tangent space is the invariant tangent subspace, and it maps surjectively to the tangent space of the line: averaging a lift of the base tangent vector gives an invariant lift. Thus this fixed locus is smooth over the line and has the same fibre codimension on nearby nonzero fibres. This contradicts quasi-étaleness of the original effective action. The action on S^0 is therefore quasi-étale. $\square$

Corollary F.3 (Lc equality and principal boundaries). *If w is h -lc, then $(S^0, \operatorname{div}(\bar{h}))$ is lc and ω^0 has the same fine torus weight and G -character as $\bar{h}$. Consequently $\omega^0/\bar{h}$ is invariant, and the quotient cone has the log-form data of Section D. If Δ is any effective rational sum of principal divisors near x , with (S, Δ) lc and $A_S(w) = w(\Delta)$, its initial boundary Δ^0 is lc on S^0 . All these assertions also hold with specified additional equivariant gradings.*

Proof. Include the finitely many boundary equations in the presentation. Their orders and discrepancy equality persist exactly under rational regrading. In (F.5), each equation f has homogenization $ft^{-v(f)}$, which is a nonzerodivisor and has no component E ; it restricts to $\bar{f}$ on E . In particular the initial boundary is rationally Cartier. The morphism $\mathcal{S} \rightarrow S \times \mathbb{A}^1$ is birational. Comparing the coefficient at its only vertical prime by (F.6) shows that

$$(\mathcal{S}, E + \Delta_S) \longrightarrow (S \times \mathbb{A}^1, \Delta \times \mathbb{A}^1 + S \times \{0\})$$

is crepant precisely when $A_S(v) = v(\Delta)$. Its source is therefore lc, and lc adjunction gives the stated central pair. At codimension-one points of the normal Cartier fibre the morphism is smooth, so the different is exactly the ordinary restriction of the homogenized boundary. Apply this to $\Delta = H$. Formula (F.3) and injectivity of evaluation on Λ_w identify the weight of ω^0 with that of $\bar{h}$. Quasi-étaleness and the finite-map discrepancy formula descend klt, lc, and the invariant log form to the quotient. A suitable power of ω^0 descends to a divisor-free pluricanonical form. $\square$

F.3 An exact bridge for inequalities

Proposition F.4 (Sign transfer). *Suppose w is common h -lc and $w(f_j) = 0$ for every j . Let $g = U/h^b$, where $0 \neq U \in \mathcal{O}_{S,x}$, $b \in \mathbb{N}_{>0}$, and $w(g) = 0$. Let $\mathcal{Q}$ be the actual common slice and $\mathcal{Q}^0$ the common slice defined by $\bar{h}$ and the $\bar{f}_j$ on S^0 . Then*

$$q(g) \leq 0 \quad (q \in \mathcal{Q}) \quad \Longleftrightarrow \quad q^0(\bar{g}) \leq 0 \quad (q^0 \in \mathcal{Q}^0). \quad (\text{F.7})$$

One may add any finite list of tests of value zero at w . There is no bound on the degree or the cost of g in this statement.

Proof. Use Lemma E.7 to encode the common slice. Choose positive rational θ_j with $\sum_j \theta_j < 1$ and sufficiently general scalars λ_j , and set

$$P_j = h^{m_j} + \lambda_j F_j, \quad \Psi = (1 - \sum_j \theta_j)H + \sum_j \frac{\theta_j}{m_j} \operatorname{div}(P_j). \quad (\text{F.8})$$

The lemma says that Ψ is effective and lc, with exactly the common rays as its nonzero zero-discrepancy rays. It also gives $q(P_j) = m_j q(h)$ on every common ray. These two properties let us express a sign inequality as log canonicity of a small boundary perturbation.

Assume the left side of (F.7). For sufficiently small positive rational δ the divisor

$$\Psi_\delta = \Psi + \delta(\operatorname{div}(U) - bH) \quad (\text{F.9})$$

is effective and lc near x . Here is the exact reason for smallness sufficing. The coefficient of H in (F.8) has a positive reserve, which preserves effectivity. On a fixed log resolution including U , every coefficient below one stays below one for small δ ; for a coefficient-one prime the discrepancy change is $-\delta q(g) \geq 0$. Checking these finitely many components checks lc on the entire resolution. Restricting the germ if necessary discards centres whose closed image does not contain x . No uniform choice of δ in U is needed.

All equations in (F.9) have their expected orders at w , and $A_S(w) = w(\Psi_\delta)$ because $w(g) = w(f_j) = 0$. Corollary F.3 gives an lc initial boundary. For every $q^0 \in \mathcal{Q}^0$,

$$q^0(\bar{P}_j) \geq m_j q^0(\bar{h}).$$

Consequently

$$0 \leq A_{S^0, \Psi_\delta^0}(q^0) \leq -\delta q^0(\bar{g}),$$

which proves the right side of (F.7).

Conversely assume that right side. Carry out exactly the same boundary construction on S^0 , choosing the λ_j general for both the actual and central constructions. The intersection of these finitely many nonempty open conditions is nonempty. The assumed sign and a log resolution of the central equations give a small rational $\delta > 0$ for which Ψ_δ^0 is effective and lc near o . Choose an integral rational regrading that preserves all the equations involved. Homogenize them in the Rees family. Their rational linear combination Δ_S is an effective rationally Cartier horizontal boundary, disjoint in components from E , and its restriction is Ψ_δ^0 .

The total space is normal, E is normal and Cartier, and its ordinary canonical divisor is Cartier by Proposition F.2. Thus $K_S + E + \Delta_S$ is $\mathbb{Q}$ -Cartier. The different is the ordinary restriction, as checked above. Inversion of adjunction for log canonicity [21, Theorem] now makes $(S, E + \Delta_S)$ lc near o . The section with all rescaled coordinates $x_i t^{-a_i}$ zero specializes to o , so this neighbourhood meets it over a nonzero parameter. Off E the family and its horizontal boundary are the product with (S, Ψ_δ) . Hence the latter pair is lc near x . For a common actual ray, the initial boundary encoding has zero discrepancy, and the last perturbation subtracts $\delta q(g)$. It follows that $q(g) \leq 0$, proving the converse.

The resolution computations give inequalities on every divisorial ray, and continuity on the monomial cones of the lc skeleton gives them on the full quasi-monomial slices. Additional finite tests are treated in the same boundary and presentation. $\square$

Corollary F.5 (Exact tests and pure weights). *Suppose w is common h -lc. If every common actual valuation satisfies $q(f_j) = 0$, then every common cone valuation satisfies $q^0(\bar{f}_j) = 0$. The symbols $\bar{f}_j$ are invariant under G and $\mathbb{T}_w$. Every nonzero pure weight valuation of the cone is h -lc and, after normalization by $\bar{h} = 1$, is common. Pure weights on the boundary of the dual weight cone are allowed under the closed-centre convention.*

Proof. Apply Proposition F.4 to $g = f_j$, and combine its upper bound with the defining lower bound. The equality $w(f_j) = 0$ and injectivity of the fine grading imply that f_j has weight zero. The invariant log form $\omega^0/\bar{h}$ has weight zero, so the pure-weight discrepancy computation of Section D gives lc equality for every pure weight. The test ratios then all have value zero. Klt implies that $\bar{h}$ is strictly positive on every nonzero such weight, so the stated normalization is possible. $\square$

F.4 Compression of two gradings

A second optimization takes place on the first cone. To iterate the construction on the original germ, we must realize these two successive gradings by one actual valuation there. The canonical-character calculation will prove its lc equality, while the exact bridge keeps the previous order constraints.

Lemma F.6 (Compression). *Suppose w is as above, is h -lc, and satisfies $w(f_j) = 0$ for every j . Let v be a centred $G \times \mathbb{T}_w$ -invariant, $\bar{h}$ -lc quasi-monomial valuation on S^0 , with normal finitely generated klt graded algebra R^{00} . Suppose also $v(\bar{f}_j) = 0$. The algebra R^{00} has its joint grading by the old torus and the v -grading. For arbitrarily small positive δ outside a countable exceptional set, it is the associated graded of an actual centred quasi-monomial valuation w_δ on S , whose grading evaluates a joint degree (a, b) as $a + \delta b$. The valuation w_δ is h -lc, is exact on all f_j , and, after $h = 1$ normalization, tends multiplicatively to w uniformly on all nonzero regular functions. If v is nonzero on an old degree-zero rational function, the rational rank of w_δ is strictly larger than that of w .*

Proof. Choose old-homogeneous elements of $\check{R}$ whose v -symbols generate R^{00} and lift them to regular functions x_i on S . Enlarge the list to generate the first graded algebra and to include h, F_j and every other specified equation. Use G -eigenvectors throughout. Put $\alpha_i = w(x_i)$ and $\beta_i = v(\bar{x}_i) > 0$. Both successive subductions of Lemma F.1 give

$$R^{00} = \mathbb{C}[X]/\text{in}_\beta(\text{in}_\alpha I). \quad (\text{F.10})$$

Completion of the first ideal does not change its initial ideal for positive β .

Choose finitely many generators of the prime ideal on the right that are iterated initials of elements of I . This is possible by first choosing α -homogeneous elements of $\text{in}_\alpha I$ with the desired β -initials; such elements lift with their prescribed α -initial. For each of these finitely many series, positivity and finiteness of the relevant leading faces imply that its iterated initial is its $(\alpha + \delta\beta)$ -initial for all sufficiently small $\delta > 0$. Hence

$$\text{in}_\beta \text{in}_\alpha I \subset \text{in}_{\alpha+\delta\beta} I.$$

The first ideal is prime and both quotients have dimension n . They are equal. Lemma F.1 supplies the actual multiplicative valuation w_δ and its quasi-monomiality. Avoiding countably many values of δ makes $(a, b) \mapsto a + \delta b$ injective on the joint lattice.

We now prove the lc equality, after constructing the valuation. The first degeneration has canonical character equal to the weight of $\bar{h}$. Apply Proposition F.2 to v on that cone, keeping the old torus action. Since v is $\bar{h}$ -lc, the resulting canonical generator has exactly the joint character of $\bar{h}$. Independently apply the same proposition to the actual valuation w_δ , whose normal finitely generated graded algebra has just been identified with R^{00} . Its discrepancy is the evaluation of this canonical character under compression. Therefore

$$A_S(w_\delta) = \langle w_\delta, \text{wt}(\bar{h}) \rangle = w_\delta(h). \quad (\text{F.11})$$

This uses no prior lc assumption about w_δ . The included coordinates identify the specified symbols, and the joint degree of each $\bar{f}_j$ is zero; hence $w_\delta(f_j) = 0$.

For the uniform comparison, if $(1 - \varepsilon)\alpha_i \leq \alpha_i + \delta\beta_i \leq (1 + \varepsilon)\alpha_i$ for every i , the same inequalities hold for every monomial weight. The threshold-ideal description, including its intersection with the actual local ring, gives

$$(1 - \varepsilon)w(f) \leq w_\delta(f) \leq (1 + \varepsilon)w(f) \quad (0 \neq f \in \mathcal{O}_{S,x}).$$

Normalization by h changes the factors by another quantity tending to one. Finally the joint lattice surjects onto the old lattice. A degree-zero rational function with nonzero v -value provides a nonzero vector in its kernel. Injective compression therefore strictly increases rational rank. $\square$

F.5 Balanced cones satisfying the angular order condition

Proposition F.7 (Rank refinement). *Consider a sequence of the preceding data in dimensions at most n_0 . Suppose that the common actual slices satisfy $q(f_j) = 0$ identically, that w is a centred point of the slice with the stipulated normal finitely generated klt graded algebra, and that the cones satisfy the following balance condition with one constant C along the sequence:*

$$q^0(s) \leq Cw^0(s) \quad \begin{array}{l} 0 \neq s \in \check{R} \text{ fine-homogeneous,} \\ q^0 \text{ common,} \quad q^0(\bar{h}) = w(h) = 1. \end{array} \quad (\text{F.12})$$

After passage to a subsequence, the w can be replaced by w' with all these properties, with a new fixed balance constant, and with

$$w'(f) = (1 + o(1))w(f) \quad (0 \neq f \in \mathcal{O}_{S,x}), \quad (\text{F.13})$$

uniformly in f in the normalization $w'(h) = w(h) = 1$. On the angular model $V_{\mathbf{u}}$ of the new cone, the notation of Section D then satisfies

$$T(\Gamma) \leq (1 + \eta_i)D(T), \quad \eta_i \longrightarrow 0, \quad (\text{F.14})$$

for every effective $\mathbb{Q}$ -divisor $\Gamma \sim_{\mathbb{Q}} D$ and every primitive divisorial T such that $a(T) = 0$ and $T(\bar{f}_j) \geq 0$ for all j . This is condition (C- α) of Section C. The initial symbols here refer to w' .

The balance hypothesis holds when w is the exact minimizer on the common slice for the height $q(h)$.

Proof. The last assertion is the central domination estimate in Corollary E.10; equivariance follows from the uniqueness of the slice minimizer. At every stage, Corollary F.5 makes all common central test values zero and makes every pure weight common. The $\bar{f}_j$ are genuine functions of the angular field $\mathcal{K}$. At a height-minimizing Gauss extension of T , regularity of F_j gives

$$T(\bar{f}_j) \geq -m_j D(T). \quad (\text{F.15})$$

Suppose (F.14) fails along a subsequence. There are a fixed rational $c > 0$, primitive witnesses T , and effective divisors

$$\Gamma = D + \frac{1}{m} \operatorname{div}(\phi), \quad m \in \mathbb{N}_{>0}, \quad \phi \in \mathcal{K}^*,$$

such that $a(T) = 0$, $T(\bar{f}_j) \geq 0$, and

$$T(\Gamma) - D(T) \geq 2cD(T). \quad (\text{F.16})$$

Here $D(T) > 0$ because $a + \mathbf{D}$ is klt. After clearing denominators, the ray-ring description in Section D makes $G_0 = \bar{h}^m \phi$ a regular homogeneous function upstairs. Choose a rational Gauss extension minimizing $\bar{h}$ at T . It is $\bar{h}$ -lc, and its lift to the finite quasi-étale cover cone remains lc. Its centre has the vertex in its closed image: this holds downstairs for a Gauss valuation, and finiteness and the unique point over the vertex give it upstairs. It is common, and (F.16) reads

$$\frac{g(q)}{q(\bar{h})} \geq 2c, \quad g(q) = \frac{1}{m} q(G_0) - q(\bar{h}).$$

Apply the one-sided tilt of Lemma E.11 to this cone. Its exact minimizer v , normalized by $v(\bar{h}) = 1$, is centred, $G \times \mathbb{T}_w$ -invariant, and has normal finitely generated klt associated

graded. Localization at the vertex does not change this graded algebra: a unit has constant initial term. The valuation v is common, hence exact on the old tests, and the tilt gives

$$v(\phi) > 0, \quad b(v) \geq 1.$$

The tilted height b is bounded above by a fixed constant on both the old and the new common slice, with constants depending only on n_0 and the fixed c at this stage. Central domination therefore gives, on the new cone,

$$q^{00}(s) \leq C_1 v^0(s) \tag{F.17}$$

for common q^{00} normalized by $\bar{h} = 1$. The joint canonical character equals that of $\bar{h}$ by Proposition F.2, including its old-torus assertion.

Compress these two gradings by Lemma F.6. Since ϕ has old weight zero and $v(\phi) > 0$, rational rank strictly increases. The resulting actual valuation is h -lc, exact on the tests, and can be chosen arbitrarily close to w uniformly on every regular function. We check balance, which is necessary to repeat the argument. A joint-homogeneous s has an old-homogeneous lift $\tilde{s} \in \check{R}$; the v -filtration splits into old weight spaces. If its joint degree is (a, b) , then

$$b = v(\tilde{s}) \leq Ca$$

by (F.12), since v is common. In the normalization $w(h) = v(\bar{h}) = 1$, the compressed degree is $(a + \delta b)/(1 + \delta)$. Thus

$$v^0(s) = b \leq C(1 + \delta)w_\delta^0(s).$$

Together with (F.17) this is the next balance estimate, with a fixed constant. A generic compression makes every fine-homogeneous element joint-homogeneous, so this checks precisely all elements required in (F.12).

If the angular order condition still fails, pass to a subsequence with a fixed positive defect and repeat. Each repetition raises rational rank, which is at most the dimension. There can be at most n_0 repetitions. At a final stage the supremum of the positive normalized defects must tend to zero. This supremum is finite: a height-minimizing Gauss witness and (F.12), applied to G_0 of grading value m , bound $T(\Gamma)/D(T)$ by the balance constant. Taking these suprema, or slightly larger positive numbers, gives the single sequence η_i in (F.14), uniformly for all its displayed tests and divisors. At each of the boundedly many compressions choose the all-functions multiplicative error to tend to zero; their product still tends to one and proves (F.13). All exact test equalities, equivariance, canonical characters, and quasi-étaleness have been preserved at every step. $\square$

G Pinning the gradings

We now combine the character supply of Section D with the optimization and transfer results of Sections E and F. The discrepancy gap in this section is always a gap for primitive divisors on the actual quotient germs. No such gap is imposed on the auxiliary central cones.

The local phase assertion of [28, Theorem 3.1] constructs centred divisorial valuations with bounded homogeneous discrepancy and cyclic-character congruences, after passage to a subsequence, under its hypotheses on Gorenstein klt germs of fixed dimension $n \geq 2$. It does not guarantee that these valuations lie on the prescribed h -lc slice $A(T) = T(h) = 1$ below, so it does not replace the pinning obstruction.

Theorem G.1 (Pinning obstruction). *Fix an integer $n > 0$ and $\epsilon > 0$. There is no sequence of the following data satisfying all the conditions below. For each index i , let $x_i \in S_i$ be an n -dimensional complex klt germ with an action of a finite cyclic group Γ_i fixing x_i , such that*

the quotient map is quasi-étale and S_i/Γ_i is ϵ -lc. Suppose K_{S_i} has a nowhere-vanishing local eigen-generator ω_i , and let h_i be a nonzero regular eigenfunction with the same character χ_i as ω_i . The character need not be faithful. Assume that $(S_i, \text{div}(h_i))$ is lc and admits a centred quasi-monomial valuation T_i with

$$A_{S_i}(T_i) = T_i(h_i) = 1.$$

The excluded additional condition is: for every fixed integer $m > 0$, there is an index $i(m)$ such that, for all $i \geq i(m)$ and every regular semi-invariant s of character χ_i^m ,

$$T_i(s) \geq m. \quad (\text{G.1})$$

All constants in the proof are uniform along the sequence under consideration; they may depend on the dimension and on the finitely many previous stages of the construction. An assertion made eventually also holds after any further subsequence. We suppress i whenever this does not obscure a limit. We may successively pass to subsequences, since the number of stages and of subsidiary row additions will be bounded.

G.1 Costs and the induction data

An invariant rational function Y has *cost at most m* if $h^m Y$ is regular, where m is a nonnegative integer. We increase costs to positive integers and to common denominators when convenient. Costs are subadditive under multiplication. The same terminology applies on a graded cone, with h replaced by its initial symbol. Every homogeneous ratio on such a cone of cost m lifts to an actual invariant ratio of the same cost: lift its regular numerator, and project to the character χ^m by linear reductivity of the finite group.

We keep the test functions and accumulated variables on the original germ, while replacing the working valuation and its cone. Algebraic independence of the accumulated symbols is the quantity that will force termination. Their weights may nevertheless become rationally proportional: suitable ratios of monomials then have weight zero and retain information in the angular field. Pinning the weights to a line therefore need not reduce the transcendence degree of the accumulated symbols.

The following data are retained at the beginning of each stage.

- (I1) There are finitely many actual invariant test ratios f_j , each of cost $O(L_-)$, where initially $L_- = 1$. The common actual slice

$$\mathcal{Q} = \{q : A_S(q) = q(h) = 1, q(f_j) \geq 0 \text{ for every } j\} \quad (\text{G.2})$$

contains a centred valuation, and every test has value exactly zero throughout this slice.

- (I2) There are accumulated pairs Y_j^+, Y_j^- of invariant ratios of cost $O(L_-)$, with opposite values on $\mathcal{Q}$. We write $y_j(q) = q(Y_j^+) = -q(Y_j^-)$. The products $Y_j^+ Y_j^-$ occur among the exact tests. Consequently the equality of the pair values also holds on every later common cone slice, by Proposition F.4.
- (I3) Either all y_j vanish on $\mathcal{Q}$, or their vector lies on a single rational line. In the latter case one accumulated variable, indexed by 0, is an *anchor*. It was chosen at an unbounded scale L_1 ; its pair has cost $O(L_1)$, and $L_1 \leq L_-$. For each other accumulated variable there are integers t_j, d_j , with $d_j > 0$ bounded, such that

$$y_j = (t_j/d_j)y_0, \quad |t_j| = O(L_-/L_1). \quad (\text{G.3})$$

These equations include the point where all values are zero and are imposed by tests of both signs. In the all-zero case there is no anchor.

- (I4) The optimization height $\mathfrak{B}$ is rational and positive on $\mathcal{Q}$, and in the normalization $q(h) = 1$ satisfies

$$\rho \leq \mathfrak{B}(q) \leq C \quad (\text{G.4})$$

for fixed positive constants. Initially $\mathfrak{B}(q) = q(h)$. It can change once, when an anchor is first chosen, to

$$\mathfrak{B}(q) = q(h) - \mu q(Y_0^{\text{sign}})/L_1, \quad \mu \geq 0, \quad (\text{G.5})$$

where either member of the anchor pair can be used and μ is bounded. This is an allowed height in Theorem E.8. On a common central slice the same expression has a uniform upper bound after normalizing $q'(h) = 1$, because both anchor ratios have cost $O(L_1)$.

- (I5) Take the exact $\mathfrak{B}$ -minimizer and apply the rank refinement of Proposition F.7, choosing the approximation error to tend to zero. This produces an equivariant centred working valuation w , with finitely generated normal graded algebra, whose values on all regular functions are multiplicatively $1 + o(1)$ times those of the exact minimizer. Its $\mathfrak{B}$ -normalized volume is within $1 + o(1)$ of the minimum. Its cone satisfies the balance estimate (F.12) and the order condition (C.5), with error $\eta_i \rightarrow 0$, on the angular tests $\mathcal{L}$ specified by the current initial tests. The symbols of all accumulated Y_j^+ are algebraically independent. If there is an anchor,

$$|w(Y_0^+)| \geq cL_1 \quad (\text{G.6})$$

for a fixed $c > 0$.

The balance hypothesis needed for the refinement in (I5) follows from the preceding height bounds. Let v be the exact $\mathfrak{B}$ -minimizer, normalized by $v(h) = 1$. For every common central q^0 with $q^0(h) = 1$, Corollary E.10 gives

$$q^0(s) \leq 2^n \frac{\mathfrak{B}^0(q^0)}{\mathfrak{B}(v)} v^0(s) \leq \frac{2^n C}{\rho} v^0(s) \quad (0 \neq s \text{ regular and homogeneous}).$$

Here only the upper bound for $\mathfrak{B}^0$ is used on the central slice; the denominator is bounded below at the actual minimizer. Proposition F.7 then preserves balance with a fixed constant.

Initially there are no tests and no pairs, so these conditions follow from the centred point T and Sections E–F. Corollary E.9 applies to $\mathfrak{B}$. Indeed, normalize two valuations by $\mathfrak{B} = 1$. The radially scaled version of Lemma E.4 fixes the averaged value of $h = A$ and gives exact interpolation of the opposite anchor pair. In particular its allowed height is at most 1. Its volume is therefore at least the minimum in height-one normalization. The same gap argument gives multiplicative comparison on all functions. The bounds (G.4) then convert that comparison back to $h = 1$. Applied to numerators and h -powers, it controls ratio values up to the corresponding cost times the relative error.

Let $M, \mathbf{u}, \sigma$ be the invariant character lattice, the weight of h , and the cone from Proposition D.1, for the working algebra. Equip $M_{\mathbb{R}}$ with

$$\|k\| = \max_{\substack{y \in \sigma \\ \langle y, \mathbf{u} \rangle = 1}} |\langle y, k \rangle|. \quad (\text{G.7})$$

Fine weights evaluate injectively under w . Thus the exact equations in (G.3) imply the same equations for the weights of the symbols. In the line case write $\kappa_0 = \text{wt}(\text{in}_w Y_0^+)$. Opposition on pure common tests and the paired costs give $\|\kappa_0\| = O(L_1)$, whereas (G.6) gives the reverse bound. Hence

$$\|\kappa_0\| \asymp L_1. \quad (\text{G.8})$$

After a stage with $L_- \rightarrow \infty$, we will additionally retain:

- (H1) Every lattice vector of width $O(L_-)$ belongs to $\mathbb{Q}\kappa_0$ if an anchor is present, and is zero otherwise. In the line case the index of $\mathbb{Z}\kappa_0$ in $M \cap \mathbb{Q}\kappa_0$ is bounded.
- (H2) The angular tests in $\mathcal{L}$, namely $a = 0$ with nonnegative orders on all initial tests, restrict trivially to the field generated by $S(L_-D)$.

Here $a, D, S(\cdot)$ have the angular meanings of Sections C–D. No assertion (H1) or (H2) is required while L_- remains bounded, and there is then no anchor. These two assertions will be proved for the next stage after the high-scale pin, without being used to construct its rank refinement.

G.2 The next character block and the terminal case

Suppose first that an anchor is present and M has no direction transverse to its line. Write $\kappa_0 = \ell k_0$, where k_0 is primitive in M ; assertion (H1) bounds ℓ . The restriction of w to the actual quotient field has rational rank one and is Abhyankar. To justify that this is the actual value group, take invariants in every level of the finite-group-stable filtration: exactness of invariants identifies the invariant graded algebra with the graded algebra of the quotient filtration. Its homogeneous fraction weights generate M , and evaluation is injective. Thus the quotient value group is generated by

$$\gamma = |w(Y_0^+)|/\ell.$$

Abhyankar equality is preserved under the finite field extension, so this rank-one valuation is divisorial up to scaling. Its primitive normalization is $w|_{\text{Frac}(S/\Gamma)}/\gamma$. Quasi-étale log pullback and $A_S(w) = 1$ give its discrepancy as $1/\gamma$. By (G.6) and $L_1 \rightarrow \infty$, this tends to zero, contradicting ϵ -lc of the actual quotient. Rank zero of M is impossible, since a finite extension of the quotient field cannot support a nontrivial valuation with trivial restriction. This disposes of the terminal possibilities.

Otherwise choose a block of lattice directions as follows. With an anchor use the quotient norm modulo $\mathbb{R}\kappa_0$ on the projected lattice; without one use (G.7). After a previous high stage its shortest nonzero length divided by L_- tends to infinity. For if a projected vector had length $O(L_-)$, lift it and subtract the nearest multiple of the primitive line vector. Its remaining line contribution is $O(L_1)$ by (G.8) and the bounded index, producing a transverse lattice vector of width $O(L_-)$, contrary to assertion (H1).

Before the first high stage, if some successive minima remain bounded, choose a nonempty maximal bounded block of independent directions, of width $O(L)$ with $1 \leq L = O(1)$. This is a *bounded stage*. In every other case choose an integral scale $L \rightarrow \infty$ comparable to the first successive minimum, and take the whole first block of successive minima comparable to L . After a subsequence, the next successive minimum, if any, divided by L tends to infinity. These are *high stages*. Lift the selected directions k_j to M by rounding along the anchor line when necessary. They have width $O(L)$, and $L_- = O(L)$.

In a high stage, let s denote the block size plus one when an old anchor is present, and just the block size otherwise. Put

$$R_* = L/L_1 \quad \text{with an old anchor}, \quad R_* = 1 \quad \text{without one.} \quad (\text{G.9})$$

For every fixed $N \geq 1$, eventually

$$\#\{k \in M : \|k\| \leq CNL\} \leq C'R_*N^s, \quad (\text{G.10})$$

with C' independent of N . Indeed, the gap to the next successive minimum confines these vectors to the first-block span modulo the line. Distinct projected lattice points have separation at least cL , so a norm-ball packing in its bounded dimension gives $O(N^{s-1})$ points when

there is an anchor, and $O(N^s)$ otherwise. Above each projected point, the number of lifts in the ball is $O(NL/L_1)$, since the lattice step on the anchor line has width comparable to L_1 . This proves (G.10).

G.3 Neutral tests and simultaneous non-cancellation

Apply Lemma C.3 at scale L . Its test-cost hypothesis uses $L_- = O(L)$; its additional field hypothesis, when $L_- \rightarrow \infty$, is precisely assertion (H2). Write its outputs as K_*, p, v_i, B . If the angular field is the constant field, set $p = 0$ and $v_i = 1$. Increase the fixed constant B if needed. Lift a basis of $S(BLD)$ to actual ratios G_a of cost $O(L)$ with the prescribed neutral initial symbols. This basis is finite for each germ, but its cardinality may grow along the sequence; the cost constant is uniform. We include 1 in the basis. We may also arrange that p basis members have algebraically independent initial symbols: the cutoff plateau has an endpoint $\beta_1 \geq c/L$, and its bounded birational system in D_W/β_1 pulls back into $S(BLD)$ and generates K_* as a field. A transcendence basis can be chosen from a field-generating set and extended to a vector-space basis.

For each selected direction k_j , Corollary D.5 supplies homogeneous invariant ratios of weights $\pm q_j k_j$, with bounded positive integers q_j , and cost $O(L)$. Align the two multipliers by taking bounded powers and lift the ratios to pairs $Y_j^\pm$ on the actual germ. Its small-height hypothesis holds: on a primitive angular test with $a = 0$ and $LD \rightarrow 0$, an old test of cost $O(L_-)$ has negative order bounded in absolute value by $O(L_- D) = o(1)$. Its order is integral, hence is nonnegative eventually. Such a test therefore belongs to $\mathcal{L}$, where the required (C.5) holds.

Add the G_a and the new products $Y_j^+ Y_j^-$ as tests, before imposing any individual new pair or proportionality constraint. Call the result the *pre-pinning slice*. All these tests have value zero at w , so this slice contains w . On the working cone, the old tests together with the G_a tests force trivial restriction on K_* by saturation: nonnegativity on a vector-space basis implies nonnegativity on its whole linear system. The assertion also holds for pure tests, which are trivial on the angular field. Lemma D.2 supplies the homogeneous Gauss description and the finite-cover lifts for all other common lc tests. Rational monomial tests suffice, and the equalities extend to their real subcones. The G_a and the new pair products have initials in K_* , since their neutral costs are $O(L)$. The bridge therefore gives value at most zero on the actual pre-pinning slice. Their test inequalities give the reverse sign, proving exactness and opposition there.

Lemma G.2 (Non-cancellation in every degree). *At the preceding stage, consider any finite linear combination of monomials in the accumulated and newly added Y_j^+ and the current G_a , with nonnegative exponents. Suppose its least- w group has a nonzero sum of initial symbols. If all its terms have the same value at an actual point of the pre-pinning slice, their sum has that value. This assertion holds eventually simultaneously in all polynomial degrees. Moreover, at a later working valuation on this slice where these terms are tied, the same non-cancellation assertion transfers to its cone at every common test where the terms remain tied and the pair values are opposite.*

Proof. Write $A_j = \text{in}_w Y_j^+$ and $B_j = \text{in}_w Y_j^-$. The pair products $A_j B_j$ lie in K_* . With an old anchor, the finitely many old relations

$$\text{wt}(A_j) = (t_j/d_j)\kappa_0$$

give neutral rational functions $A_j^{d_j} A_0^{-t_j}$. Each is a quotient of two neutral functions of cost

$O(L)$. More explicitly, for $t_j \geq 0$ write it as

$$\frac{A_j^{d_j} B_0^{t_j}}{(A_0 B_0)^{t_j}};$$

for $t_j < 0$ its numerator is simply $A_j^{d_j} A_0^{|t_j|}$. The required costs are bounded by a fixed multiple of $d_j L_- + |t_j| L_1 = O(L_-) = O(L)$. There are at most n accumulated variables, and their denominators d_j are bounded. Thus one fixed constant works for this entire finite collection. The inclusion $S(CLD) \subset K_*$ puts their numerators and denominators in K_* at one eventual index. Without an anchor all the old A_j themselves are neutral and of bounded cost, and belong to K_* instead.

The new weights are independent modulo the old line, or modulo zero. Consequently a monomial relation of total weight zero has zero net exponent in each new variable. Clearing the bounded old denominators expresses a power of every remaining weight-kernel monomial in the finite neutral generators just described. Every such monomial is therefore algebraic over K_* . This is a simultaneous algebraic statement: once the finite generators have entered K_* , there is no further eventual index depending on the exponents.

Choose a term in the least- w group, and multiply the whole sum by the opposite factors Y_j^- corresponding to its Y_j^+ factors. Its G_a factors already have value zero. The complemented sum has a nonzero w -initial of weight zero. Each term of that initial is algebraic over K_* by the preceding kernel argument, so the initial itself is algebraic over K_* . Every common cone test is trivial on this element. Here we use the elementary fact that a valuation trivial on a field is trivial on its algebraic extension inside a valued field: a polynomial equation with nonzero constant term would otherwise have a unique term of smallest value.

The complemented sum is an invariant ratio with a regular numerator after multiplication by some power of h . Its cost is allowed to depend on the chosen polynomial. Proposition F.4 has no uniform cost restriction, and yields an actual upper bound zero on the pre-pinning slice. At a point where the original terms tie, opposition and the zero values of G_a give value zero for every complemented term. The valuation inequality gives the lower bound zero. The complemented sum thus has exact value zero, which proves non-cancellation.

At a later working valuation w' where the terms tie, this establishes exact value zero before any new bridge is invoked. Its initial symbol there is nonzero. Apply the bridge in the other direction, using only tests exact at w' . At a common test on the new cone where the terms remain tied, the transferred upper bound and the same termwise lower bound again give equality. This proves the onward assertion. $\square$

The enlarged collection of positive symbols is algebraically independent at w : the new weights are independent modulo the old line, and the old symbols were independent. It remains independent at every later working valuation on the pre-pinning slice. Indeed, a proposed polynomial relation among the later symbols can be grouped by their later grading. Each nonempty group has a nonzero least-part initial at w , and Lemma G.2 forbids cancellation of that group at the later valuation.

At a bounded stage, add both new members $Y_j^\pm$ individually as tests. Their values must then be zero. The original centred point T remains in the slice by (G.1), because every cost used so far is bounded. This invocation is uniform over the finite interval of possible integral costs, and applies to every numerator in each of the corresponding semi-invariant spaces. Thus the bounded stage is complete: replace L_- by L and optimize and refine again. No assertion of (G.1) at an unbounded cost has been used.

G.4 The count at a high scale

The neutral tests now prevent cancellation and preserve the independent symbols when the working valuation changes. At an unbounded scale, the input saturation condition does not directly control all newly chosen costs. We instead bound the number of symbols available at that scale. The high-scale pin will force enough exact relations to avoid violating this bound.

We now work at a high stage. For each fixed integer $N \geq 1$, each integer $0 \leq t \leq CNL$, and the character χ^t , the span of regular homogeneous symbols of that character of grading value at most CNL has dimension at most

$$C' R_* v_i N^{p+s}. \quad (\text{G.11})$$

The constant C' is independent of N and t . To see this, divide a symbol by h^t . It becomes an invariant homogeneous fraction with weight in M . Balance (F.12), tested on rational pure rays, bounds its width by $O(NL)$. Those rays are common tests because all current test weights are zero. For each weight, Corollary D.6 bounds its dimension by $C'' v_i N^p$. Sum over (G.10).

The same upper bound, with only a fixed change of constant, holds on the cone of any other actual equivariant centred valuation v with

$$v \geq cw \quad (\text{G.12})$$

on all regular functions, in $h = 1$ normalization, for fixed $c > 0$. Indeed the order ideals satisfy $\mathfrak{a}_{a/c}(w) \subseteq \mathfrak{a}_a(v)$. The resulting surjection of finite colength quotients is equivariant. Taking each finite-group eigenspace is exact, so the cumulative dimensions of its associated graded pieces have the asserted comparison. The strict or weak endpoint of a grading cutoff can be handled by slightly increasing its fixed constant. In particular we transfer (G.11) from the original working cone; we never replace it by a count with a constant depending on a later tilt.

We record the elementary product estimate used below.

Lemma G.3 (Uniform product growth). *Let V be a finite-dimensional subspace of a function field over an algebraically closed field, containing 1. Put $g = \dim V$ and $a = \text{trdeg } k(V) \leq n$. If $V^{(N)}$ is the span of N -fold products of members of V , there is $c_n > 0$, depending only on n , such that*

$$\dim V^{(N)} \geq c_n g N^a \quad (N \geq 1).$$

Proof. The rational map defined by V has a nondegenerate integral projective image of dimension a in $\mathbf{P}^{g-1}$. Its degree- N homogeneous coordinate space is $V^{(N)}$. Successively cut by general hyperplanes. In positive dimension the reduced hyperplane section is nondegenerate in its hyperplane. One way to see the required nondegeneracy is to pass to the normalization: a general hyperplane divisor is reduced in codimension one, so another linear form vanishing on its reduced support has a regular ratio with the cutting form, and the ratio is constant. The reduced section is integral until the last curve section by Bertini; the final reduced set of points spans its hyperplane and has Hilbert function at least its dimension in degree one in every higher degree. For the homogeneous coordinate ring, multiplication by a nonzero cutting linear form is injective, and the quotient surjects onto the coordinate ring of the reduced cut. Thus successive differences of Hilbert functions dominate the Hilbert functions of these cuts. Summing these inequalities gives, for $a > 0$, the sufficient bound

$$\dim V^{(N)} \geq (g - a) \binom{N + a - 1}{a}.$$

Since $g \geq a + 1$, one has $g - a \geq g/(a + 1)$, giving the asserted constant. For $a = 0$ the image is a point, $g = 1$, and the assertion is immediate. $\square$

G.5 The high-scale pin

At a high stage the new costs grow with L , so the original input (G.1) no longer supplies a centred point after individual values are forced to zero. We instead constrain their proportions while retaining a centred point close enough to the current minimum.

Proposition G.4 (High-scale pin). *Consider a high stage with pre-pinning slice, working valuation w , baseline height $\mathfrak{B}$, scale L , and s selected values as above. After passage to a subsequence, finitely many invariant tests of cost $O(L)$ can be added with the following properties.*

- (i) *The resulting slice contains a centred valuation. The added tests are exact on that slice, all old tests and proportionalities remain in force, and the accumulated positive symbols remain algebraically independent at the exact minimizer and working valuation in (iii) below.*
- (ii) *The selected values either vanish or lie on a single rational line. In the line case their proportions have the bounds (G.3), with $L_- = L$. An old anchor remains the anchor; if a new anchor is chosen, its scale is $L_1 = L$.*
- (iii) *The baseline height is retained unless an anchor is first chosen, in which case it may change to the rational height (G.5). It satisfies (G.4) on the new actual slice. At its exact minimizer v' , and at an arbitrarily close rank refinement w' , the anchor, when present, satisfies $|v'(Y_0^+)|, |w'(Y_0^+)| \geq cL_1$. Both valuations dominate a fixed positive multiple of w on every regular function. In particular,*

$$w' \geq cw.$$

Consequently (I1)–(I5) hold with $L_- = L$. The next-scale assertions (H1)–(H2) are not assumed in constructing the pin or its rank refinement.

Domination preserves the counting bound on the original working cone. The rational proportions permit equations of cost $O(L)$, and the nonzero anchor size will detect a small primitive quotient discrepancy if only its line remains. We first construct this pin, then prove that it also gives the next-scale field and lattice assertions.

G.6 Finding all necessary relations

We begin the proof of Proposition G.4. Within this stage, progress is measured by the rank of the integral relations imposed on the s selected values. This row rank is separate from the transcendence degree used to terminate the outer induction.

For the s selected variables put $L_j = L_1$ for an old anchor, and $L_j = L$ for every new variable, and set

$$x(q) = (y_j(q)/L_j)_{j=1}^s.$$

When there is an anchor we include its index 0 in this notation. Opposition and the paired costs bound all coordinates uniformly on the actual pre-pinning slice. Lemma E.4 and Corollary E.5 give exact convex interpolation of the coordinates, so their image is compact and convex. Its points can be realized by centred valuations whenever the interpolation assigns positive weight to a centred input.

We successively impose integral rows

$$\sum_j k_j y_j = 0, \quad \hat{k} = (k_j L_j / L)_j, \quad \|\hat{k}\|_{\text{Eucl}} = O(1). \quad (\text{G.13})$$

A row is imposed by testing two opposite monomials. For its positive sign use

$$M_k = \prod_{k_j \geq 0} (Y_j^+)^{k_j} \prod_{k_j < 0} (Y_j^-)^{-k_j};$$

exchange the signs to obtain the other monomial. Each has cost $O(\sum_j |k_j| L_j) = O(L)$, and opposition makes the two inequalities equivalent to the row equality.

Let $\mathcal{R}_i \subset \mathbb{R}^s$ denote the span of the normalized rows already imposed. We retain a comparator v_0 on the current slice whose $\mathfrak{B}$ -normalized volume is within $1 + o(1)$ of the minimum on the actual slice at the start of this high stage. Initially use w . The gap estimate then gives $v_0 = (1 + o(1))w$ on all regular functions. With an old anchor, $|y_0(v_0)|/L_1$ stays bounded away from zero. Pass to subsequences so the row spaces converge and the limits of the chosen rows are independent.

For clarity, independence of their transverse integer rows is equivalent to independence of these normalized row limits in the present situation. The non-anchor coefficients k_j are bounded integers, hence become fixed on a subsequence. If a rational combination of their transverse rows vanished, the same combination of full rows would be supported only on the anchor. Its exact value at v_0 must vanish. Since the anchor coordinate is nonzero, its anchor coefficient vanishes as well. Thus any dependence in the transverse rows is a dependence of the full rows. Conversely independent transverse rows keep the full normalized rows independent in the limit. With no anchor, all coefficients are bounded integers and the statement is immediate.

Let K_i be the projection image $\{x(q)\}$ of the current constrained slice, and let

$$C_i = \text{conv}(\{0\} \cup K_i) = [0, 1]K_i.$$

After a subsequence these bounded compact convex sets converge in Hausdorff distance. Their limit lies in $\mathcal{R}_\infty^\perp = \lim \mathcal{R}_i^\perp$. We first handle the case

$$\text{span}(\lim C_i) \neq \mathcal{R}_\infty^\perp. \quad (\text{G.14})$$

Choose a unit vector $u \in \mathcal{R}_\infty^\perp$ perpendicular to the left-hand span, and rational vectors $u_i \rightarrow u$. Then

$$u_i \cdot x(q) = o(1) \quad (\text{G.15})$$

uniformly on the current slice. Fix a sufficiently small $\delta > 0$, with its size chosen below, and consider the rational height

$$b(q) = \delta \mathfrak{B}(q) - \sum_j u_{ij} y_j(q) / L_j. \quad (\text{G.16})$$

We take the scales integral, so all coefficients can be rational. According to the sign of u_{ij} , use Y_j^+ or Y_j^- , and absorb its h -denominator into the coefficient of $q(h)$. This puts b in the allowed form with nonnegative coefficients on the subtracted regular numerator values. Equations (G.4) and (G.15) show that it is positive and equals $(1 + o(1))\delta \mathfrak{B}$ uniformly on this slice.

Let v be its exact minimizer. Comparison with v_0 shows that the $\mathfrak{B}$ -normalized volume of v is within $1 + o(1)$ of the original stage minimum. Consequently

$$v = (1 + o(1))w \quad (\text{G.17})$$

on all regular functions, and v is exact on all present tests. On its cone every nonzero pure test, and every Gauss test over a primitive angular divisor with $a = 0$ and $LD \rightarrow 0$, is common. Indeed all test weights are zero, their costs are $O(L)$, and the integral angular

orders cannot be negative of size $o(1)$. All pair equalities transfer to this cone. Positivity of the induced height, supplied by Theorem E.8, and the uniform central upper bound for $\mathfrak{B}$ imply

$$\sum_j u_{ij} y'_j / L_j \leq C \delta q'(h) \quad (\text{G.18})$$

on these tests.

Apply Corollary D.7. For a fixed $M_0 \gg 1/\delta$, choose integers

$$r_j = M_0 u_{ij} L / L_j + O(1). \quad (\text{G.19})$$

It gives a homogeneous invariant ratio Z on the cone of v , of weight

$$-q \sum_j r_j \text{wt}(\text{in}_v Y_j^+), \quad (\text{G.20})$$

where q is a bounded positive integer, and of cost at most

$$CqM_0\delta L. \quad (\text{G.21})$$

The constant C is fixed before δ and M_0 are chosen. The bound on q may depend on these fixed choices. This distinction follows from the proof of the tilted-character result: its height correction has size $O(M_0\delta L)$; the $O(L)$ rounding error and the fixed constant in (G.18) are absorbed by first taking $M_0 \gg 1/\delta$. Only the later regularizing multiplier need depend on these parameters. The absolute grading value of Z has the same bound. In fact (G.20) evaluates to

$$-qM_0L u_i \cdot x(v) + O(qL),$$

which is bounded by (G.21) eventually, by (G.15) and the chosen size of M_0 . No assumption (C.5) on this newly tilted cone is used.

Claim G.5. *After a subsequence there is an additional row of the form (G.13), exact at v , whose transverse part is independent of the previously imposed transverse rows.*

Proof. Suppose otherwise. Then, for each fixed bound on normalized coefficient size, every exact integral tie at v eventually belongs to the old row span. To verify this assertion, a tie with transverse part outside the old span would give the desired row after fixing its bounded integer transverse coefficients. If its transverse part lies in the old span, subtract the corresponding rational combination of the fixed old transverse rows. The remainder is a pure-anchor tie. It is zero because (G.17) preserves the nonzero anchor value at v . Without an anchor there is no remainder. The claimed eventual assertion for all ties with a given bound follows by choosing a failing tie along a subsequence if necessary.

Let $\tilde{u}_i$ be a unit vector in $\mathcal{R}_i^\perp$ tending to u . Choose a large fixed integer $N \gg qM_0$. In the exponent box

$$0 \leq e_j \leq \lfloor NL/L_j \rfloor, \quad \hat{e} = (e_j L_j / L)_j,$$

there are at least $cR_* N^s$ integer points. The range of $\tilde{u}_i \cdot \hat{e}$ has length $O(N)$. Partition this range into intervals of width $qM_0/4$. One slab contains at least

$$cR_* N^{s-1} qM_0 \quad (\text{G.22})$$

points. For each exponent in this slab, consider on gr_v its positive-variable monomial multiplied by Z^l , where

$$0 \leq l \leq \lfloor N/(qM_0\delta) \rfloor, \quad (\text{G.23})$$

and by a basis of the products of N of the G_a whose w -initials are independent. Saturation and counting supply at least $v_i N^p$ such products.

Different powers l have disjoint grading values in this construction. A collision would give the exact integral tie

$$e - e' - q(l - l')r.$$

Its normalized size is bounded with N, M_0, δ, q fixed. Under our supposition it belongs to the old row span, so its normalized dot product with $\tilde{u}_i$ is zero. The slab bounds the contribution from $e - e'$ by $qM_0/4$. But (G.19) gives

$$\hat{r} = M_0 u_i + O(1), \quad |\tilde{u}_i \cdot \hat{r}| \geq M_0/2$$

eventually, after increasing the fixed M_0 . For $l \neq l'$ the other contribution has magnitude at least $qM_0/2$, a contradiction. It is essential here that $\tilde{u}_i$ is orthogonal to the entire old normalized row span.

At any one power l , the slab size times $v_i N^p$ symbols are linearly independent. Before changing to v , the selected variable weights separate the exponent vectors, and the chosen G_a products have independent initials at w . Any possible relation after the change can be grouped by its v -grading, and Lemma G.2 rules out cancellation in each group. Multiplication by the nonzero Z^l preserves this independence. Combining with disjoint grading values for different l gives at least

$$c\delta^{-1} R_* v_i N^{p+s} \tag{G.24}$$

independent ratios.

Every ratio just counted has cost and absolute grading value at most $C_0 NL$, where C_0 is independent of δ, M_0, q . The slab monomials cost $O(NL)$; the G_a products do too; and (G.21) multiplied by the exponent bound (G.23) costs $O(NL)$. A common h -denominator gives regular numerators of one finite character and degree $O(NL)$. By (G.17), eventually $v \geq w/2$, so the transferred upper bound (G.11) has a constant independent of the tilt. It contradicts (G.24) if δ is sufficiently small.

The order of choices is as follows. First fix the current stage constants, the old rows, the saturation constant and the original counting constant. Then fix $\delta > 0$ small enough for this last contradiction. Next fix M_0 sufficiently large compared with $1/\delta$ and the rounding constants. The character construction then gives a bound Q for q ; pass to a subsequence fixing q , or choose using Q . Only now choose a sufficiently large fixed N , and finally let i be large enough for all of these fixed-parameter assertions. Thus δ never tends to zero with i , and no eventual count is used at a variable product degree. This proves the claim. $\square$

Impose the new row by both signs and retain v as the new comparator. It is centred and still within $1 + o(1)$ of the original stage minimum. The limits of the enlarged rows stay independent because their transverse integer projections are fixed and independent. Each time (G.14) holds, Claim G.5 increases this rank. There are at most s such increases, and at most $s - 1$ with a retained nonzero anchor. Therefore this procedure reaches the alternative

$$\text{span}(\lim C_i) = \mathcal{R}_\infty^\perp. \tag{G.25}$$

This is strict finite progress within a stage.

G.7 Rational directions and anchor size

Fix a small positive number λ . Mix the comparator projection with the full current projection set:

$$K_i^\lambda = (1 - \lambda)x(v_0) + \lambda K_i.$$

These points have actual centred realizations q with $q \geq (1 - \lambda)v_0$ on all regular functions, by Lemma E.4. Their baseline heights interpolate exactly because the height depends only on h and the anchor coordinate. The bounds on $\mathfrak{B}$ show, for sufficiently small fixed λ , that their $\mathfrak{B}$ -normalized volumes are within any prescribed small fixed relative error of the original stage minimum. For example domination gives the volume factor $(1 - \lambda)^{-n}$, and the height changes by $O(\lambda)$ relative to its positive lower bound.

The radial hull $[0, 1]K_i^\lambda$ still has a full-dimensional limit in $\mathcal{R}_\infty^\perp$. Indeed K_i^λ contains $x(v_0)$, and differences from that point span $\lambda(K_i - x(v_0))$. Its linear span is therefore the same as that of K_i . The same statement holds in the limit. A compact convex set of full linear span containing zero has nonempty relative interior, so by Hausdorff convergence these radial hulls contain relative balls of radius bounded below in $\mathcal{R}_i^\perp$, unless that space has dimension zero. If there is an old anchor, choose such a ball away from anchor-coordinate zero: the comparator has nonzero anchor coordinate, and a convex combination with an interior point gives an interior point retaining that property. Radial truncation is used only to choose a direction; every nonzero direction chosen in it has an actual mixed representative, with the centred domination just established.

With an old anchor, write an integral row matrix as (b_i, A) , where A is its transverse part. After the subsequences already taken, A is a fixed integral matrix of full row rank, and the entries of b_i are $O(R_*)$. The feasible ratios $z_j = y_j/y_0$ satisfy

$$Az = -b_i. \quad (\text{G.26})$$

Directions in the ball away from $x_0 = 0$ give a relative ball of feasible ratios, in coordinates z/R_* . We can choose a point there with bounded common denominator and $z_j = O(R_*)$. To make the denominator assertion explicit, a fixed invertible minor of A gives a particular solution of (G.26) with denominator dividing its determinant and size $O(R_*)$. The rational kernel of A has a fixed lattice basis. Round in a fixed sufficiently fine denominator refinement of that lattice. Since $R_* \geq 1$, its mesh after division by R_* is uniformly small enough to enter the specified ball. If the affine solution space has dimension zero, the particular solution is already the required exact ratio. Thus the denominator is bounded and the numerator bounds are $O(L/L_1)$.

Without an old anchor, the full row matrix is fixed integral. If its orthogonal space is nonzero, the ball contains a nonzero rational direction of bounded complexity. Choose a nonzero coordinate as the new anchor and put $L_1 = L$. Choose the ball and direction so that this coordinate has absolute value at least a fixed positive constant; its actual mixed representative has that same lower bound or a larger one, since the radial factor lies in $(0, 1]$. If the orthogonal space has dimension zero, all selected values are already zero.

Impose the chosen proportions, or all zero values, by both-sign monomial tests. Their costs are $O(L)$: with an old anchor a row $d_j y_j - t_j y_0 = 0$ has cost $O(d_j L + |t_j| L_1)$ with d_j bounded and $|t_j| = O(L/L_1)$; with a new anchor both coefficients are bounded. The selected actual mixed representative survives all these equations, so the new slice contains a centred point. Old proportionalities are retained by the old tests. In the cases with no new anchor, the exact baseline minimum on the pinned slice is as close to the preceding working valuation w on all functions as desired, by choosing λ sufficiently small and applying the gap estimate on the original stage slice. In particular an old anchor still has $|y_0| \geq cL_1$, and domination by a fixed multiple of w holds.

When a new anchor is chosen, a further step is needed: equations on ratios alone do not prevent the minimizing vector from moving to zero. The prior baseline is then $q(h)$. Let z denote the centred mixed comparator on the pinned slice. Choose the sign of the anchor so that

$$p_0(q) = q(Y_0^{\text{sign}})/L_1, \quad p_0(z) \geq c_1 > 0.$$

Its ordinary volume is within a prescribed small fixed error of the prior ordinary minimum V_* . Let $M = \max p_0$ on this compact, $h = 1$ -normalized slice. Thus $M \geq c_1$ and $M = O(1)$. For $0 \leq \mu \leq (1 - \rho)/M$, consider

$$b_\mu(q) = q(h) - \mu p_0(q), \quad (\text{G.27})$$

where $\rho > 0$ is a small fixed number. These heights are bounded below by ρ and uniformly above. The exact minimizing ray varies continuously with μ , by Theorem E.8.

At the endpoint $\mu_* = (1 - \rho)/M$ the minimizer v_* has $p_0(v_*) \geq M/2$ for small enough ρ . Here are the estimates. Interpolate a maximizing witness for p_0 with z , assigning weight ρ to z . The interpolant u is centred, has $b_{\mu_*}(u) \leq 2\rho$, and (E.13), applied with $f = h$ and both inputs normalized by $h = 1$, gives

$$\text{vol}(u) \leq \rho^{1-n} \text{vol}(z).$$

Minimality and $\text{vol}(v_*) \geq V_*$ yield

$$b_{\mu_*}(v_*) \leq 2\rho^{1/n} (\text{vol}(z)/V_*)^{1/n} < \frac{1}{2}$$

for sufficiently small fixed ρ . Therefore $p_0(v_*) > M/[2(1 - \rho)] \geq M/2$.

If the minimizer at $\mu = 0$ already has $p_0 \geq c_1/4$, use $\mu = 0$. Otherwise continuity gives a nonempty open interval of parameters where $c_1/4 < p_0(v_\mu) < c_1/3$, and we choose a rational μ in that interval. Since $p_0(z) \geq c_1$, one has $b_\mu(z) < b_\mu(v_\mu)$. Tilted minimality consequently gives

$$\text{vol}(v_\mu) \leq \left(\frac{b_\mu(z)}{b_\mu(v_\mu)} \right)^n \text{vol}(z) \leq \text{vol}(z). \quad (\text{G.28})$$

The same final bound is immediate at $\mu = 0$. Thus the new minimizer still lies as near the original ordinary minimum as required, while its anchor value is bounded below by $c_1 L_1/4$. The gap estimate gives the required all-functions domination. Adopt this rational height as $\mathfrak{B}$ for all subsequent stages. It has the form (G.5) and bounds (G.4).

Finally take the rank refinement with error tending to zero, sufficiently small to preserve the anchor lower bound. This preserves the pinning equalities, the exact pair and test identities, and the independent symbols by Lemma G.2. It supplies balance and (C.5) and does not use the next-scale field and lattice assertions. Thus (I1)–(I5) hold with $L_- = L$, proving Proposition G.4. It remains to verify the two next-scale assertions.

G.8 Field and lattice consequences of the pin

Lemma G.6. *The working valuation w' furnished by Proposition G.4 satisfies (H1)–(H2) at the next scale $L_- = L$.*

Proof. The proposition supplies the new exact tests, rational proportions and, when present, the anchor lower bound. It also gives $w' \geq cw$ on all regular functions, so the transferred count (G.11) remains available with a fixed constant.

Use double bars for the initial symbols at w' . Let E be the field generated by the p chosen algebraically independent $\overline{G_a}$ and the s selected variable symbols. These have transcendence degree $p + s$, by Lemma G.2: they were independent together at w , since the selected variable weights were independent and the chosen G_a were neutral. Write E^0 for its subfield of weight zero. Every common test on the new cone is trivial on E^0 . Indeed, the both-sign pinning equations also hold on that cone. If there is an anchor, its nonzero value at w' ensures that two selected monomials with equal weight at w' have equal value at every common test; in the all-zero case this is immediate. Lemma G.2 then gives the predicted exact value for

each nonzero homogeneous polynomial. Ratios of homogeneous polynomials of equal weight generate E^0 , proving the assertion.

There is also a space V_0 of invariant ratios on this new cone, of cost and absolute grading value $O(L)$, with

$$\dim V_0 \geq cR_*v_i. \quad (\text{G.29})$$

Take the double-bar symbols of the whole G_a basis, and, if an old anchor was present, multiply by its powers with exponent from 0 to $\lfloor R_* \rfloor$. Their costs remain $O(L)$. They are linearly independent at w by their distinct anchor weights and the basis property, and remain so by Lemma G.2.

Choose an enlargement V of V_0 , with the same type of fixed cost and grading bounds, containing 1, the specified generators of E , and every neutral function in $S(LD_{\text{next}})$. We may also include in V a nonzero homogeneous ratio of weight a bounded positive multiple of any chosen new lattice vector k of width $O(L)$. For the latter use Corollary D.5 at w' : the refined cone satisfies (C.5), and tiny LD_{next} forces all new tests by the same integrality and cost argument as before. The space V is spanned by homogeneous ratios. Give all products of N members of V one common denominator h^t , where $t = O(NL)$; their regular numerators then have a single finite character and degree $O(NL)$. The transferred upper bound (G.11) applies.

If the field generated by V had transcendence degree $a > p + s$, Lemma G.3 and (G.29) would give at least $cR_*v_iN^a$ independent products, against $C'R_*v_iN^{p+s}$. Choose one sufficiently large fixed N and then let i be sufficiently large. This contradiction shows that the field generated by V is algebraic over E . Uniformity for the chosen k follows by applying this argument to any alleged sequence of failing choices.

A neutral element algebraic over E is algebraic over E^0 . In fact, clear denominators in an algebraic equation so that its coefficients are polynomials in the homogeneous generators of E , take a nonzero homogeneous part of the equation, and divide by a nonzero coefficient of that weight. All resulting coefficients have weight zero. Common tests are trivial on E^0 , and hence on these algebraic elements. The Gauss description and finite-cover lifting translate this back to every angular test in $\mathcal{L}$, proving assertion (H2) for the next scale.

The same algebraic equation argument shows that the weight of an algebraic homogeneous element lies in the rational span of the weights of E . The latter is the anchor line or zero. This proves the first part of assertion (H1). To bound its index in the line case write $\kappa'_0 = \ell k'_0$ with k'_0 primitive. Its width is $O(L_1)$ by the paired anchor costs. If $\ell \rightarrow \infty$, apply the bounded-multiple construction at the current scale L to all jk'_0 , $1 \leq j \leq \ell$. Their widths are $O(L_1)$ and $L_1 \leq L$. There are symbols of weights $q_j k'_0$, with $1 \leq q_j \leq Q$ for fixed Q , and cost and absolute grading value $O(L)$. These bounds are uniform over all these choices: otherwise a sequence of failing j_i would contradict the arbitrary-sequence version of Corollary D.5.

Let D_0 be a fixed common multiple of the bounded denominators of the finitely many pinned variable weights. The subgroup generated by those weights has primitive step $h_i \geq \ell/D_0$, and h_i divides ℓ . Among the integers $q_j j$, an integer has at most Q preimages, and each residue modulo h_i has at most $QD_0 + 1$ representatives in $[1, Q\ell]$. Hence at least

$$\frac{\ell}{Q(QD_0 + 1)}$$

distinct residues occur. All weights of the original space V_0 in (G.29) lie in this subgroup. With an old anchor its basis consists of neutral symbols multiplied by powers of that anchor; when the pin has just chosen a new anchor, V_0 consists only of neutral symbols. Multiplying this original V_0 by symbols representing distinct residues yields a direct sum, since the weight supports are disjoint. Its dimension is an unbounded multiple of R_*v_i , while its costs and

degrees are $O(L)$. The fixed- N version of (G.11) forbids this. Thus ℓ is bounded, completing assertion (H1). $\square$

G.9 Termination

Proof of Theorem G.1. Assume the excluded sequence exists. The initial data satisfy (I1)–(I5). At every stage, either the sole remaining anchor line gives the primitive-discrepancy contradiction above, or a nonempty independent block of new directions can be chosen. A bounded block is pinned directly by (G.1); a high block is pinned by the finite row procedure, the rational-direction choice, and, if required, the anchor-height adjustment. The next-stage invariants then hold by the field and lattice argument. Each nonterminal stage adds at least one new positive variable, and their initial symbols remain algebraically independent at the next working valuation. A function field of dimension n cannot contain more than n algebraically independent elements. Thus the process cannot have more than n nonterminal stages, whereas all its terminal alternatives have been excluded. This contradiction proves the theorem. $\square$

H From complement failure to the pinning input

We prove Theorem 2 by translating a failure of bounded klt complements into the sequence excluded by Theorem G.1. The construction uses a canonical cover when the birational morphism is an isomorphism, and a signed anti-canonical torsor otherwise. We then apply Chen’s equivalence and justify the passage between algebraically closed fields of characteristic zero.

H.1 A prime saturating the complete systems

Fix a dimension $d > 0$ and $0 < \epsilon \leq 1$, and suppose the birational complement assertion fails over $\mathbb{C}$. Birkar’s relative complement theorem gives an integer $b = b(d) > 0$ such that every example has an lc b -complement near its specified point [3, Theorem 1.8]. The hypotheses are precisely Fano type, the zero boundary, and nef $-K_X$; in particular no generalized nef-part index is being imposed. Choose a sequence with no klt M_i -complement, where $M_i > 1$ tends to infinity, $b \mid M_i$, and every fixed positive integer divides M_i eventually. For example, sufficiently large factorials have these properties. Failure of a uniform index allows this choice for each prescribed M_i .

We may assume $-K_X$ is ample over Z . On a variety of Fano type, the relative Mori dream space consequences of the klt MMP imply that nef $-K_X$ is semiample [8]. Since f is birational, this divisor is relatively big. Its ample model is thus a birational contraction $g : X \rightarrow X^a$ over Z . A divisible multiple of $-K_X$ is the pullback of its ample model divisor. Pushing compatible canonical divisors forward identifies the latter with the corresponding multiple of $-K_{X^a}$, so $K_X = g^*K_{X^a}$ after compatible choices. Equivalently, a possible rational principal adjustment descends because the function fields agree. Thus the contraction is crepant, X^a remains ϵ -lc and of Fano type over Z , and klt complements on it pull back to klt complements on X [29]. Applying the lc complement theorem on this model and replacing X by it preserves the failure under consideration.

Shrink Z to an affine neighbourhood of the specified point. For every integer k , a section of $\mathcal{O}_X(-kK_X)$ is represented by a rational function s with

$$\Delta_s = \operatorname{div}(s) - kK_X \geq 0. \quad (\text{H.1})$$

The degree- b complement is represented by s_b and satisfies $(X, \Delta_{s_b}/b)$ lc. Here and below all section systems are complete relative systems on this neighbourhood.

Proposition H.1 (Factorial saturation). *For each example there is a primitive divisorial order w over X whose image centre on Z contains the specified point and for which*

$$w(\Delta_s) \geq kA_X(w) \quad (k > 0, k \mid M_i), \quad w(\Delta_{s_b}) = bA_X(w). \quad (\text{H.2})$$

After replacing the example by a geometric generic slice, one may assume that the image centre is a closed point z , that $1 \leq \dim X = \dim Z \leq d$, and that all the inequalities hold for the complete local systems at z .

Proof. The coherent sheaf $f_*\mathcal{O}_X(-M_iK_X)$ has finitely many generators on the affine base. Resolve their fractional ideal of ratios to a fixed section, together with the relevant divisors. On the resulting smooth model $p : Y \rightarrow X$, their pulled-back divisor system has a common fixed $\mathbb{Q}$ -divisor F and a basepoint-free residual system. This construction is valid even if M_iK_X is not Cartier: each section divisor is $\mathbb{Q}$ -Cartier, their differences are principal, and principalizing the fractional ratio ideal makes the moving differences Cartier. The fixed divisor has order equal to the minimum of the section orders at every prime on the model.

The section $s_b^{M_i/b}$ belongs to this system. Therefore $F/M_i \leq p^*(\Delta_{s_b}/b)$, and the fixed-order discrepancy

$$A_X(E) - \frac{\text{ord}_E(F)}{M_i}$$

is nonnegative for every prime E over X . Take the support on Y to be simple normal crossings. A general residual member, with coefficient $1/M_i < 1$, is transverse to this support. If every fixed-order discrepancy were strictly positive over a neighbourhood of the specified point, this general member would give a klt M_i -complement there. The only possible zero discrepancies of the fixed log smooth data come from strata containing a coefficient-one component. There are finitely many such components, and their images in Z are closed because $Y \rightarrow Z$ is proper. Thus the absence of a klt complement on every neighbourhood implies that one component E with image containing the specified point has $\text{ord}_E(F) = M_iA_X(E)$. Set $w = \text{ord}_E$.

For $k \mid M_i$ and any section s in degree k , its power $s^{M_i/k}$ is in degree M_i . Hence $(M_i/k)w(\Delta_s) \geq \text{ord}_E(F) = M_iA_X(w)$. The same inequality for s_b , together with lc of the chosen complement, gives equality in degree b . These statements also apply to local sections: coefficients in the local base ring have nonnegative w -order, and shrinking the base only inverts functions of order zero whenever the centre still contains the specified point.

Write W for the image centre of w on Z . It is a proper closed subset: $w(\Delta_{s_b}) > 0$ and a divisorial centre cannot dominate the birational base. If $q = \dim W > 0$, choose a general linear projection $Z \rightarrow \mathbf{A}^q$ that is dominant and generically finite on W . Pass to the geometric generic fibre and follow a component of E . The image of that component in the new base is a closed point, and the new dimension is $\dim X - q \geq 1$.

Here are the required preservation details. The prime E is horizontal over $\mathbf{A}^q$. After shrinking its base, the chosen resolution and its finitely many relevant strata are smooth wherever they dominate this base. Dividing a top form by a base volume form therefore restricts compatible canonical divisors to the generic fibre, preserving the log discrepancies and section-divisor orders. Extending the generic constant field separably does not ramify these prime orders: the nonzero discriminant of a finite separable constant extension is a unit in their valuation rings. Geometric integrality can be obtained by choosing the corresponding components after this extension.

Divisorial sheaves restrict to the corresponding divisorial sheaves by localization and separable scalar extension, as is seen on the smooth locus and then in codimension one. Flat

base change for the proper morphism shows that the original generators generate the new complete systems; only the finitely many degrees dividing M_i and the degree b are needed here. Their inequalities hence hold for every local section at the new image point. Normality, birationality, the discrepancy bound, relative ampleness of $-K_X$, and a Fano type witness are preserved by this operation. Thus all the conclusions survive the slice. The algebraically closed generic field of the complex base is isomorphic to $\mathbb{C}$; alternatively the finite data can again be embedded into $\mathbb{C}$. We retain the notation X, Z, z, w for these new data. $\square$

After a subsequence the new dimension and which of the following two constructions applies are fixed. We keep the original bounded integer b , even if the dimension has decreased.

H.2 The canonical cover and the signed torsor

Lemma H.2 (The isomorphic case). *Suppose f is an isomorphism near z . There is a normal integral cyclic cover $S_0 \rightarrow X$, with a unique point o over z , a quasi-étale cyclic action fixing o , and a divisor-free canonical generator of character one. It has a regular eigenfunction $f_b = s_b t^b$ of character b and a divisorial valuation w_0 centred at o such that, with $A_0 = A_{S_0}(w_0) > 0$,*

$$(S_0, \operatorname{div}(f_b)/b) \text{ is lc}, \quad w_0(f_b) = bA_0, \quad w_0(a) \geq kA_0 \quad (\text{H.3})$$

for every regular eigenfunction a of character k when $k > 0$ and $k \mid M_i$. The variety S_0 is klt.

Proof. Let r be the local Cartier index of K_X and shrink so that $rK_X = \operatorname{div}(c)$. Normalize X in

$$t^r = c^{-1},$$

with the cyclic group acting on t by its standard character. The cover is integral of degree r . Indeed a smaller Kummer degree would make c^{-1} a p th power for a prime $p \mid r$, and then $(r/p)K_X$ would be principal, contrary to minimality. At a prime of X , K_X is Cartier and the exponent of c is divisible by r . The codimension-one Kummer calculation shows that the cover is quasi-étale. If $\operatorname{div}(\omega_X) = K_X$, the form $t\omega_X$ has divisor zero upstairs. Finite crepant pullback proves that S_0 is klt.

A fibre of this finite quotient is a single orbit: invariant functions separate distinct finite orbits. If the fibre over z had more than one point, regular functions separating its points would have a nontrivial eigenspace nonzero at one, hence at every, point of that orbit. After shrinking by its invariant norm, it would be an eigenunit. Write it st^j with $0 < j < r$. The codimension-one calculation gives $\operatorname{div}(s) = jK_X$, contradicting the minimal index. Thus the fibre is one fixed point.

For every integer k , the regular functions of character k are precisely st^k with $s \in \mathcal{O}_X(-kK_X)$; changing k by r is accommodated by $t^r = c^{-1}$. Their divisors are the pullbacks of Δ_s . Extend w to a primitive divisorial valuation w_0 on the cover. Its centre is o . If its restriction is ew , finite crepant pullback gives $A_{S_0}(w_0) = eA_X(w)$ and $w_0(st^k) = ew(\Delta_s)$. Equation (H.2) proves all the assertions. $\square$

For the second construction, retain the grading variable t as an indeterminate, and write $K = \mathbb{C}(X) = \mathbb{C}(Z)$. Choose a rational top form ω_X with $\operatorname{div}(\omega_X) = K_X$. The positive degrees record the anti-canonical section inequalities. We also keep negative degrees: after restriction to a finite scalar group, one character space contains degrees of both signs. Bounds on both signs will therefore be needed to control every section of a fixed character.

Lemma H.3 (The signed anti-canonical torsor). *Suppose f is not an isomorphism near z . The signed algebra*

$$R = \bigoplus_{k \in \mathbb{Z}} H^0(X, \mathcal{O}_X(-kK_X))t^k, \quad S_0 = \operatorname{Spec} R, \quad (\text{H.4})$$

over the affine base is a finitely generated normal domain of dimension $\dim X + 1$. Write $R_k = H^0(X, \mathcal{O}_X(-kK_X))t^k$ for its homogeneous part of degree k . Its positive chamber is the open

$$S_0^+ = \bigcup_{k > 0} \bigcup_{0 \neq a \in R_k} D_{S_0}(a),$$

where $D_{S_0}(a)$ is the principal open on which a does not vanish. This chamber has complement of codimension at least two. The form

$$\Omega_0 = t\omega_X \wedge d \log t \quad (\text{H.5})$$

is a divisor-free canonical generator of grading weight one, and every finite scalar grading subgroup acts quasi-étalely.

The variety S_0 is klt, and $(S_0, \operatorname{div}(s_b t^b)/b)$ is lc. There is a grading-fixed closed point o over z and a rational Gauss valuation w_0 centred there which satisfies (H.3) for every actual positive degree $k \mid M_i$. Moreover there are numbers $\alpha > 0$ and $F_w > \alpha$, depending on the example, for which every nonzero homogeneous section satisfies

$$w_0(st^k) \geq \begin{cases} k\alpha, & k > 0, \\ |k|(F_w - \alpha), & k < 0. \end{cases} \quad (\text{H.6})$$

Proof. Finite generation and normality. The positive algebra is finitely generated by relative ampleness of $-K_X$. For the negative algebra, take a small $\mathbb{Q}$ -factorialization and a K -MMP over Z . Fano type gives termination and a semiample output, since every divisor is relatively big for a birational contraction [8]. The integral-degree section spaces are preserved in these steps. To check this without assuming a Cartier index in degree one, on a common model write $p^*K_{X_1} = q^*K_{X_2} + E$ for a K -MMP step, where E is effective and q -exceptional. An effective section divisor $\operatorname{div}(s) + mK_{X_1}$ pushes to an effective section divisor on X_2 . Conversely, the pullback of an effective section divisor on X_2 , plus mE , is effective, and pushes to its section divisor on X_1 . These are rational pullbacks, valid for every integral $m > 0$ because the divisors involved are $\mathbb{Q}$ -Cartier; one can equivalently verify the inequalities after taking a Cartier power. The initial small modification identifies the divisorial sheaves in codimension one. On the semiample output Y , let $g : Y \rightarrow Y^c$ be the semiample contraction and choose $rK_Y = g^*L$ with L relatively ample. For each $0 \leq j < r$, the projection formula identifies the residue module with

$$\bigoplus_{a \geq 0} H^0(Y^c, g_* \mathcal{O}_Y(jK_Y) \otimes \mathcal{O}_{Y^c}(aL)).$$

The pushforward is coherent by properness, so this is finite over the ample section ring. The Cartier Veronese is finitely generated and the finitely many residue modules are finite over it. The negative algebra is therefore finitely generated. The union of generators of the two half-algebras generates R .

The ring R is the intersection of $K[t, t^{-1}]$ with the Gauss valuation rings specified, for every prime D of X , by

$$\operatorname{ord}_D(s) - k \operatorname{coeff}_D(K_X) \geq 0 \quad \text{on } st^k.$$

This follows from the codimension-one description of a divisorial sheaf, and proves normality. Its fraction field is $K(t)$: the fraction field of degree zero is K , and the generic fibre contains every integral degree. In particular $\dim S_0 = \dim X + 1$.

The positive chamber and the canonical form. We prove that every prime divisor meets the positive chamber, where the canonical form can be computed on a Laurent bundle. Suppose that a prime divisor P lies outside it. The connected grading torus preserves every irreducible component of this complement, so P is homogeneous. It cannot contain all nonzero graded pieces. Indeed the locus where they all vanish is a closed subscheme of Z missing its generic point, where the signed algebra is $K[t, t^{-1}]$. That locus has dimension at most $\dim Z - 1 = \dim S_0 - 2$.

Choose a negative homogeneous element q of degree $-m$ not in P . Set

$$A = R[q^{-1}] = R_{\leq 0}[q^{-1}], \quad C = A_0.$$

The equality follows by multiplying any positive-degree element by sufficiently many powers of q . The ring $C[q, q^{-1}]$ is a Laurent polynomial ring: its nonzero powers have different degrees. For a homogeneous $a \in A_k$ the identity

$$a^m q^k \in C, \quad a^m = (a^m q^k) q^{-k} \tag{H.7}$$

proves integrality over this Laurent ring. Both formulas are valid for negative k , because q is a unit on this chart. Finite generation then makes A finite over $C[q, q^{-1}]$. The ring C contains the affine base ring R_0 and lies in K , so $\text{Frac } C = K$.

The height-one homogeneous prime induced by P contracts to a height-one homogeneous prime of $C[q, q^{-1}]$. Since q is invertible, that contracted prime is generated by its intersection with C and lies over a prime divisor of $\text{Spec } C$, a chart of the negative Proj. This also excludes trivial restriction to K : there is no homogeneous prime divisor solely in the invertible Laurent direction. The negative Proj is the relatively ample K -model obtained by the small modification, the K -MMP and its semiample contraction. None of these operations extracts divisors. The prime in question therefore has a strict transform D on X ; its restriction has the form $w' = e \text{ord}_D$, $e > 0$.

There are sections of both sufficiently divisible signs whose Δ -orders at D are zero. For the positive sign this is relative generation of a multiple of $-K_X$. For the negative sign use a generated multiple of K on its ample model avoiding the surviving prime. The birational map identifies the two generic DVRs and their compatible canonical coefficients. Its section lifts through the preserved section space, and the effective MMP correction has zero coefficient at this strict divisor. Thus its Δ -order on X is also zero. Relative generation supplies both sections globally over the affine base with the indicated degree signs.

Let $v = \text{ord}_P$, and put $\beta = v(t) + w'(K_X)$, where $w'(K_X)$ means rational pullback order. For a homogeneous element,

$$v(st^k) = w'(\Delta_s) + k\beta. \tag{H.8}$$

Nonnegativity of v on the two zero-multiplicity sections forces $\beta \geq 0$ and $\beta \leq 0$. Hence $\beta = 0$, and the positive section just chosen has value zero, contrary to P being outside the positive chamber. This proves the codimension assertion.

On the positive chamber the signed algebra is the usual Laurent divisorial bundle over X . To see this on a Proj chart, localize at a divisible positive section. Poles along its complementary Cartier divisor can be cleared by its powers, identifying every degree piece with the corresponding localized divisorial sheaf on X . A Cartier Veronese is a line bundle with its zero section removed, and the full chart is finite over that bundle by taking powers. In particular the inverse image of a codimension-at-least-two subset of X has codimension at least two. At a regular codimension-one point write $K_X = \text{div}(c)$; then ct is an invertible Laurent coordinate and

$$t\omega_X \wedge d \log t = (ct)(\omega_X/c) \wedge d \log(ct).$$

This is a nowhere-vanishing top form, and every nonidentity finite scalar acts freely there. The established codimension-two complements extend these conclusions to the assertions about Ω_0 and quasi-étaleness on S_0 .

Discrepancies and the lc boundary. The form also controls valuations whose centres lie outside the positive chamber. We will use its discrepancy estimate both for the boundary defined by $s_b t^b$ and for the finite quotients in Lemma H.4. If a divisorial multiple v centred on S_0 has nontrivial restriction w' to K , the relative Abhyankar inequality makes w' a divisorial multiple: its value group is discrete, and adjoining t increases residue transcendence degree by at most one, forcing its residue field to have transcendence degree $\dim X - 1$ over $\mathbb{C}$. Its centre exists on X by properness. Relative generation of a positive Cartier multiple gives $\beta = v(t) + w'(K_X) \geq 0$. The top form gives

$$A_{S_0}(v) \geq A_X(w') + \beta, \quad (\text{H.9})$$

with equality for a Gauss extension. One can compute this on a smooth model at the prime defining w' . Choose a uniformizer x and separating residue coordinates $y_1, \dots, y_{\dim X - 1}$. The ratio of ω_X to $d \log x \wedge dy_1 \wedge \dots \wedge dy_{\dim X - 1}$ has w' -order $A_X(w') + w'(K_X)$. After wedging with $d \log t$, the remaining logarithmic differential has nonnegative log order at v . Indeed, at an upstairs uniformizer ρ , every logarithmic differential has at most a simple $d\rho/\rho$ pole, at most one such factor survives a wedge, and the differentials of the residue-coordinate units are regular. At a Gauss valuation its residues are independent and this log order is zero. Multiplication by t yields (H.9).

If w' is trivial, work on the smooth generic fibre $\text{Spec } K[t, t^{-1}]$; a centred valuation there has $v(t) = 0$ and positive discrepancy. Thus S_0 is klt. For a nontrivial restriction, (H.8), (H.9), and lc of the actual degree- b complement give

$$A_{S_0}(v) - \frac{v(s_b t^b)}{b} \geq A_X(w') - \frac{w'(\Delta_{s_b})}{b} \geq 0.$$

For trivial restriction, $v(s_b t^b) = 0$. Consequently the displayed boundary pair is lc. It has the lc place constructed next and is not being asserted to be klt.

A centred order with two slopes. It remains to construct the closed point and its saturated valuation. The positive slope will come from a choice of Gauss weight; the negative slope comes from vanishing on the fibre. Every irreducible component of the reduced fibre over z has positive dimension. An isolated zero-dimensional component would disconnect the fibre unless the whole fibre were finite; a proper birational morphism to a normal base with a finite fibre is an isomorphism after shrinking, the excluded case. A negative-degree section, raised to a Cartier power, restricts on each such projective component to a section of an antiample line bundle. It is therefore zero on every component. Its effective divisor contains the entire reduced fibre, so its order at the specified w is strictly positive. This argument only concerns reduced supports and also covers nonreduced fibres.

Choose finitely many negative-ring generators $s_j t^{-m_j}$ over the degree-zero ring and set

$$F_w = \min_j \frac{w(\Delta_{s_j})}{m_j} > 0.$$

Every negative homogeneous section is a homogeneous polynomial in these generators with base coefficients of nonnegative w -order. It follows that $w(\Delta_s) \geq |k| F_w$ for $k < 0$. Choose a rational $0 < \alpha < F_w$, and let w_0 be the Gauss extension of w with $w_0(t) + w(K_X) = \alpha$. Formula (H.8) defines it, and gives (H.6); for the positive sign use effectivity of Δ_s . All nonzero degrees have strictly positive value, while the degree-zero maximal ideal is $\mathfrak{m}_z$. Thus w_0 is centred at the fixed closed point with ideal generated by these elements. A rational Gauss extension of a divisorial valuation is itself divisorial up to scaling. Equality in (H.9) gives $A_0 = A_X(w) + \alpha$, and (H.2) now yields (H.3) in actual positive degrees. $\square$

H.3 The quotient gap and matching the canonical character

In the canonical-cover case retain its cyclic group. In the torsor case choose any finite scalar subgroup μ_r ; its order will be fixed below, after the individual slopes in (H.6) are known. All characters in the next lemma are expressed relative to the standard character of this cyclic group.

Lemma H.4 (Padding and the actual quotient discrepancy). *In either construction, set*

$$S = S_0 \times \mathbf{A}^{b-1}, \quad (o, 0) \in S,$$

and give every added coordinate u_j character one. Then S is klt, the finite quotient $Y = S/\mu_r$ is ϵ -lc, and the canonical generator on S has character b . There is a regular function h of character b such that $(S, \text{div}(h))$ is lc and a centred quasi-monomial valuation T_0 such that

$$A_S(T_0) = T_0(h) = bA_0. \quad (\text{H.10})$$

Proof. The product is klt. The action remains quasi-étale: the fixed locus of every nonidentity element has codimension at least two already in S_0 . Wedging its canonical generator with $du_1 \wedge \cdots \wedge du_{b-1}$ gives a divisor-free generator of character b . A suitable power descends through the finite quotient, so K_Y is $\mathbb{Q}$ -Cartier and the quotient map is crepant.

Quasi-étaleness alone would not preserve a prescribed discrepancy gap. Let E be an actual prime divisor over Y , and normalize an extension v to $\mathbb{C}(S)$ so that its restriction to $\mathbb{C}(Y)$ is ord_E . Finite crepant pullback, with homogeneous discrepancies, gives $A_Y(E) = A_S(v)$. Although v may take fractional values upstairs, the field $K = \mathbb{C}(X)$ is pointwise invariant and is contained in $\mathbb{C}(Y)$. If $w' = v|_K$ is nontrivial, its values are integral. The relative Abhyankar inequality then gives

$$w' = e \text{ord}_F \quad \text{for a prime } F \text{ over } X \quad \text{and an integer } e \geq 1.$$

For the torsor, the log-form calculation above, with the additional coordinate forms, yields

$$A_S(v) \geq A_X(w') + \beta + \sum_j v(u_j) \geq e\epsilon \geq \epsilon.$$

Here $\beta \geq 0$ by positive generation, each u_j is regular, and the non-Gauss logarithmic differential defect is nonnegative. For the canonical cover, first restrict to $\mathbb{C}(S_0)$ and use the product log-form estimate; finite crepant pullback from S_0 to X then gives the same lower bound $A_X(w')$.

If w' is trivial, the valuation lies over the common generic point of the birational varieties X and Z . The generic quotient is smooth: the canonical cover there is a cyclic torsor, and the signed construction is a Laurent line with a free scalar action; their products with the coordinate representation have smooth quotients. Generic restriction of the ordinary prime E is still a primitive prime, since localization at the trivially valued base field does not change its value group or uniformizer. Its discrepancy on this smooth generic quotient is at least one, hence at least ϵ . This proves the required gap for every actual quotient prime.

Put $f_b = s_b t^b$ and take a general combination

$$h = a_0 f_b + a_1 u_1^b + \cdots + a_{b-1} u_{b-1}^b. \quad (\text{H.11})$$

We check lc using the ideal $I = (f_b, u_1^b, \dots, u_{b-1}^b)$. For any divisorial valuation v with nontrivial restriction v_0 to $\mathbb{C}(S_0)$, that restriction is divisorial up to scaling and the product differential calculation gives

$$A_S(v) \geq A_{S_0}(v_0) + \sum_j v(u_j) \geq \frac{v(f_b)}{b} + \sum_j v(u_j) \geq \min\{v(f_b), bv(u_1), \dots, bv(u_{b-1})\}.$$

If the restriction is trivial, $v(f_b) = 0$ and the same ideal inequality holds because S is klt. Therefore (S, I) is lc. On a resolution of the ideal, a general residual member is smooth and transverse to the fixed log smooth boundary; with coefficient one this proves $(S, \text{div}(h))$ lc. All summands of h have character b .

Finally extend w_0 by the joint Gauss valuation giving every u_j weight A_0 . It is centred at $(o, 0)$ and quasi-monomial, and its product discrepancy is $A_0 + (b-1)A_0 = bA_0$. Every summand of h has that value. Their initials do not cancel for a general choice of the a_j (the added coordinates are independent), giving (H.10). When $b = 1$ there are no added coordinates and h is simply a nonzero scalar multiple of f_b . $\square$

Proposition H.5 (The pinning input). *The failure sequence produces, after a subsequence, a fixed-dimensional sequence satisfying every hypothesis of Theorem G.1.*

Proof. Let χ be character b of the chosen cyclic group; it need not be faithful. Normalize $T = T_0/(bA_0)$. The preceding lemma gives

$$A_S(T) = T(h) = 1, \quad (S, \text{div}(h)) \text{ lc}, \quad S/\mu_r \text{ is } \epsilon\text{-lc},$$

and h and the divisor-free canonical generator both have character χ . It remains to check, for each fixed positive integer m , that eventually

$$T(a) \geq m \quad \text{for every regular eigenfunction } a \text{ of character } \chi^m. \quad (\text{H.12})$$

Expand a polynomial eigenfunction in monomials in the added coordinates. A monomial of total degree j has T_0 -value jA_0 . If $j \geq bm$, this already suffices, since its coefficient is regular and has nonnegative value. Otherwise its coefficient on S_0 has character

$$k = bm - j > 0.$$

In the canonical-cover case, that entire eigenspace is the degree- k section space described in Lemma H.2. Since every fixed k divides M_i eventually, its coefficients have value at least kA_0 . The monomial therefore has value at least $(j+k)A_0 = bmA_0$.

In the signed torsor, coefficients decompose into actual degrees $k + \ell r$, $\ell \in \mathbb{Z}$. For $\ell = 0$ use the same factorial saturation. For $\ell \neq 0$ use both strict slopes in (H.6). At example i , choose $r_i > i$ so large that

$$(r_i - i) \min\{\alpha, F_w - \alpha\} \geq iA_0. \quad (\text{H.13})$$

For $1 \leq k \leq i$, every nonzero residue shift has $|k + \ell r_i| \geq r_i - i$, so its value is at least $iA_0 \geq kA_0$. This also gives the desired monomial bound. The choice is made after α, F_w, A_0 are known; no uniform lower bound on either slope is needed. For fixed m , eventually $bm \leq i$, and every required unshifted degree divides M_i .

A finite sum cannot have value smaller than the minimum of its terms. The argument therefore proves (H.12) in the affine coordinate ring. It also proves it in the local ring at the fixed point. Indeed any local function can be written with an invariant unit denominator: multiply an initial denominator by its other group translates and use its invariant norm. For an eigenfunction the resulting numerator has the same character. The invariant denominator has value zero, so the inequality is unchanged.

The dimensions satisfy

$$\dim S = \begin{cases} \dim X + b - 1, & \text{canonical-cover case,} \\ \dim X + b, & \text{signed-torsor case,} \end{cases} \quad \text{and hence} \quad \dim S \leq d + b.$$

They are thus bounded independently of i , and a subsequence has fixed dimension. The valuations are centred and quasi-monomial, the covers are finite cyclic and quasi-étale, and every remaining requirement of Theorem G.1 was checked above. $\square$

H.4 Chen's reduction and change of field

We use the following form of Chen's reduction. For every positive integer ℓ , Chen's Theorem 1.6, assertions (2) and (3), identifies the following two assertions [10]:

- (i) bounded monotonic klt complements for birational Fano type contractions with zero boundary and $\dim Z \leq \ell$;
- (ii) the Cartier-section assertion for Fano type contractions with $\dim Z \leq \ell$, with a constant depending on the total-space dimension and the discrepancy gap.

In (i), the total-space and base dimensions are equal. Consequently, Theorem 2 in each of the finitely many positive dimensions at most ℓ , for every positive discrepancy parameter, supplies (i); its dimension-zero case is trivial. One may take a common multiple of the finitely many complement indices for each fixed discrepancy parameter. Thus the birational inputs have dimension at most ℓ , whereas assertion (ii) also allows total spaces of higher dimension.

The real-pair conventions in Chen's statement cause no extra hypothesis in (i). If an integral Weil divisor K_X is $\mathbb{R}$ -Cartier, express it locally as a finite real combination of Cartier divisors. Comparing Weil coefficients gives a finite linear system with rational coefficients and rational right-hand side. A real solution implies a rational solution, so K_X is $\mathbb{Q}$ -Cartier. Similarly, a real effective log Fano witness can be replaced by a rational one. Keep its finite boundary support and its zero coefficients. On a finite open cover, choose Cartier representations of its log canonical class, using the same boundary coefficient variables on every chart. These impose finitely many rational affine equations. Rational points are dense in their real solution space. Every rational solution defines a global rational divisor whose log canonical divisor is locally $\mathbb{Q}$ -Cartier; clearing denominators on the finite cover makes it $\mathbb{Q}$ -Cartier on X . A rational affine basis of the solution space therefore places all its log canonical divisors in a fixed finite-dimensional real span of global $\mathbb{Q}$ -Cartier divisors. Relative ampleness is open in that span. On a fixed log resolution, the klt condition and positivity of the remaining boundary coefficients also persist under a sufficiently small perturbation. Hence the rational birational statement proves the full zero-boundary assertion needed in Chen's real-pair formulation.

Lemma H.6 (Point ideals and change of field). *It suffices to prove Theorem 1 over $\mathbb{C}$. More precisely, a uniform positive lower bound for the log canonical threshold of $\mathfrak{m}_z \mathcal{O}_X$ implies the Cartier-section assertion after replacing the bound by a smaller number less than one. Both the hypotheses and this ideal threshold are preserved by algebraically closed field extensions of characteristic zero.*

Proof. Shrink Z to an affine neighbourhood of z , choose generators $g_1, \dots, g_N$ of $\mathfrak{m}_z$, and resolve both (X, B) and the pulled-back ideal. If $p : Y \rightarrow X$ is the resulting log resolution, write

$$K_Y + B_Y = p^*(K_X + B), \quad \mathfrak{m}_z \mathcal{O}_Y = \mathcal{O}_Y(-F).$$

The ideal threshold is the minimum of $A_{X,B}(E)/\text{ord}_E(F)$ over the finitely many components of F having positive order. Its interpretation is lc on the inverse image of a neighbourhood of z . Indeed the ideal is the unit ideal away from $f^{-1}(z)$, and the original pair is already lc there.

For a general linear combination $g = \sum a_j g_j$, the pullback of $\text{div}(g)$ is $F + H$, where H is a general member of a basepoint-free system. On a further restriction of the base, H is smooth and transverse to the fixed simple normal crossings support. If $0 < t < 1$ is at most the ideal threshold, then $(Y, B_Y + tF + tH)$ is lc, hence so is $(X, B + tf^* \text{div}(g))$. Choose $g \neq 0$. Since $g \in \mathfrak{m}_z$, $D = \text{div}(g)$ is a nonzero effective Cartier divisor containing z . Conversely, a Cartier divisor through z has a local equation $g \in \mathfrak{m}_z$, so that $v(\mathfrak{m}_z \mathcal{O}_X) \leq v(f^*g)$ for every valuation. Thus an admissible coefficient for that divisor is also a lower bound for the ideal threshold.

All the varieties, morphisms, rational boundaries, a rational Fano type witness, generators, and the chosen resolutions descend to an algebraically closed subfield k_0 of finite transcendence degree over $\mathbb{Q}$. Such a field embeds in $\mathbb{C}$. Normality, geometric integrality and the identity $f_*\mathcal{O}_X = \mathcal{O}_Z$ are unchanged by algebraically closed extension; the latter also follows from flat base change for the proper morphism. Projectivity and relative ampleness are geometric properties. Relative nefness is geometric as well: any curve on an extension can be spread over a finite-type parameter scheme and specialized to a curve cycle over the smaller algebraically closed field, with the same intersection number. Thus a negative intersection after extension would already yield a negative intersection before extension.

For discrepancies one can use the same log resolution. In the nonvacuous range $0 < \epsilon \leq 1$, a log smooth sub-boundary with all coefficients at most $1 - \epsilon$ is ϵ -lc. At a stratum this follows by comparing with its reduced simple normal crossings support: a primitive monomial order has at least one positive integral weight, and each boundary weight contributes at least ϵ ; a nonmonomial prime has a further nonnegative discrepancy, with an integral contribution at least one when needed away from the strata. This also shows preservation of the discrepancy bound after extension. The coefficients and orders in the displayed resolution formula are unchanged, so the ideal threshold is unchanged. The complex assertion therefore gives the same lower bound over k_0 and over the original field, where the preceding general-member construction finishes the proof. $\square$

H.5 Conclusion of the two theorems

Proof of Theorem 2. Over $\mathbb{C}$, a failure in any fixed dimension and with any fixed $0 < \epsilon \leq 1$ would, by Proposition H.5, give the sequence prohibited by Theorem G.1. Therefore a uniform positive integral complement index exists. The range $\epsilon > 1$ in positive dimension is vacuous, since a prime divisor on the smooth locus has log discrepancy one.

The birational conclusion also descends to every algebraically closed field of characteristic zero. Increase the complex uniform index to a multiple greater than one, which preserves the complement assertion. Descend a given example, its Fano type witness, and a resolution to an algebraically closed field embeddable in $\mathbb{C}$. For the complex uniform index n , the complete relative system $f_*\mathcal{O}_X(-nK_X)$ commutes with these field extensions. Resolve its finitely generated fractional base ideal as in Proposition H.1. A klt member over $\mathbb{C}$ implies that every fixed-order discrepancy relevant to the fibre is strictly positive. Those discrepancies are unchanged by extension. A general member over the original infinite field consequently gives a klt n -complement after shrinking, since the general residual coefficient $1/n$ is less than one. Its section divisor Δ_s defines $\Delta = \Delta_s/n \geq 0$, and $n(K_X + \Delta) = \text{div}(s)$ is Cartier and linearly trivial on the chosen neighbourhood, as required. $\square$

Proof of Theorem 1. First work over $\mathbb{C}$ and fix the total-space dimension d and gap $\epsilon > 0$. Theorem 2 in every dimension $1, \dots, d$ and for every positive discrepancy parameter, together with its trivial dimension-zero case and the rationalization above, verifies Chen's assertion (2) for base dimension bound $\ell = d$. His Theorem 1.6 therefore gives assertion (3), with a positive constant depending only on d and ϵ [10]. Every base of a contraction from a d -dimensional variety has dimension at most d , so this applies to the given f . It supplies lc on the full inverse image of a neighbourhood of z . Lemma H.6, decreasing the coefficient below one if necessary, transports the uniform constant to an arbitrary algebraically closed characteristic-zero field and produces a nonzero effective Cartier divisor through z there. This proves the stated neighbourhood assertion. $\square$

Proof of Corollary 3. The same application of Chen's implication (2) $\Rightarrow$ (3) over $\mathbb{C}$ gives his full real-boundary assertion. Only the zero-boundary birational inputs to (2) were rationalized

above. The resulting assertion (3) already allows arbitrary effective real B with $K_X + B$ real Cartier, and supplies the same dependence on d and ϵ . No perturbation of the nef real boundary is involved. $\square$

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